PP101
Introduction to Pulp Making

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Topics

1. Introduction to Pulp industry
2. Characteristic of wood and wood pulp
3. Wood and chip handling
4. Overview of Pulping methodology
5. Bleaching sequences
6. Principle of chemical recovery
7. Properties and Testing of pulp

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1. Introduction to Pulp industry

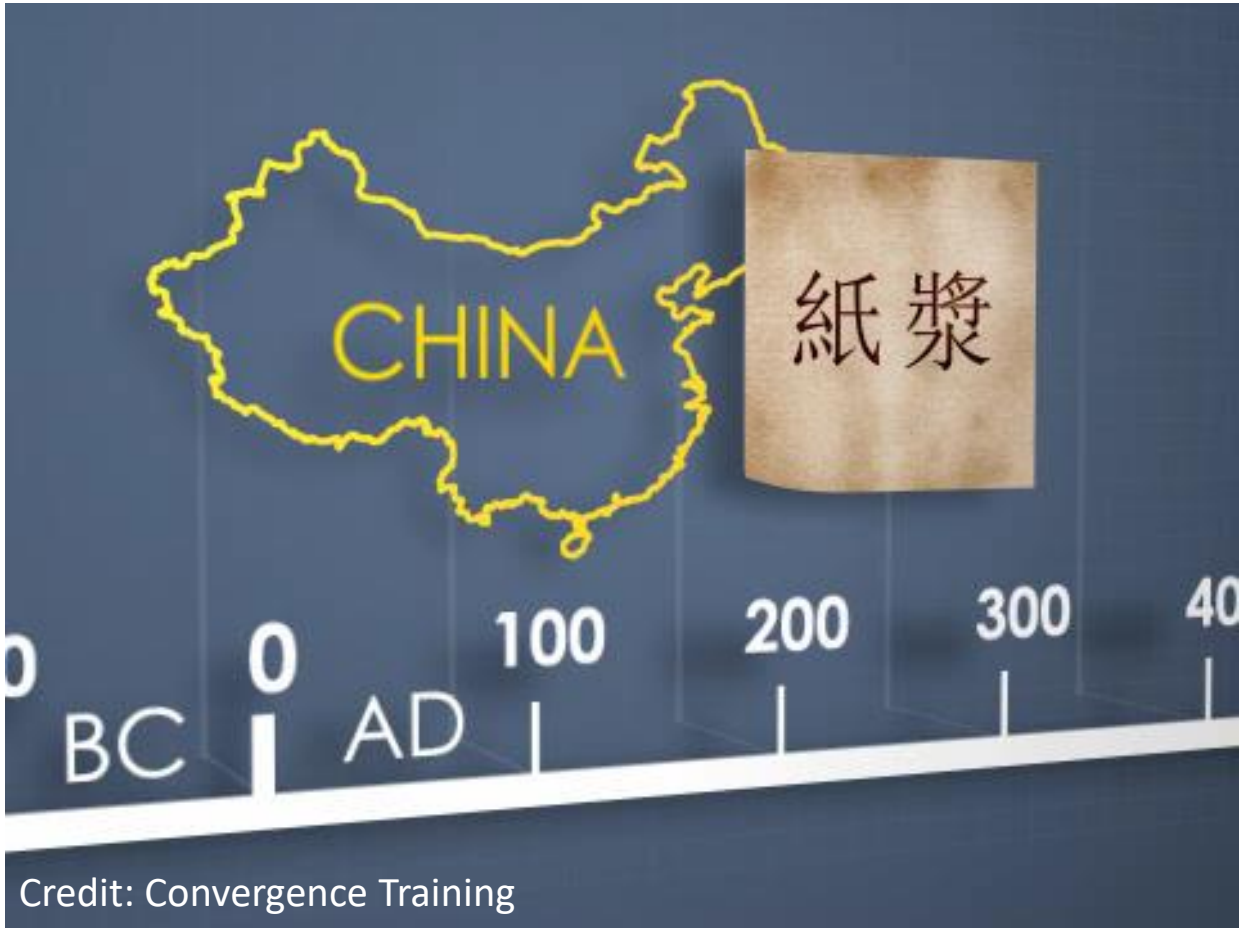
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1. Introduction to Pulp industry



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1. Introduction to Pulp industry



Credit: Convergence Training

The papermaking process was first documented in China over 1900 years ago, and for many centuries, paper was used primarily as a writing surface.



Today, thousands of different products are made partially or entirely from paper. From the fuel, oil, and air filters in your car, to the paper money in your wallet, to the roofing paper under the shingles on your home, paper is everywhere.

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1. Introduction to Pulp industry



Paper is made mostly from cellulose fibers, but the source of those fibers has changed over time.



The first paper made in China was made by hand from bamboo or mulberry bark fibers.

Credit: Convergence Training

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1. Introduction to Pulp industry

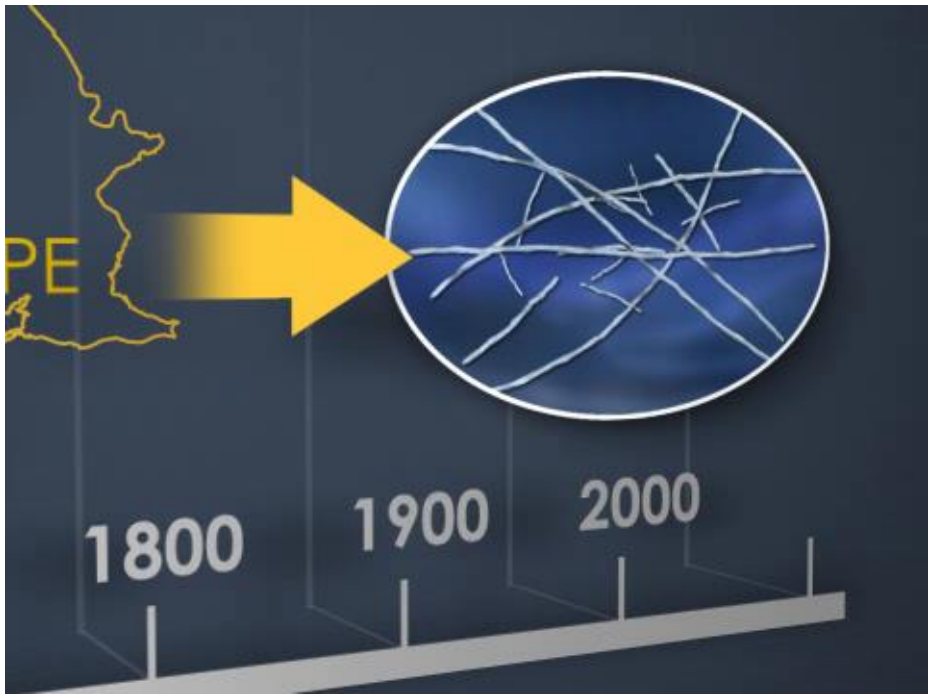


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1. Introduction to Pulp industry



When the first continuous paper machines were developed in Europe 300 years ago, cotton rags were the main fiber source, though they were soon in short supply.



This led to the search for a new source of papermaking fibers.

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1. Introduction to Pulp industry



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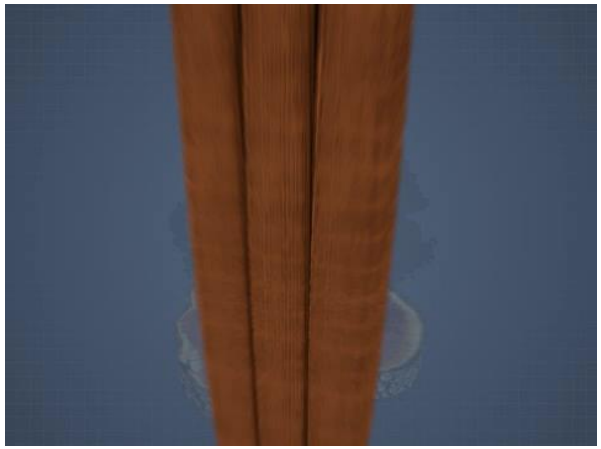
1. Introduction to Pulp industry



1. Introduction to Pulp industry







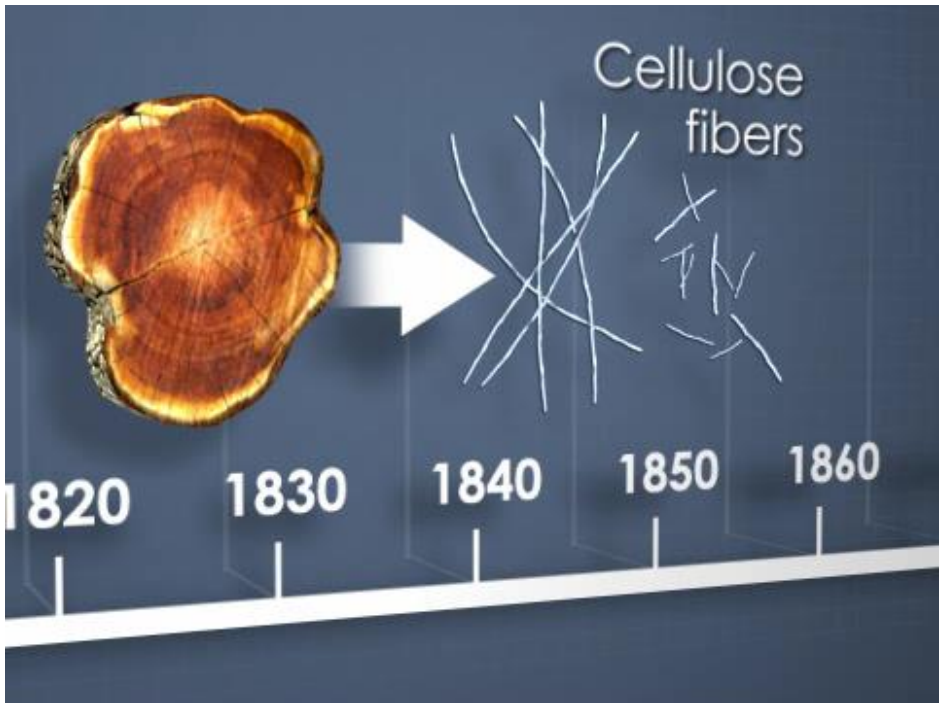


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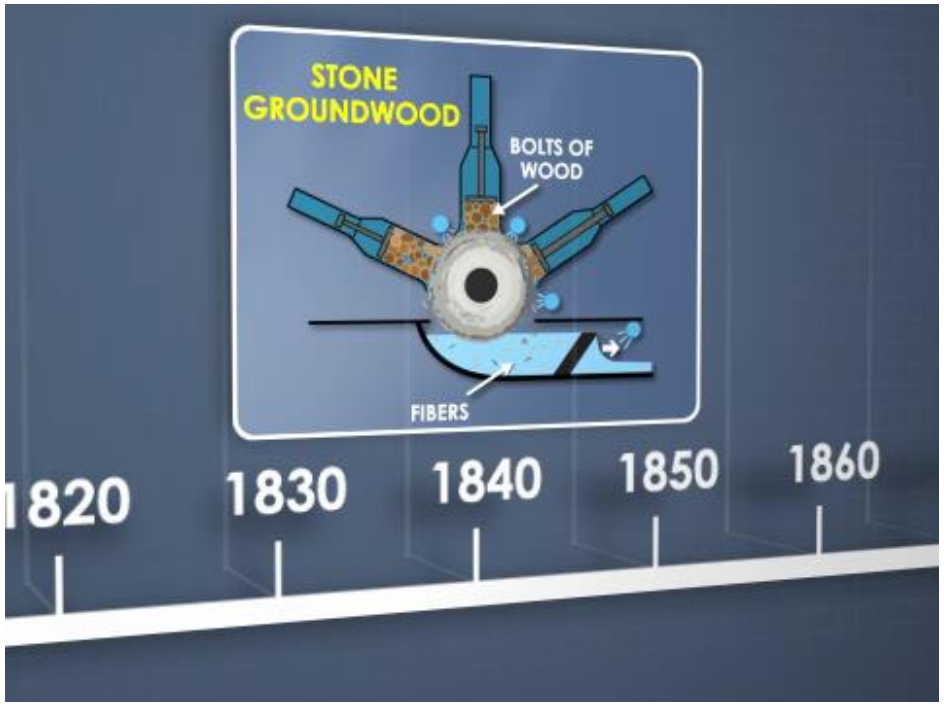
Wood is made up of **cellulose fibers** held together by a naturally-occurring, chemical compound called **lignin**.

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1. Introduction to Pulp industry



In the mid-1800's, several wood pulping processes were invented and a seemingly endless supply of cellulose fibers became available for papermaking.

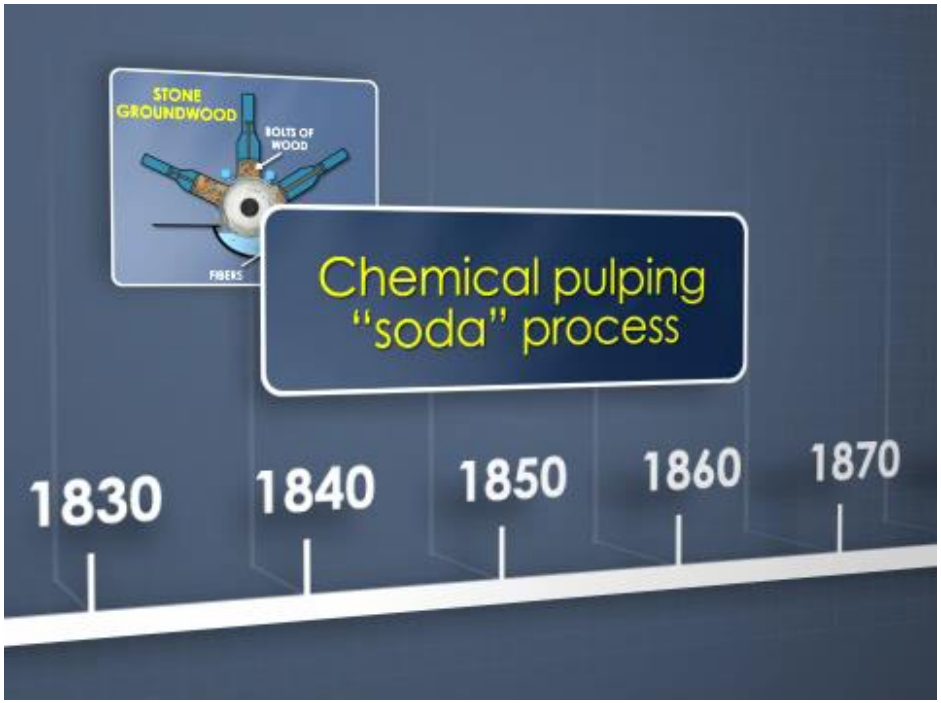


Wood fibers were first separated from each other mechanically in the stone groundwood process in 1840,

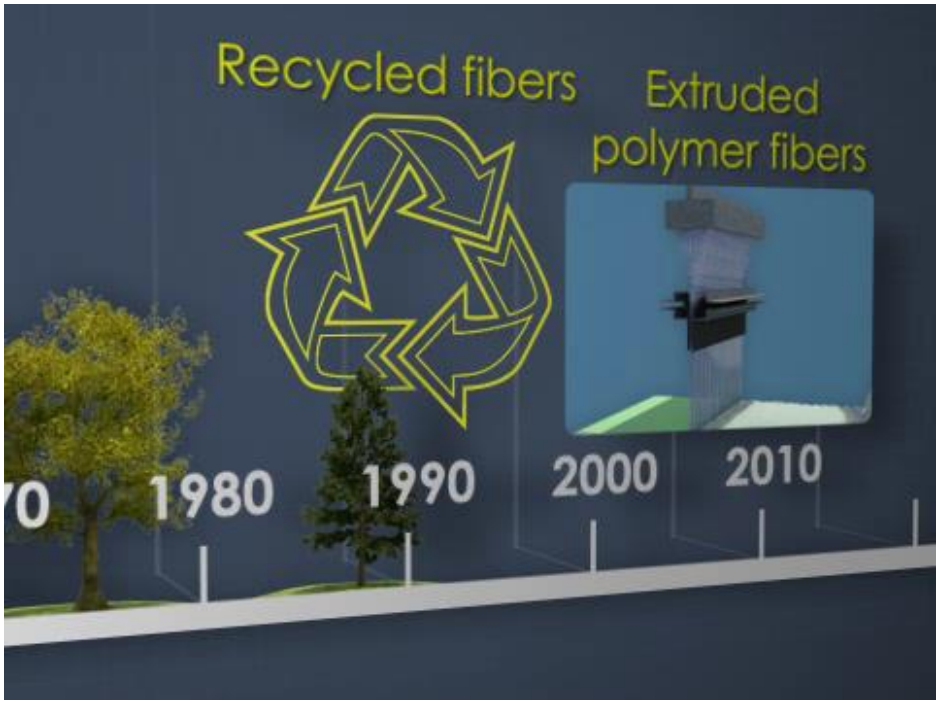
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1. Introduction to Pulp industry



the first chemical pulping process, the soda process, was invented in 1851. The kraft chemical process, currently the most popular pulping process, was invented in 1884.



More recently, as wood has become increasingly scarce, paper producers have incorporated recycled fibers into their products. Also, nonwovens made from extruded polymer fibers have taken over some of paper's market share.

Credit: Convergence Training

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2. Characteristic of wood and wood pulp

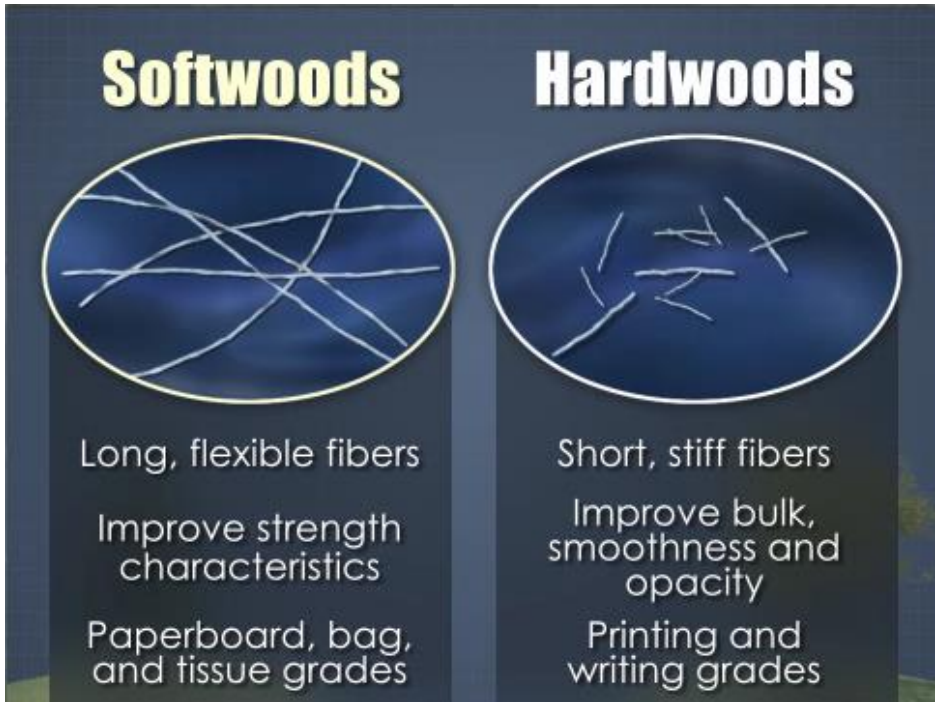
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2. Characteristic of wood and wood pulp



2. Characteristic of wood and wood pulp





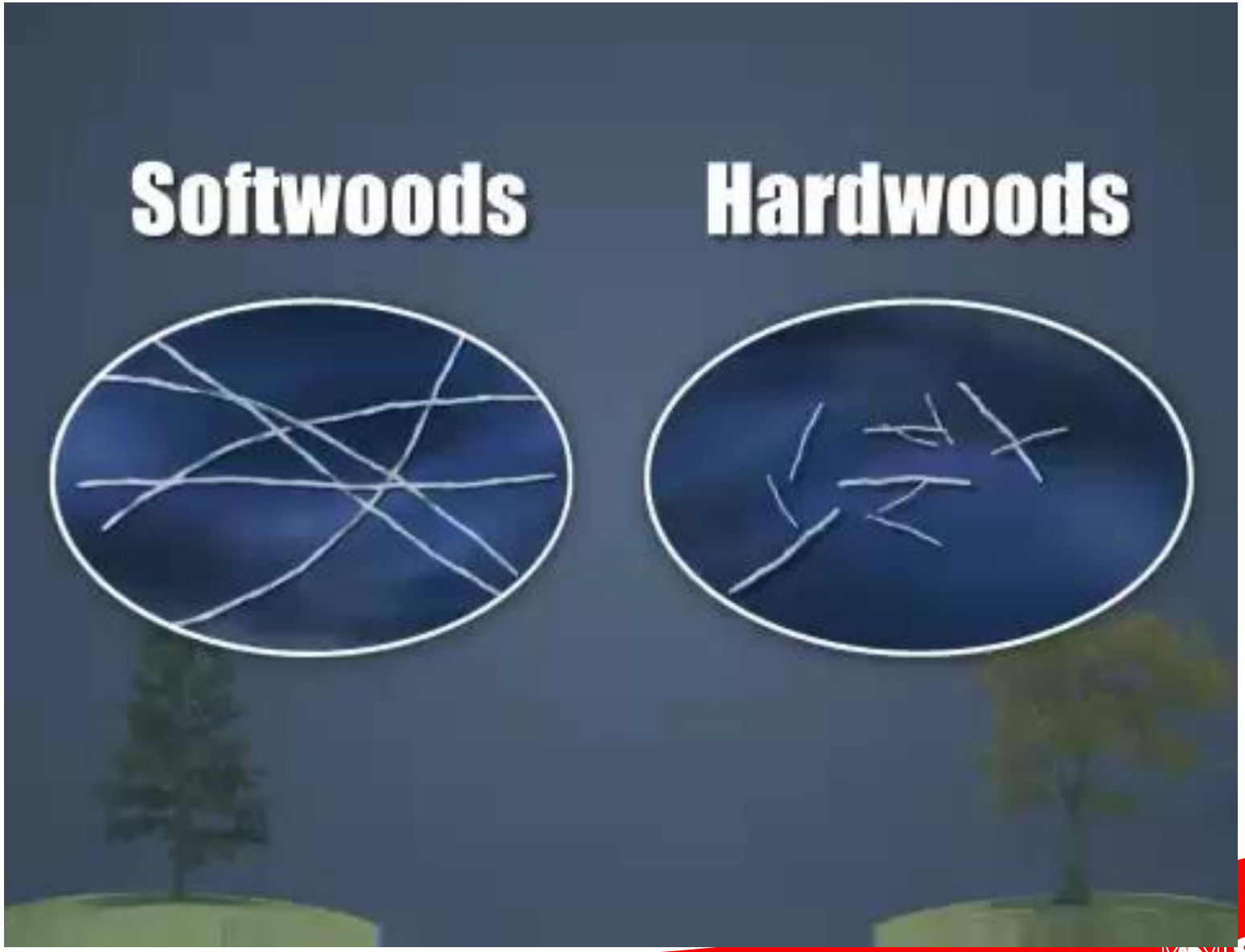
The two main classifications of trees - softwoods and hardwoods - contain fibers that are physically quite different from each other.

In general, hardwood fibers tend to be short and stiff, which contribute to the bulk, smoothness, and opacity of paper. Therefore, they are often used in printing and writing grades. Softwood fibers tend to be long and flexible, which contribute to the strength characteristics of paper. They are used in paperboard, bag, and tissue grades.

Credit: Convergence Training

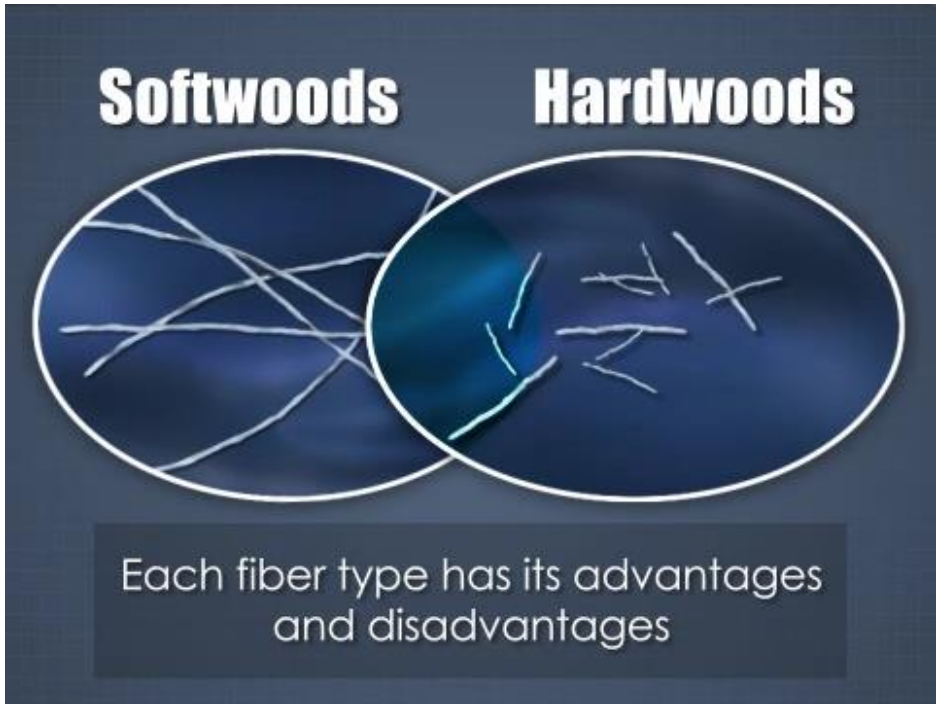
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2. Characteristic of wood and wood pulp



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2. Characteristic of wood and wood pulp





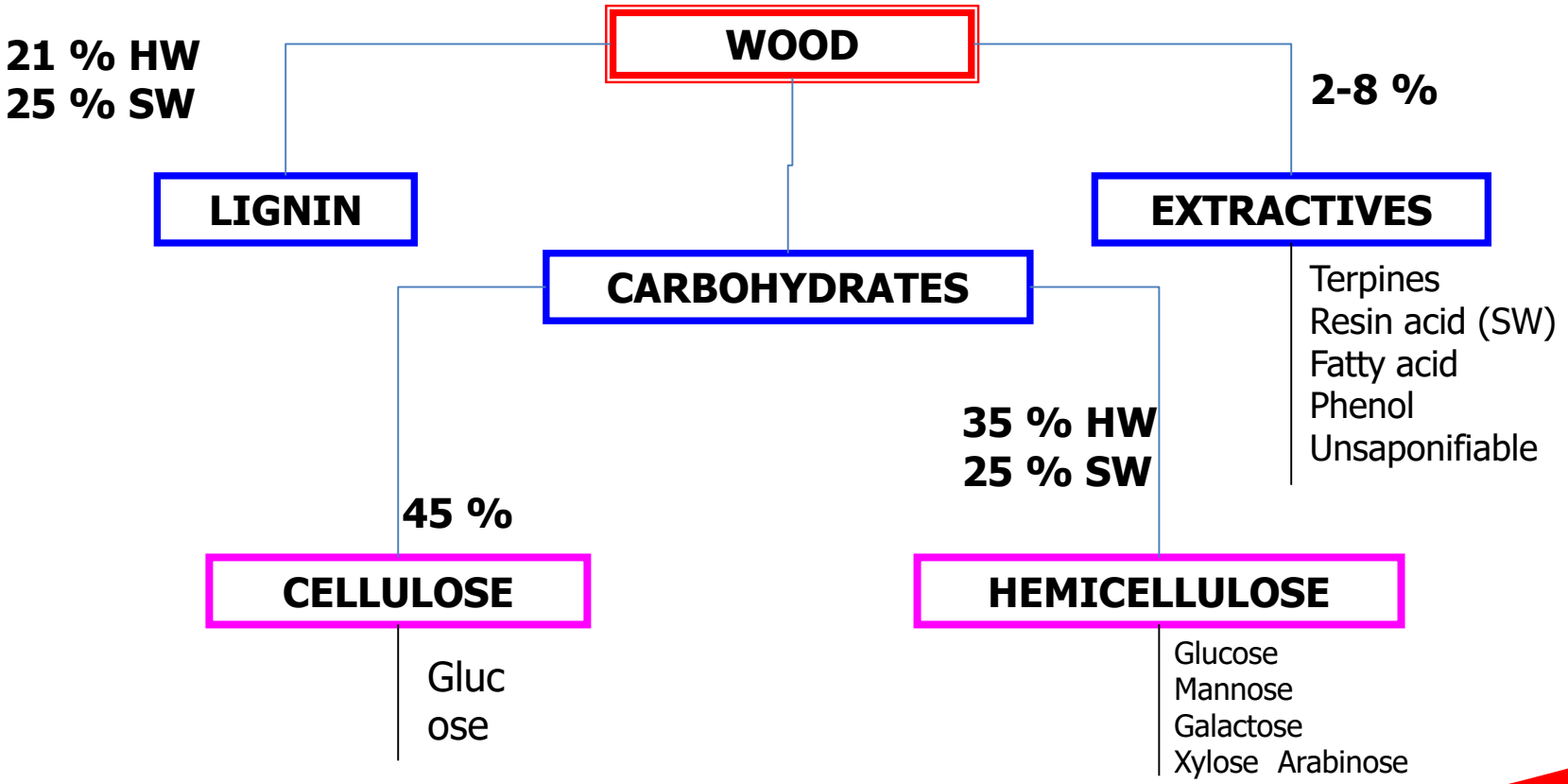
It should be noted that fiber characteristics vary widely even within the softwood and hardwood classifications, and there is some overlap. Each fiber type has its advantages and disadvantages. In the end, the choice of which wood fibers, or which blend of fibers, to use in the papermaking process depends on the grade of paper being produced and its end-use requirements. Fortunately, all wood fibers are structurally and chemically similar, so they can be pulped and made into paper by the same basic processes.

Credit: Convergence Training

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2. Characteristic of wood and wood pulp

Chemical components of wood



Credit: Suebthip Pongpaiboon

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2. Characteristic of wood and wood pulp

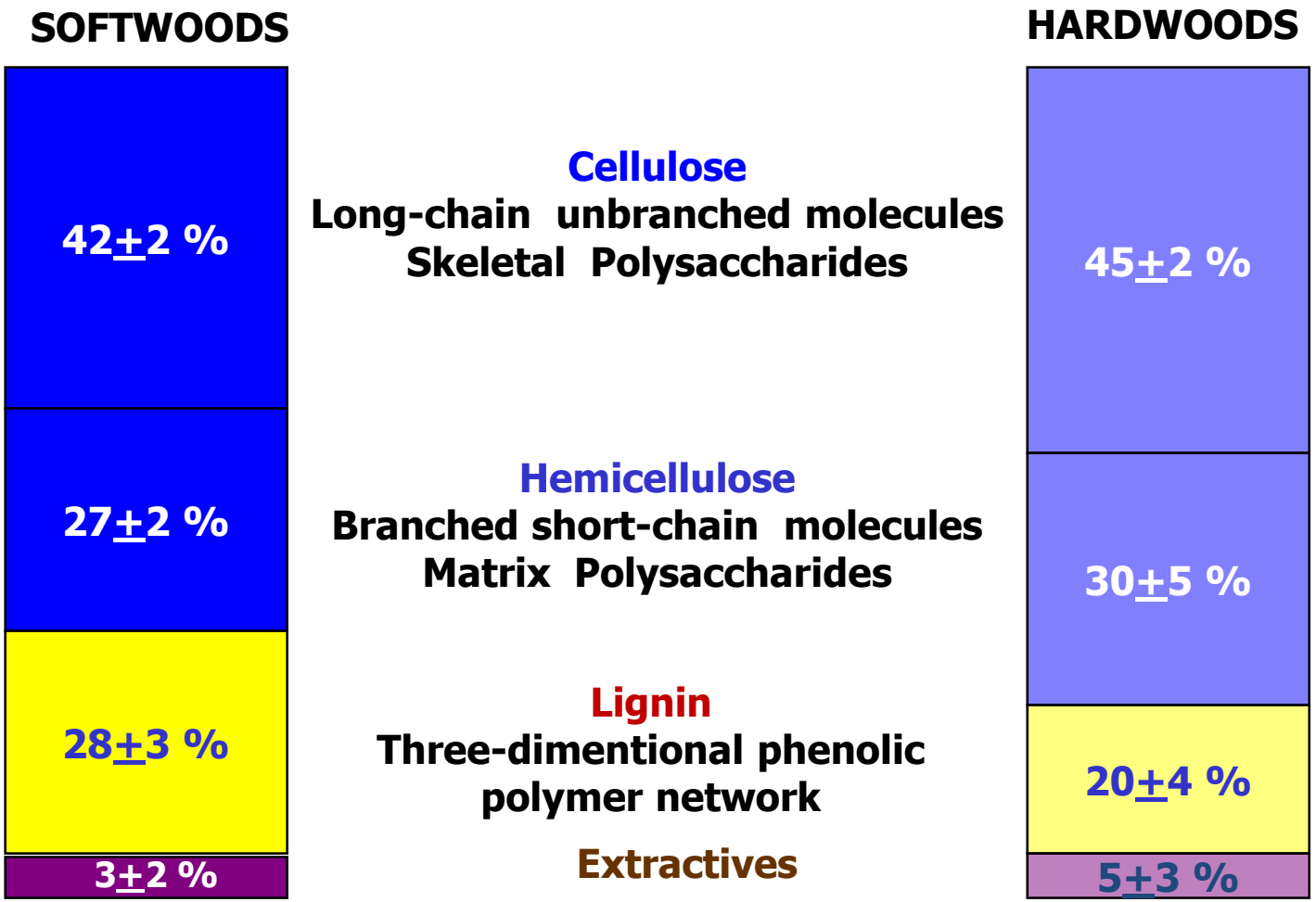


Figure 1.1 Average compositions of softwoods and hardwoods

Credit: Suebthip Pongpaiboon

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2. Characteristic of wood and wood pulp

WOOD AGE

Wood properties are affected by wood age ....

INCREASING AGE...

- Log size	Increase	- Spiral grain	Increase
- Density	Increase	- Resin pockets	Increase
- Fiber length	Increase	- Heartwood	Increase
Shrinkage		- Moisture	Decrease
- Longitudinal	Decrease	- Extractive	Increase
- Radial	Increase		
- Tangential	Increase		

Did you know?: Most of Eucalyptus species in Thailand is Camaldulensis
The most suitable wood age is 4 years

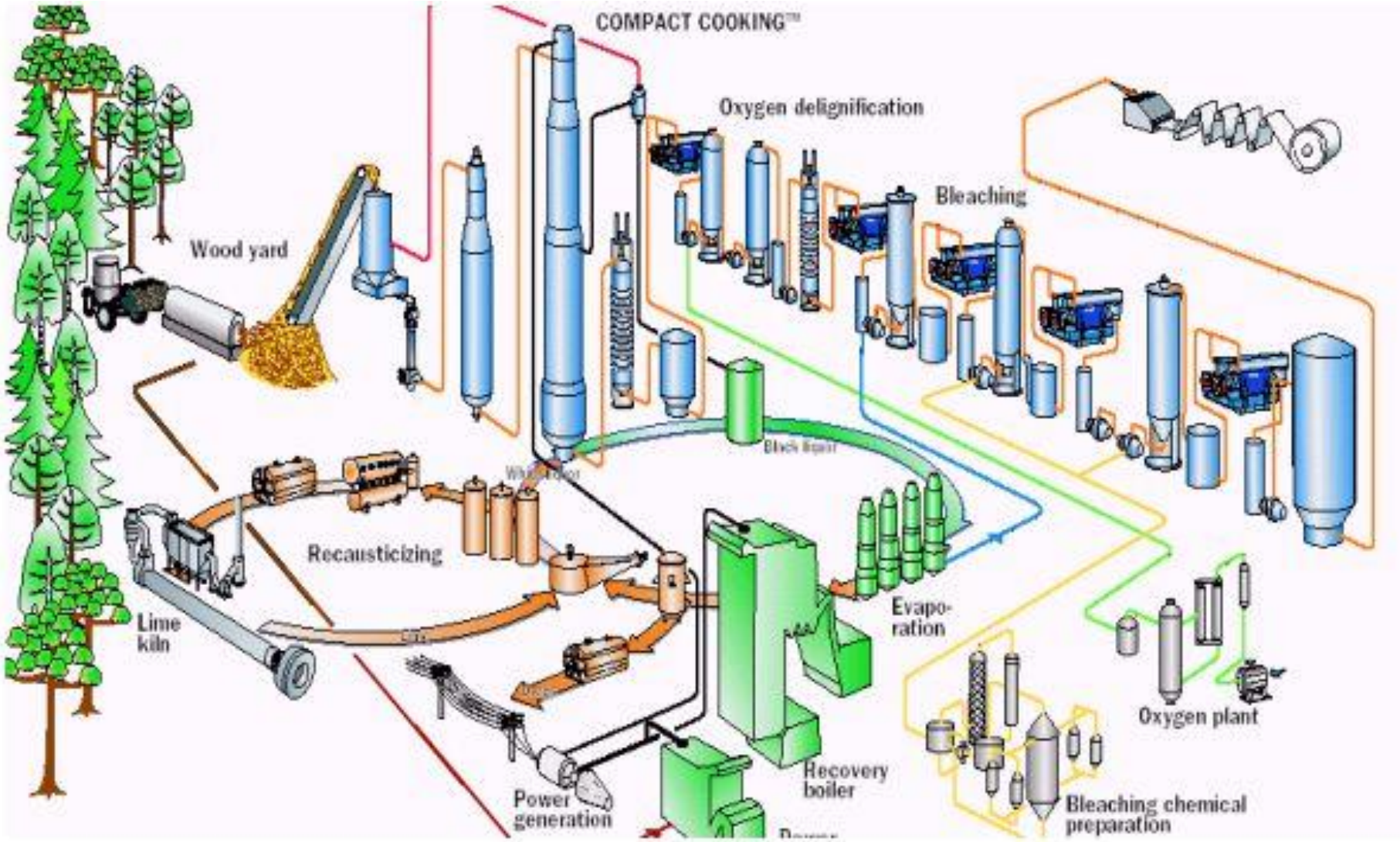
Credit: Suebthip Pongpaiboon

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3. Wood and chip handling sequences

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3. Wood and chip handling



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3. Wood and chip handling



Ref: http://www.tappsa.co.za/archive2/APPW_2004/Title2004/Applying_best_sustainable_prac/2004-27fig1a.jpg

วัตถุดิบใช้ผลิตเยื่อ
ยูคาลิปตัส



ชั่งน้ำหนัก



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LOG INDEX



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ไม้ยูคาลิปตัสที่เราไม่ต้องการ



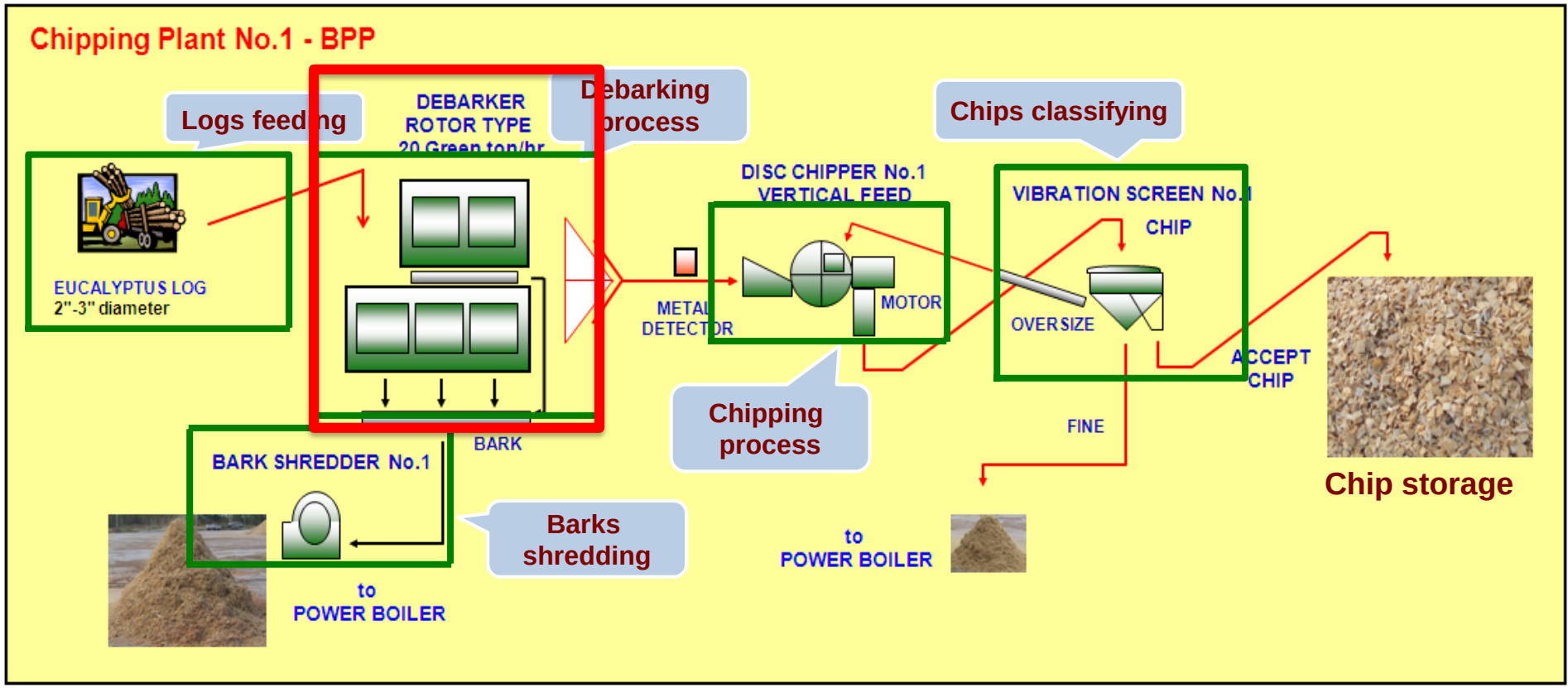






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3. Wood and chip handling



Credit: Weerachai Seachao

Debarking Process : remove bark



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Debarking Process : remove bark



1) Rotary Drum Debarker

Use gravity and weight of log
Log itself to break the bark out of wood log skin. Suitable for
Std.Log diameter = > 3 inch

Log diameter = > 2.0 inch



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Debarking Dum



ลักษณะภายใน **Debarking Drum**



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Video การทำงาน Debarking Drum



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Debarking Process : remove bark

2) Mechanical Debarker

Use mechanical energy drive comb Shredder to take out the bark from wood log skin. Suitable for

Std.Log diameter = < 3 inch

Log diameter = > 1.5 inch



08E50 Type





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ลักษณะภายใน **Super barker**





Video การทำงาน Super Barker



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ตัวอย่างไม้ยูคาลิปตัสที่ผ่านการลอกเปลือกออกแล้ว

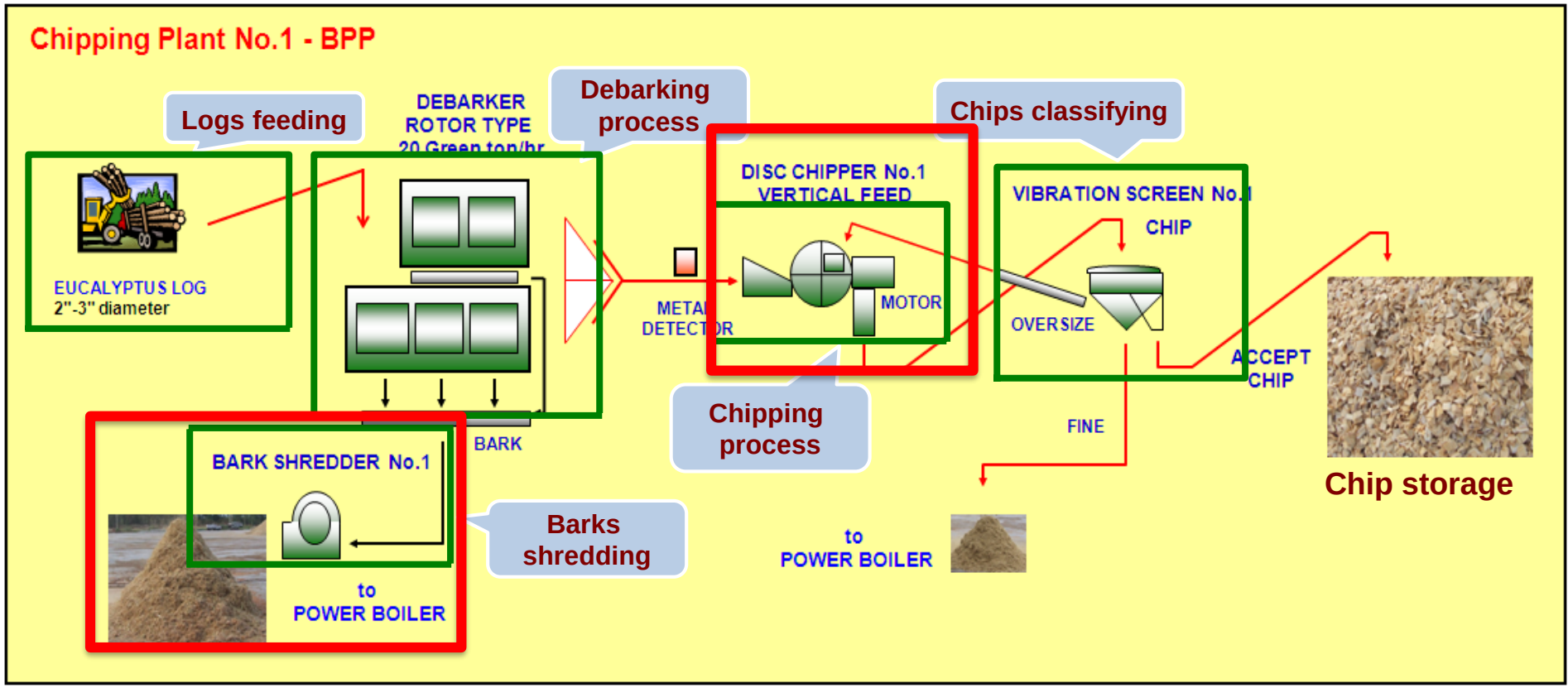


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BARK



3. Wood and chip handling

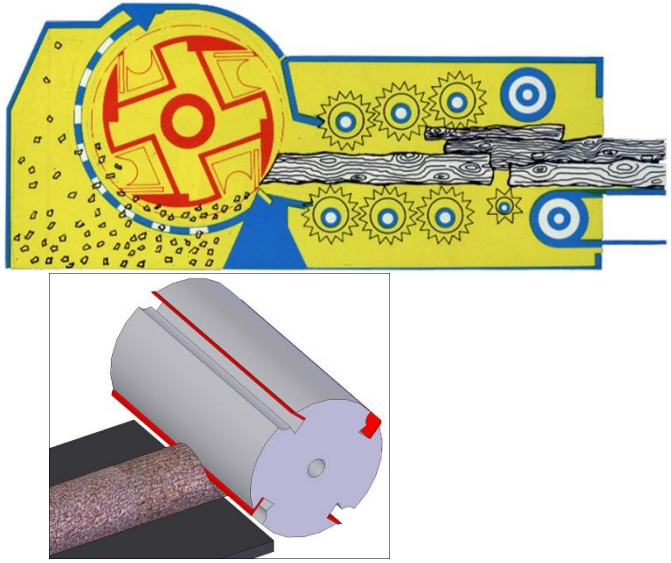


Credit: Weerachai Seachao

3. Wood and chip handling

Chipping Process: Make Chip





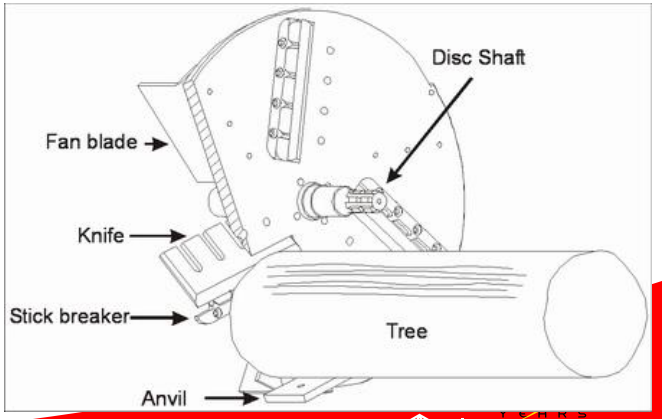
1) Drum Chipper

Drum rolling with a long knife to chop the log in to small chip
Adv: Can handle big log, tough
Disadv: Consume more energy, High maintenance cost,

2) Disc Chipper

Rotating Disc with a set of shorter knives. The disc rotate to chop the log into chips
Adv: Lower power, Lower manintenance cost
Disadv: less tough than Drum





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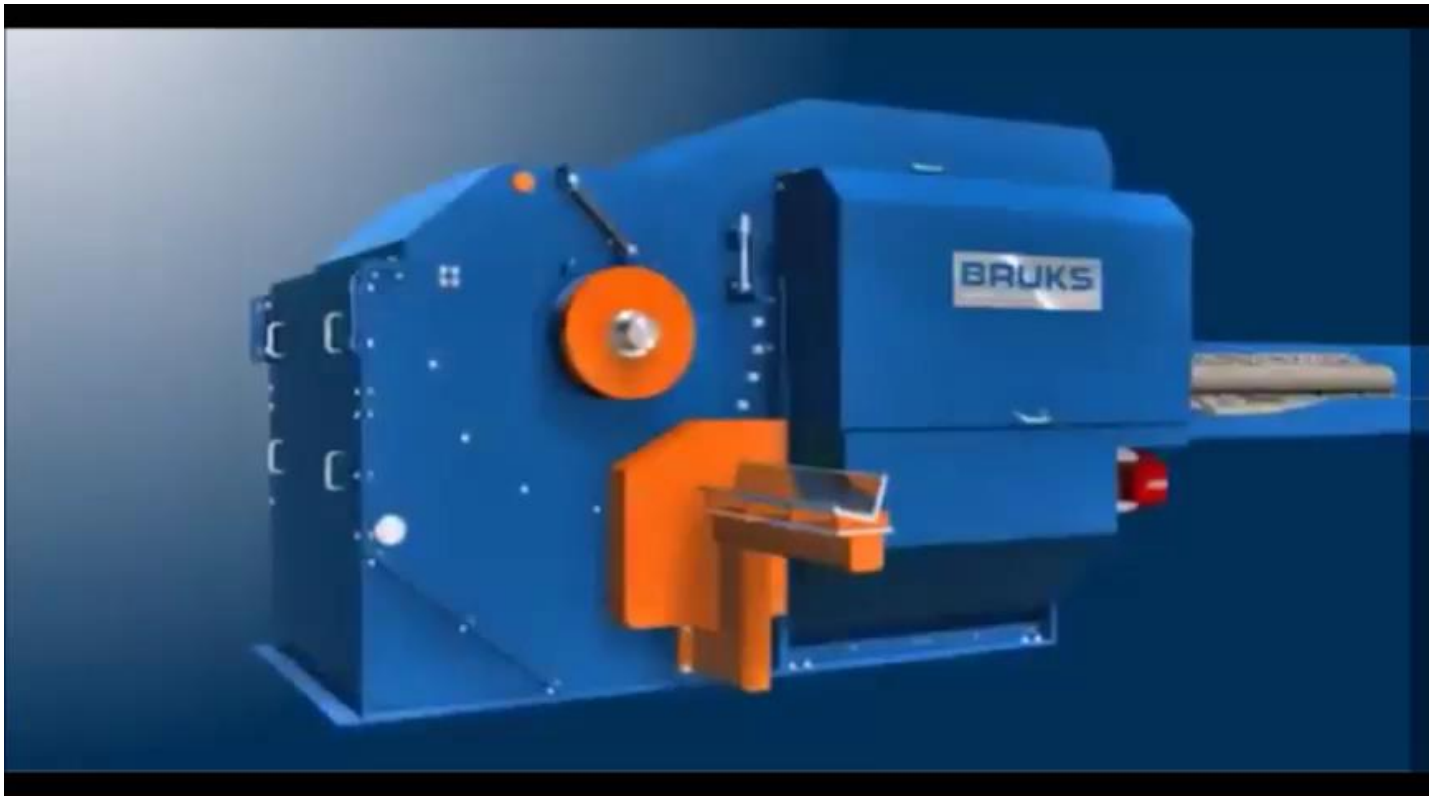
Metal Detector



What is it for?

3. Wood and chip handling

Bark Handling: Make bark



Bark Shredder: use drum chipper to make bark size smaller

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Machine Specification



Disc Chipper



Disc Chipper #1

- Gravity Feed
- Manufacturer : "Nicholson"
- No. of Knife : 8 Pcs.
- Disc RPM : 420 rpm.
- Disc Dia. : 2200 mm.
- Motor : 2x315 kW.

Disc Chipper #2

- Horizontal Feed
- Manufacturer : "Kone Wood"
- No. of Knife : 16 Pcs.
- Disc RPM 400 rpm.
- Disc Dia. : 2200 mm.
- Motor : 770 kW.

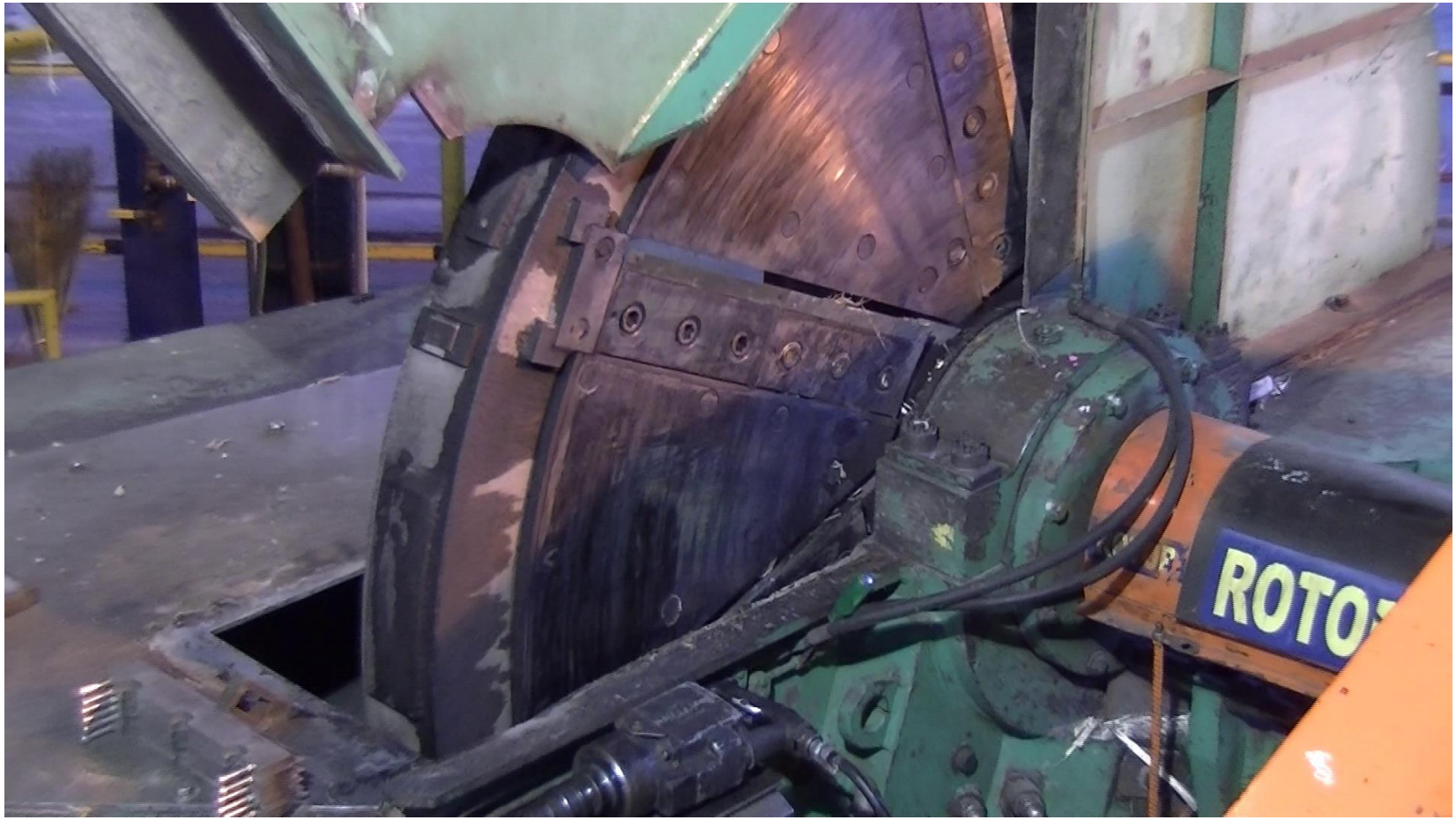
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Disc Chipper



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Video การทำงาน Disc Chipper

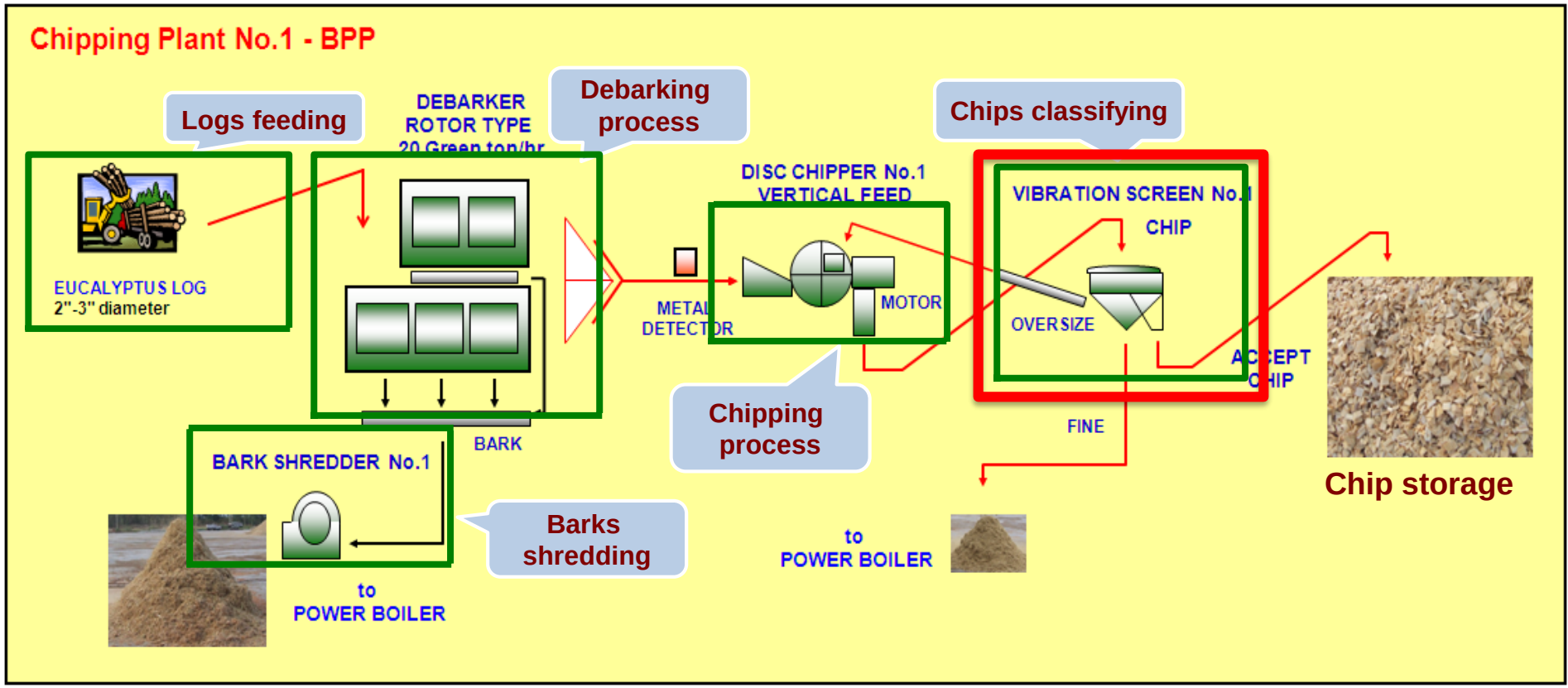


Chip Eucalyptus



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3. Wood and chip handling



Credit: Weerachai Seachao

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CHIP SCREEN (Vibration Screen)



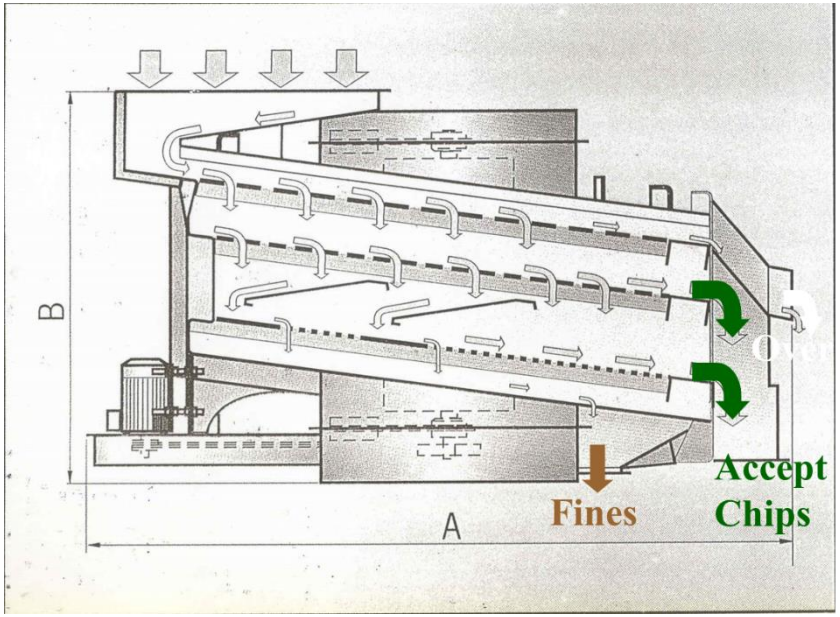


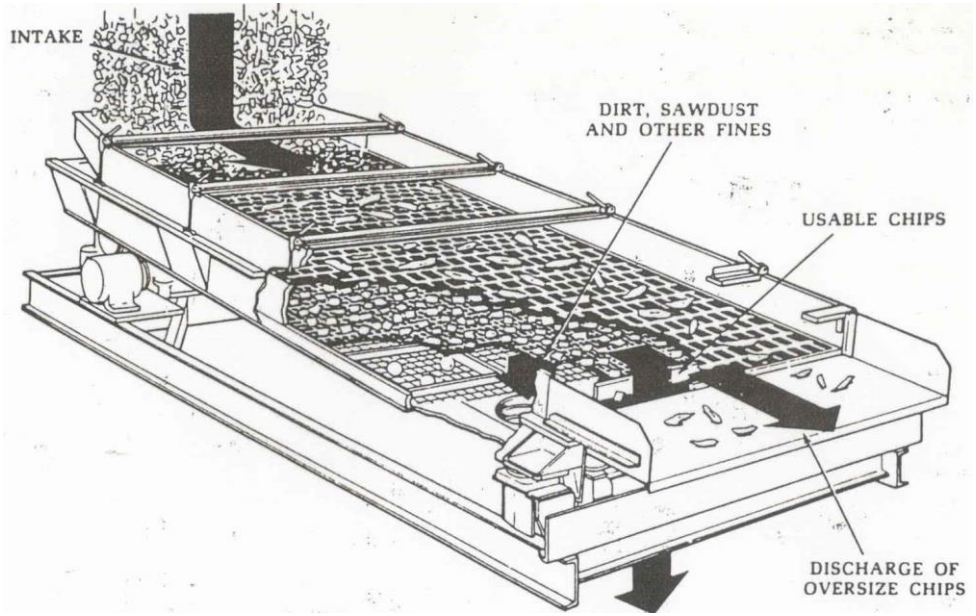


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3. Wood and chip handling

Chip Classifying: Classify Chip Size





Vibration Screen:

Chip is convey to the top of Vibration screen that vibrate the acceptable chip size will fall down through the series of screens until reach the bottom conveyor to Chip Silo or deliver to Chip Yard. The oversize remain on the screen and fall down on the discharge chute.

Did you know?: Evenly chip size can help process stability

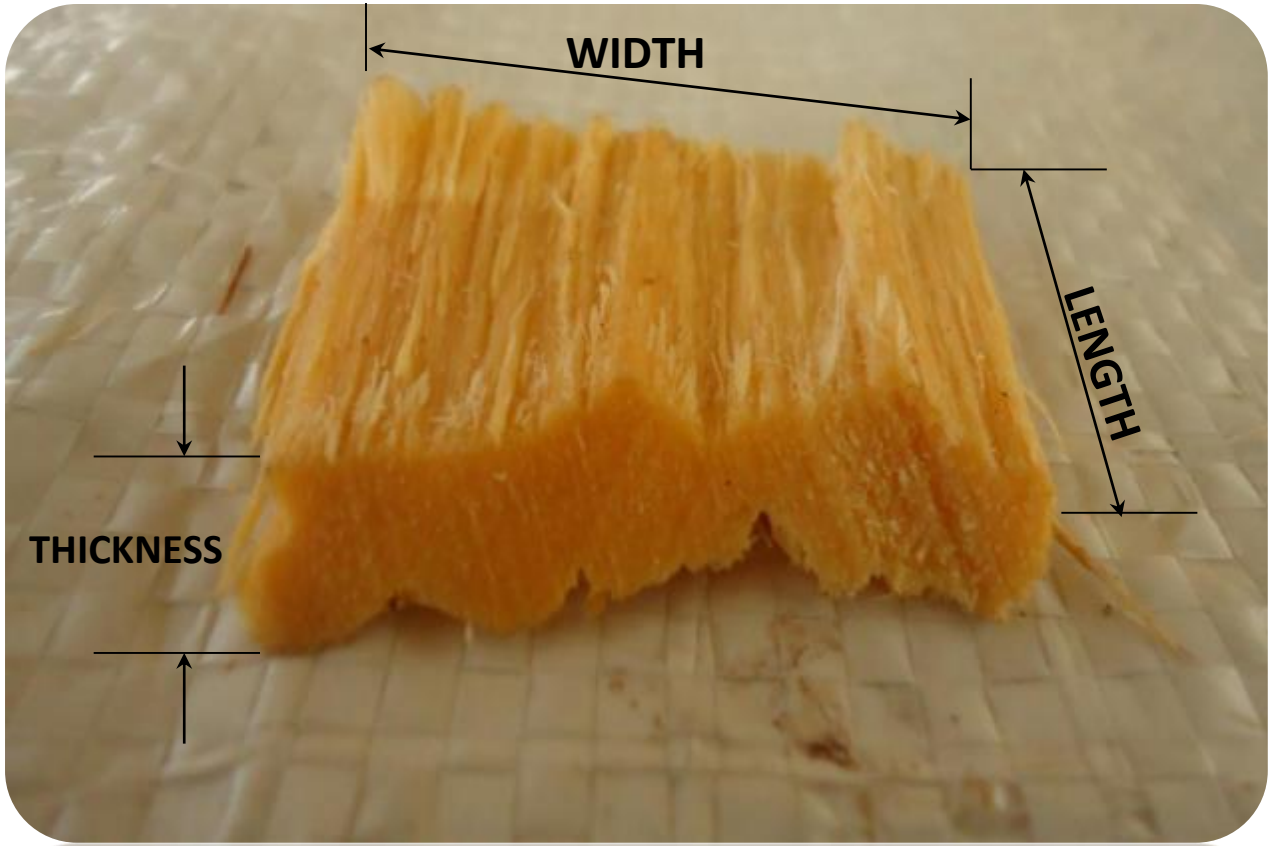
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Chip Screen (Vibration Screen)



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CHIP SIZE



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CHIP OVERSIZED



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PIN CHIP



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FINES



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Chip Classification Lab





เป็นการสุ่มตรวจคุณภาพ Chip ว่าได้ตามมาตรฐานหรือไม่ โดยการสุ่มตัวอย่าง Chip จาก Disc Chipper เพื่อมาค้นหาว่าได้ปริมาณ Chip ตามมาตรฐานหรือไม่ หากมี Oversize หรือ Fine เกินกว่ามาตรฐานแสดงว่า ใบมีดนั้นเสื่อมคุณภาพและต้องวางแผนเปลี่ยนตามลำดับ

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Chip Quality

CHIP CLASSIFICATION

SAMPLING DATE: 12 ก.พ.56

TYPE OF CHIPS: Eucalyptus

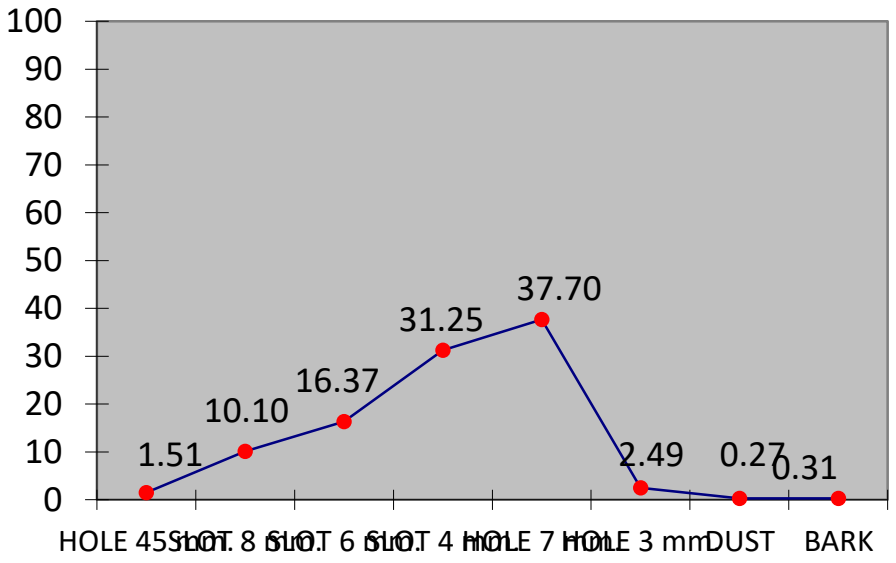
SAMPLING POSITION: SPPC

2120.99 g. (ASIS)

HOLE ON	WEIGHT OF FRACTION	% FRACTION	% STD. FRACTION	
HOLE 45 mm.	31.96	1.51	max. 1%	Oversize
SLOT 8 mm.	214.24	10.10	-	Overthick
SLOT 6 mm.	347.24	16.37	-	Accept
SLOT 4 mm.	662.82	31.25	-	
HOLE 7 mm.	799.64	37.70	-	
HOLE 3 mm.	52.73	2.49	max. 4%	Pin chip
DUST	5.83	0.27	-	
BARK	6.53	0.31	max. 0.7%	
TOTAL ACCEPT FRACTION		85.32	min. 85%	

BULK DENSITY:	VOLUME OF CHIPS:	-	M^3
	WET WEIGHT:	-	Kg
	DRY WEIGHT:	-	Kg
	MOISTURE CONTENT:	-	%
	BULK DENSITY:	-	Kg/M^3

DISTRIBUTION CURVE



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ลานกองเก็บ **Wood Chip**



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คลุมผ้าใบกอง CHIP



Chip จากลานกองเก็บส่งเข้า **Chip Silo**



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CHIP SILO


SCL Chip Silo

Chip Silo is use for storing chip that is ready to for cooking in the Digester. It use chip discharge screw to distribute the chip via a belt conveyor to convey chip continuously and constantly to the Chip Bin. The speed can be adjust due to the capacity demand


Chip Discharge Screw


Chip Discharging

3. Wood and chip handling

Chip Storage/Chip Yard: to store chip / chip aging





This is where the chips are store and feed to Cooking process. There are 2 purpose of chip storage:

1) To create chip inventory in case that log are in short supply
2) To improve the Chip properties, this apply to tropical climate country. The moisture in chip are evaporate which make it suck chemical easily at cooking stage

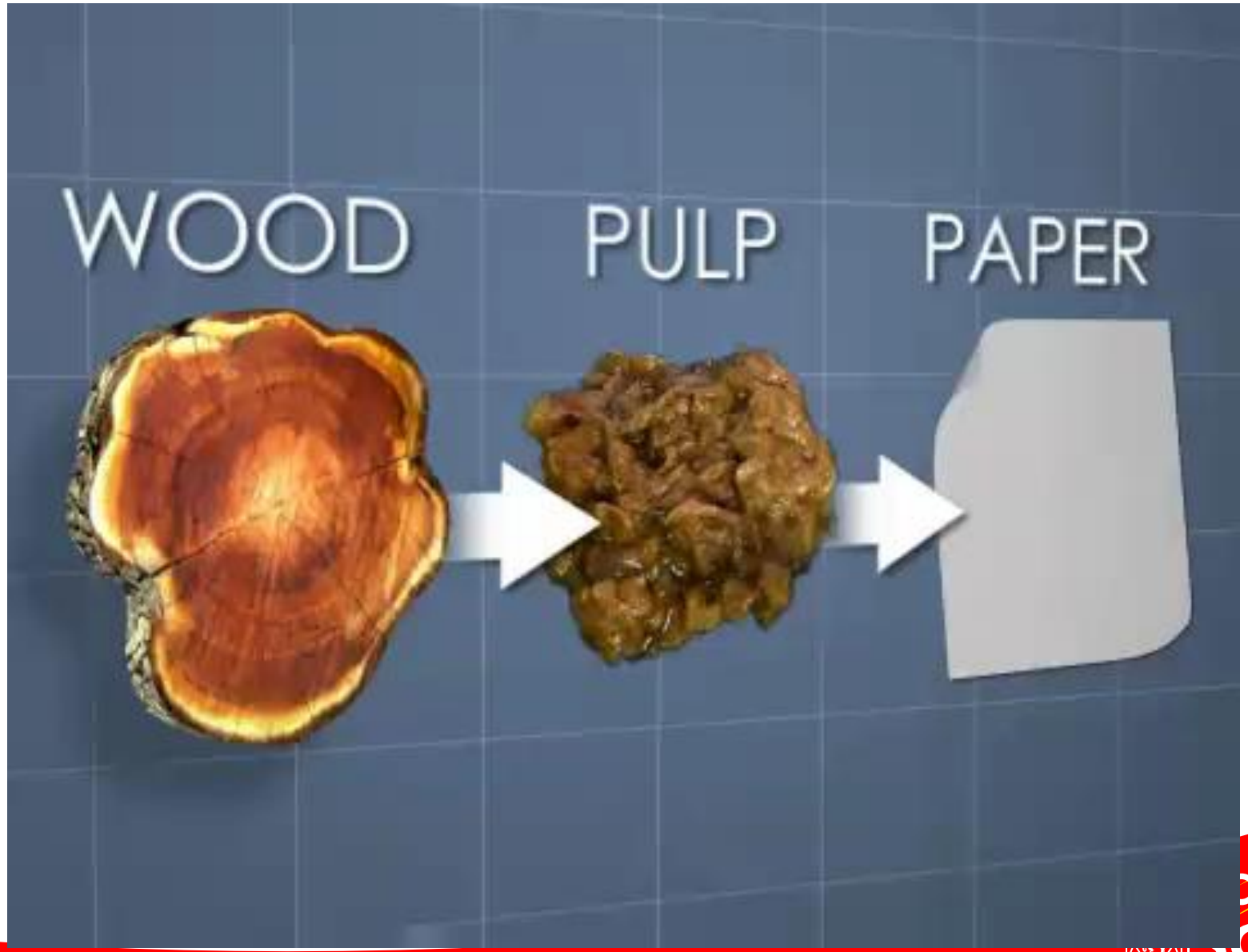
Chip aging is a number of days the chip is keeping on the Chip yard. It is normally keep for 20 – 45 days before cooking. If keep it longer the chip will be very dry which will effect to the cooking parameter

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4. Overview of Pulping methodology

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4. Overview of Pulping methodology

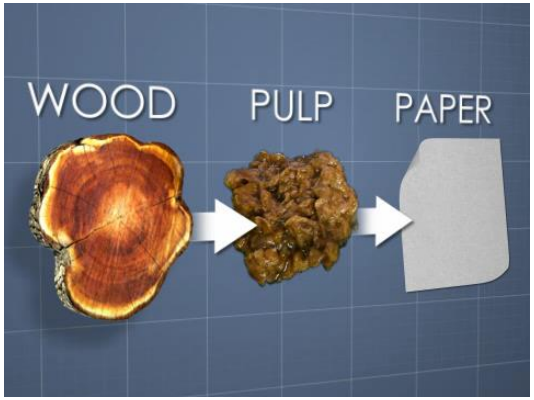


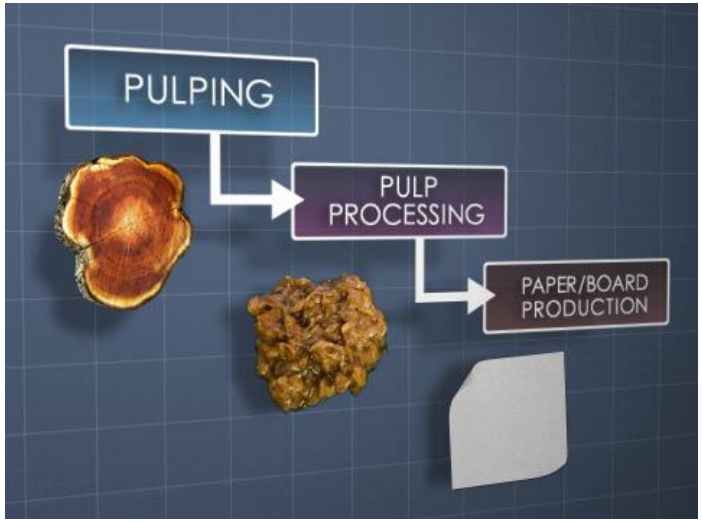
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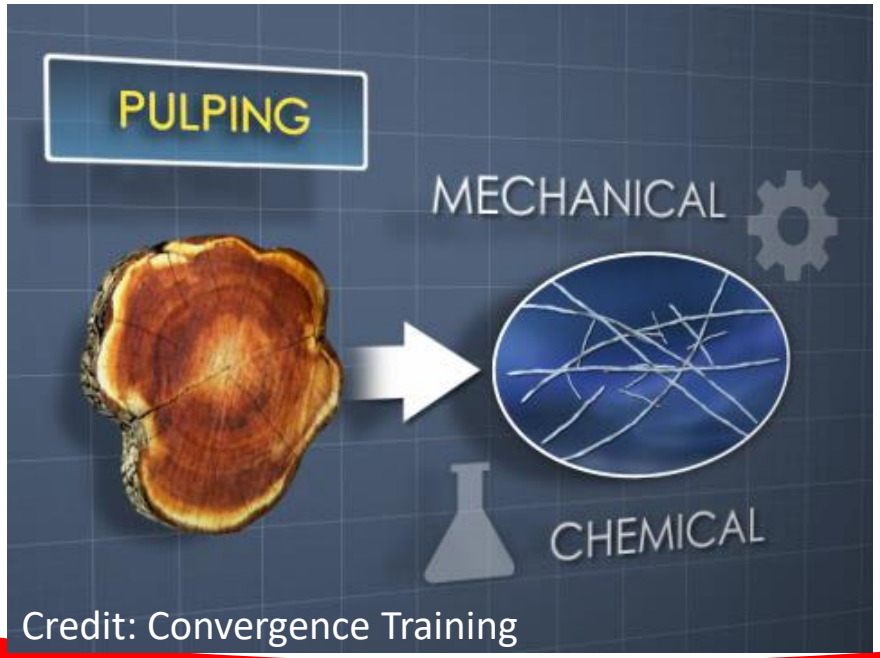
4. Overview of Pulping methodology

Today, the manufacture of pulp and paper from wood involves three major operations:

1. Pulping
2. Pulp processing
3. Paper or board production





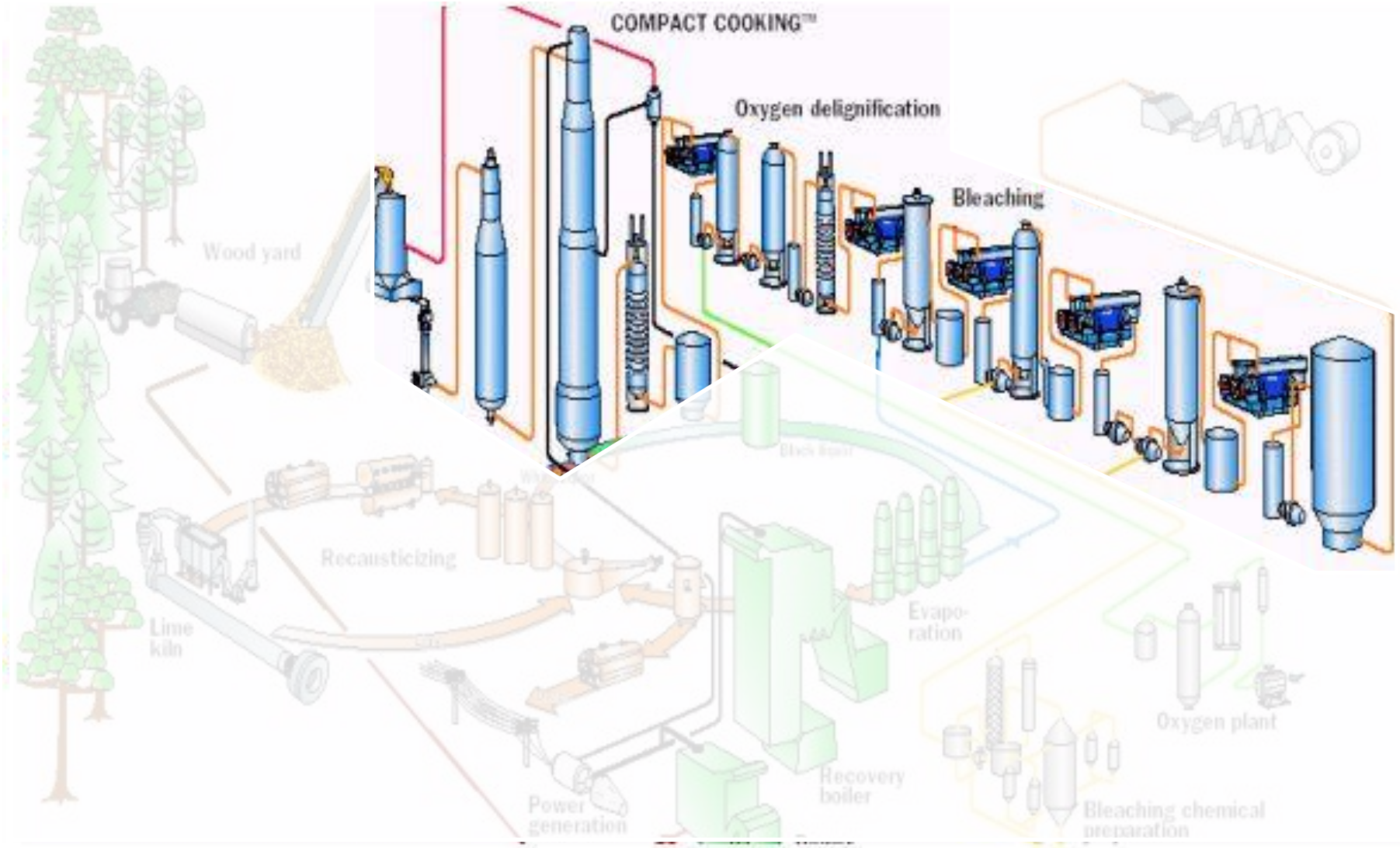


Credit: Convergence Training

The first operation, pulping, separates the cellulose fibers in wood from each other by mechanical, chemical, or a combination of mechanical and chemical means.

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4. Overview of Pulping methodology



Ref: http://www.tappsa.co.za/archive2/APPW_2004/Title2004/Applying_best_sustainable_prac/2004-27fig1a.jpg

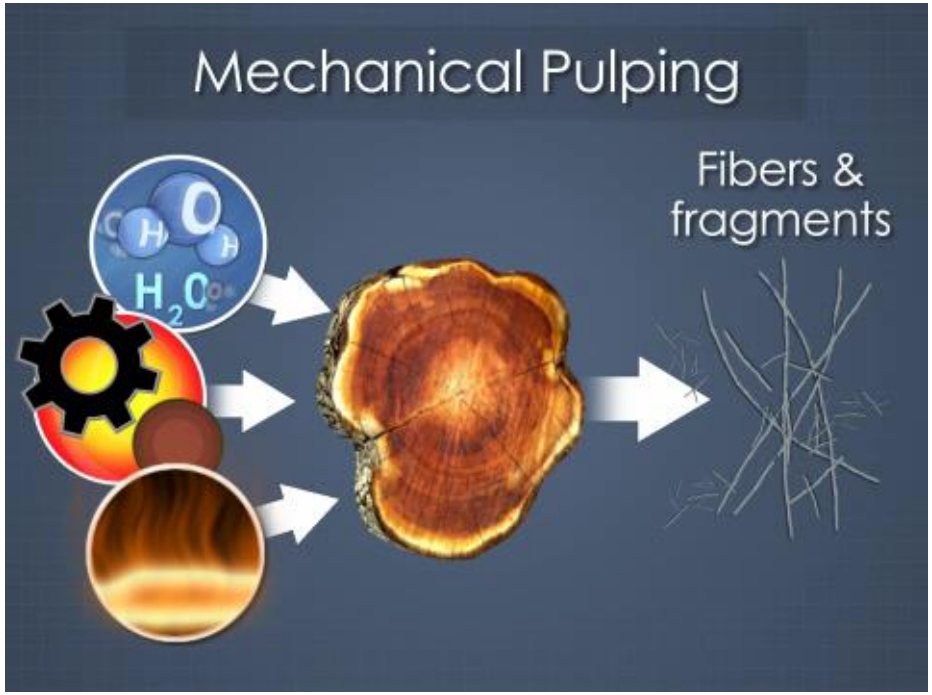
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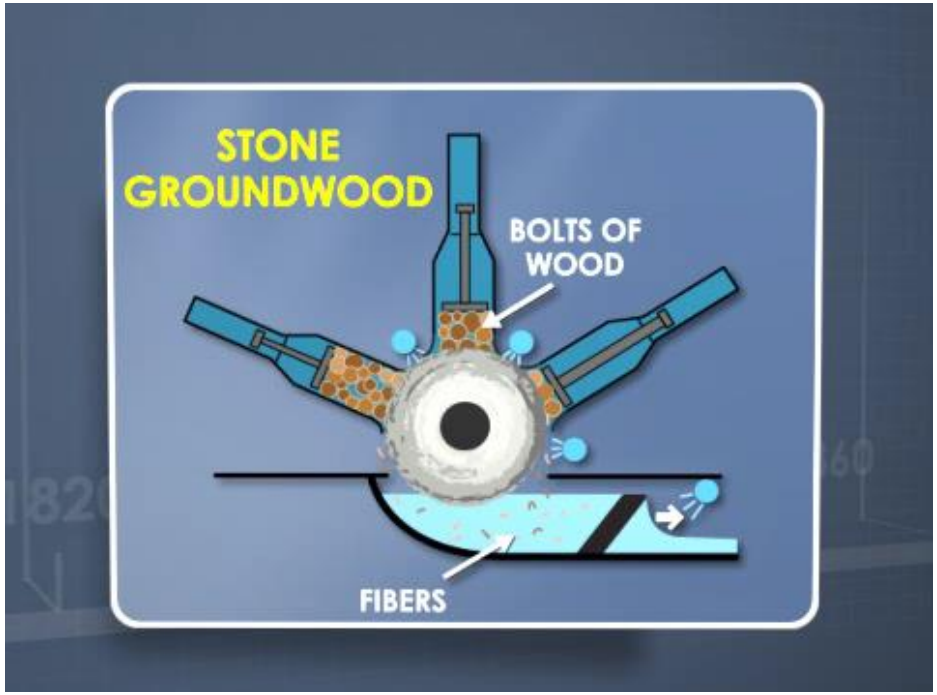
4. Overview of Pulping methodology



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4. Overview of Pulping methodology





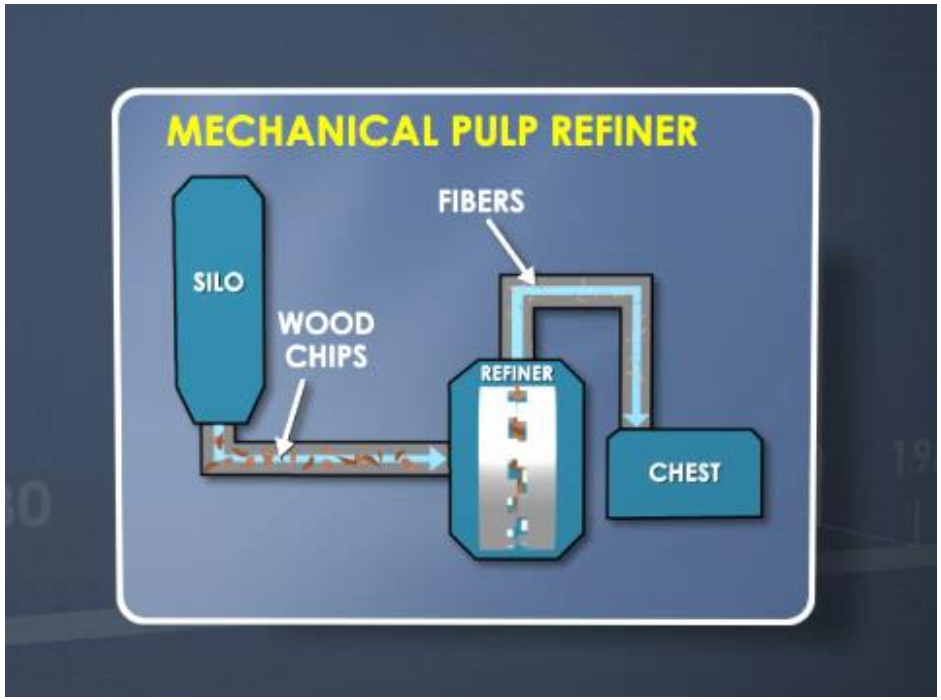
Mechanical pulping processes use water, shear forces, and heat to liberate the fibers in wood, and the pulp that is produced consists of a mix of fibers and fiber fragments.

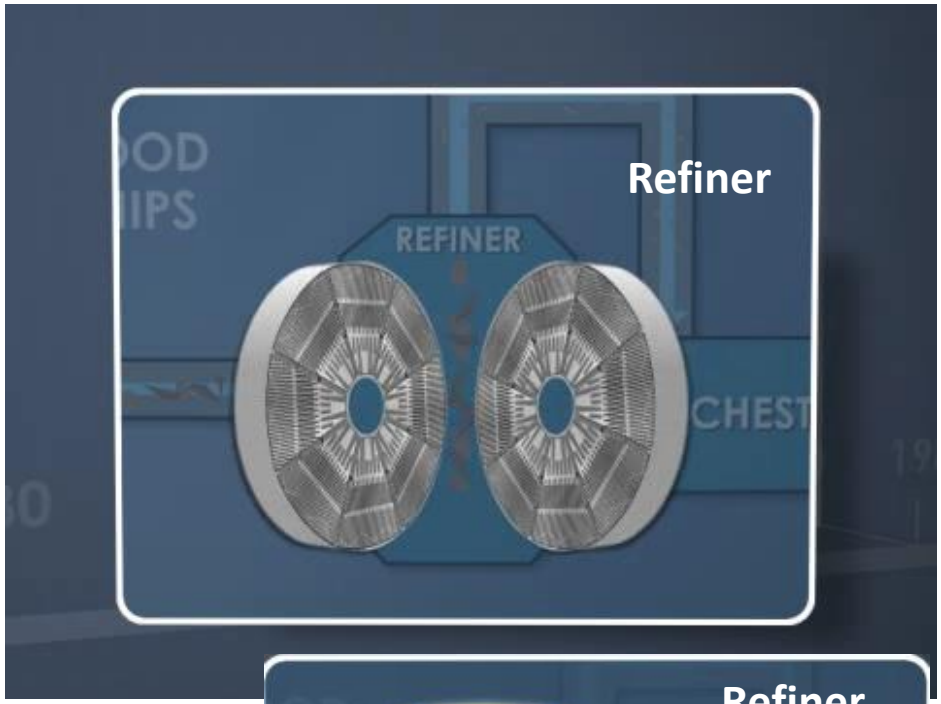
In the very first mechanical pulping process, called **stone groundwood**, bolts of wood were pressed up against a large, wet, revolving grindstone.

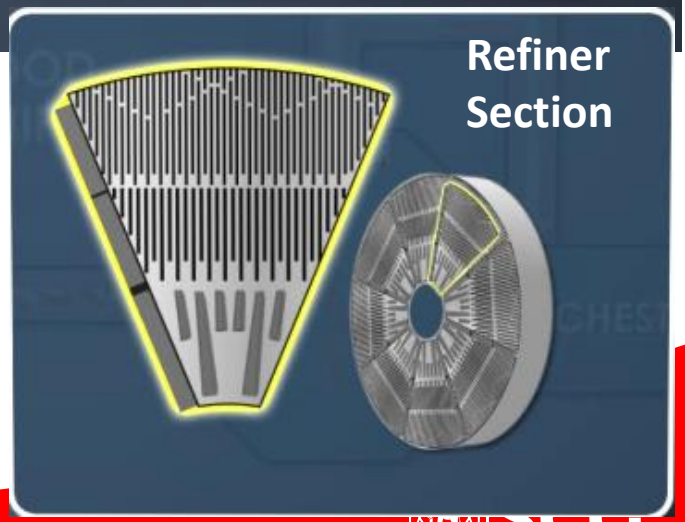
Credit: Convergence Training

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4. Overview of Pulping methodology





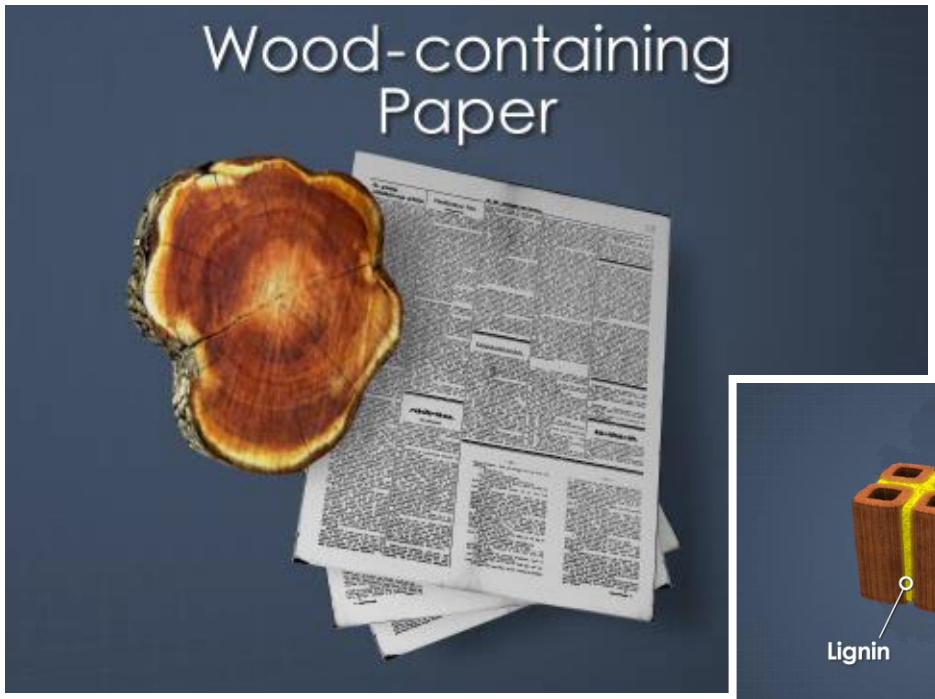


In the 1950's, a new device called a "**refiner**" was developed. In a refiner, wood chips are shredded and ground up in water, between rotating discs with raised bars on their surfaces. Today, almost all mechanical pulp is produced by refiners.

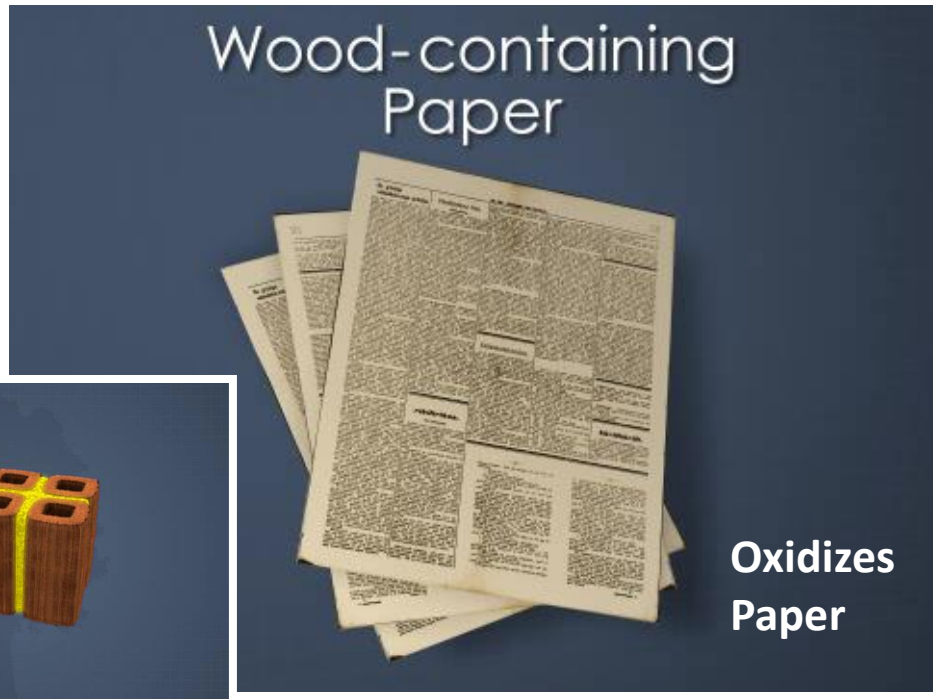
Credit: Convergence Training

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4. Overview of Pulping methodology





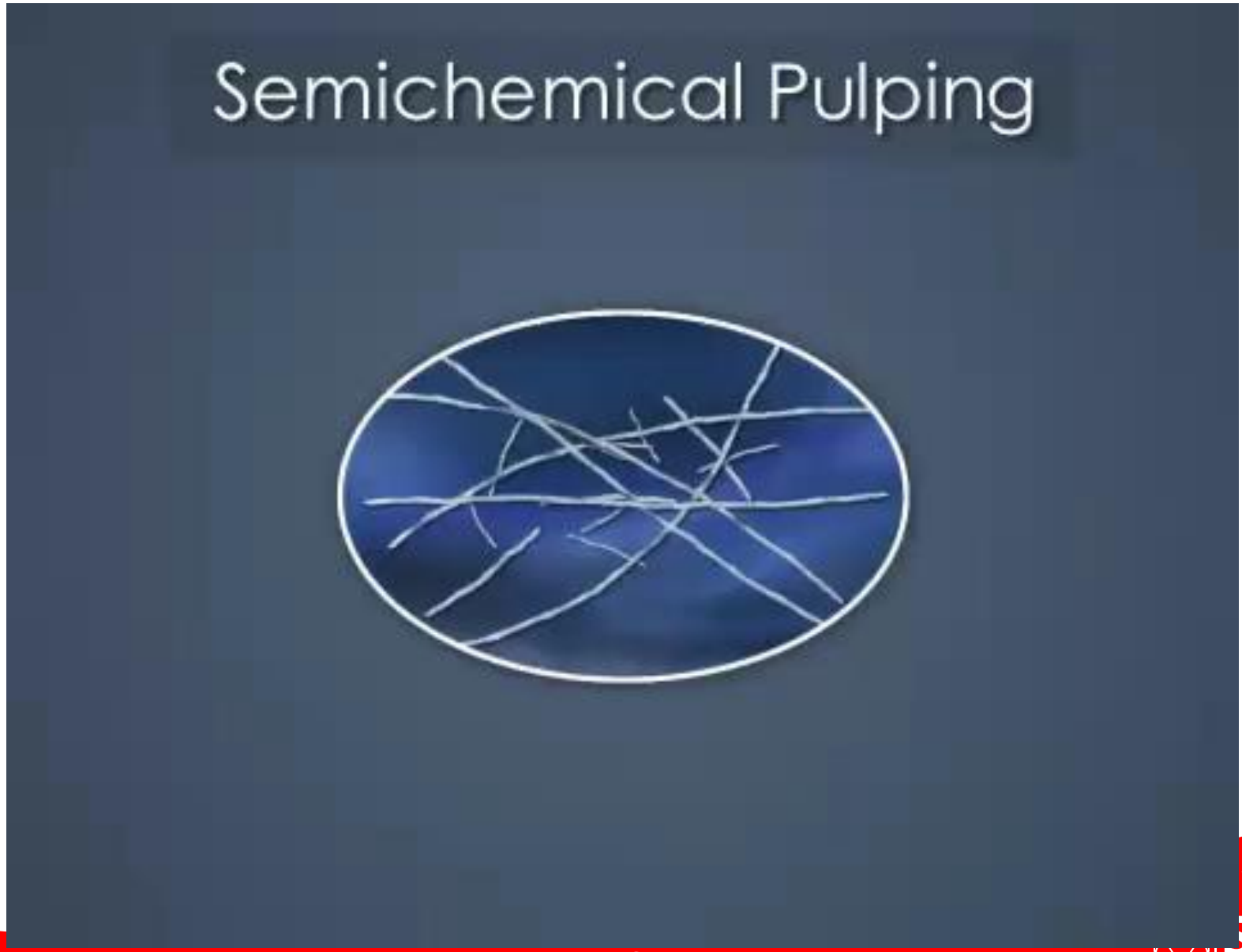


Because paper made from mechanical pulp contains all of the constituents of the wood, it is called "**wood-containing paper**." Wood-containing papers, like newsprint, are usually grey in color with high opacity and a smooth surface for printing. One disadvantage of these papers is that they tend to yellow with exposure to air and sunlight because the **lignin** in the paper darkens as it **oxidizes**.

Credit: Convergence Training

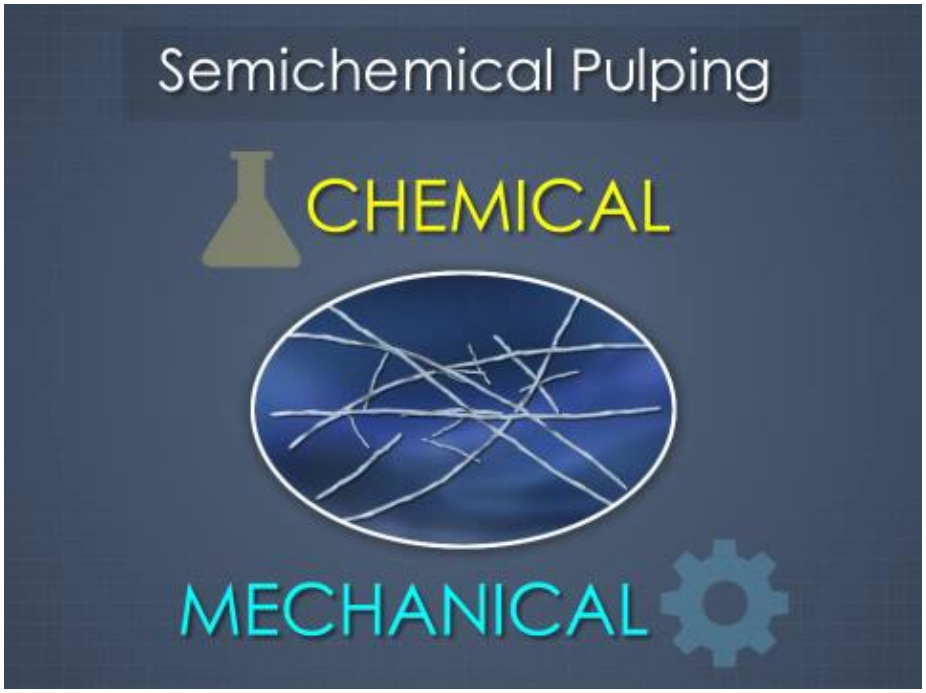
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4. Overview of Pulping methodology



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4. Overview of Pulping methodology





Semichemical pulping (SCP) processes use a combination of chemical and mechanical means to separate the fibers in wood. Two examples are:

Neutral sulfite semichemical, or NSSC, process. This process produces unbleached pulp at a relatively **high yield**.

Credit: Convergence Training

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4. Overview of Pulping methodology





"**Yield**" refers to the dry weight of the pulp produced as a percentage of the original dry wood weight.

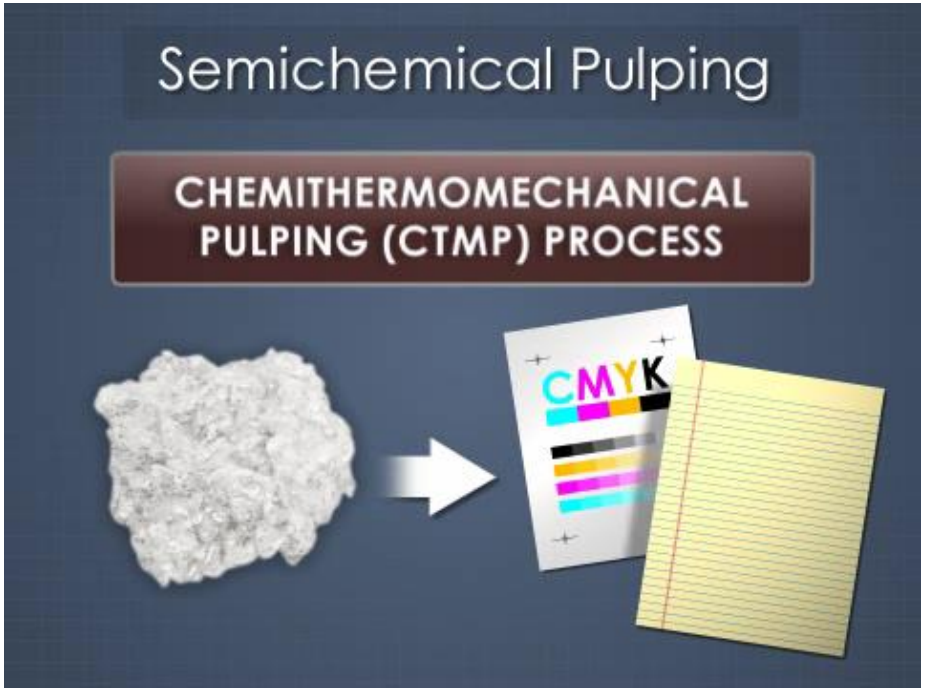
NSSC pulp is used primarily to produce **corrugating medium** for **corrugated board**.

Credit: Convergence Training

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4. Overview of Pulping methodology





Chemithermomechanical pulping, or CTMP, process. In this process, wood chips are pretreated with chemicals and steam before they are refined to improve the strength and brightness of the pulp produced.

Credit: Convergence Training

CTMP pulp is mostly used in printing and writing grades.

Remarks: There are many other semichemical pulping processes, mostly involving pretreatments and post-treatments in mechanical refining.

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SCG - SCP Mill

(Semi-Chemical Pulp Mill)

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SCP Pulp

ผลิตภัณฑ์	:	เยื่อ SCP (BP : 64,000 tpy, WS : 75,000 tpy)
วัตถุดิบ	:	ชิ้นไม้สับยูคาลิปตัส
กระบวนการผลิต	:	Green Liquor Semi-Chemical Pulping (70-75% Pulp Yield)
การใช้งาน	:	ใช้ในการผลิตกระดาษลอนลูกฟูก (CM)
ลูกค้า	:	SKIC



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SCP – Raw Material



ชิ้นไม้สับจาก

ไม้ “ลอก” เปลือก

- Eucalyptus
- Leucaena
- Acacia
- Bamboo



ชิ้นไม้สับจาก

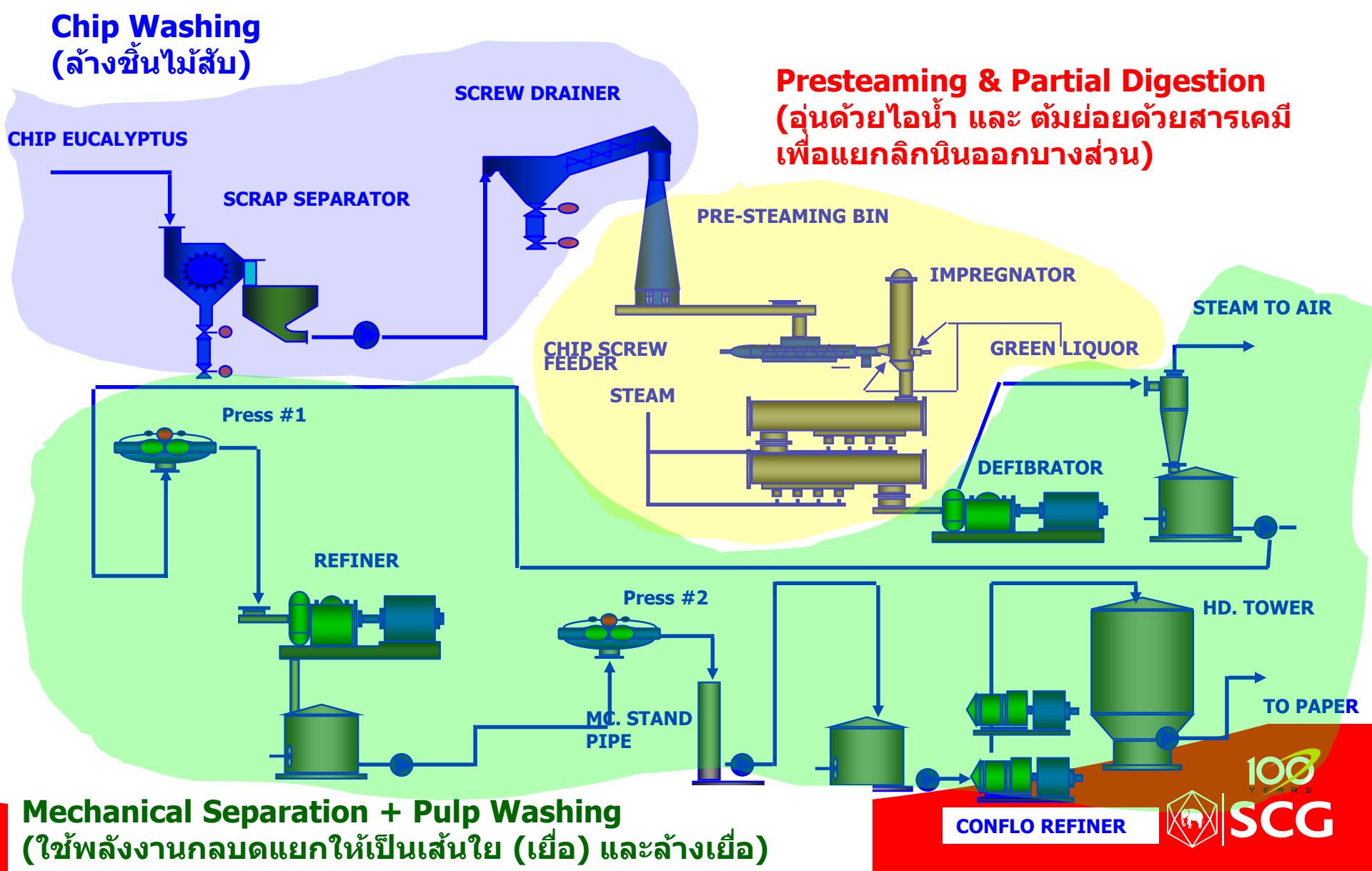
ไม้ “รวม” เปลือก



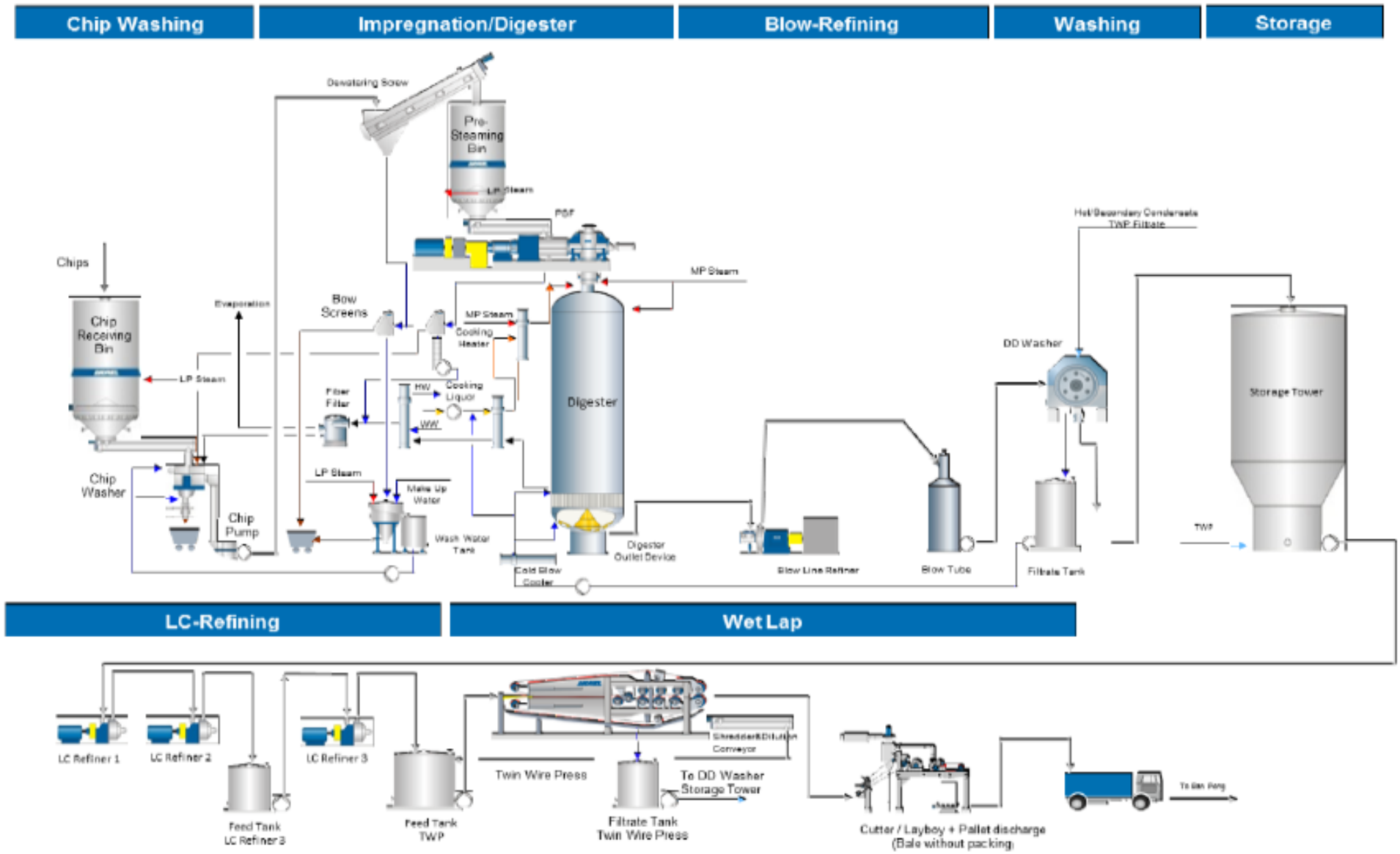
Pin Chips

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SCP Process Diagram (Banpong Mill)



SCP Process Diagram (Wangsala Mill)



SCP – Chip Washing

ระบบลำเลียง Chips เข้าสู่กระบวนการผลิต



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SCP – Chip Washing

แยกหิน / เหล็กออกจาก Chips



ดักหิน (QC)



แยกเศษเหล็กด้วยแม่เหล็กถาวร (ความแรง ~1500 Gauss)



ก้อนหิน

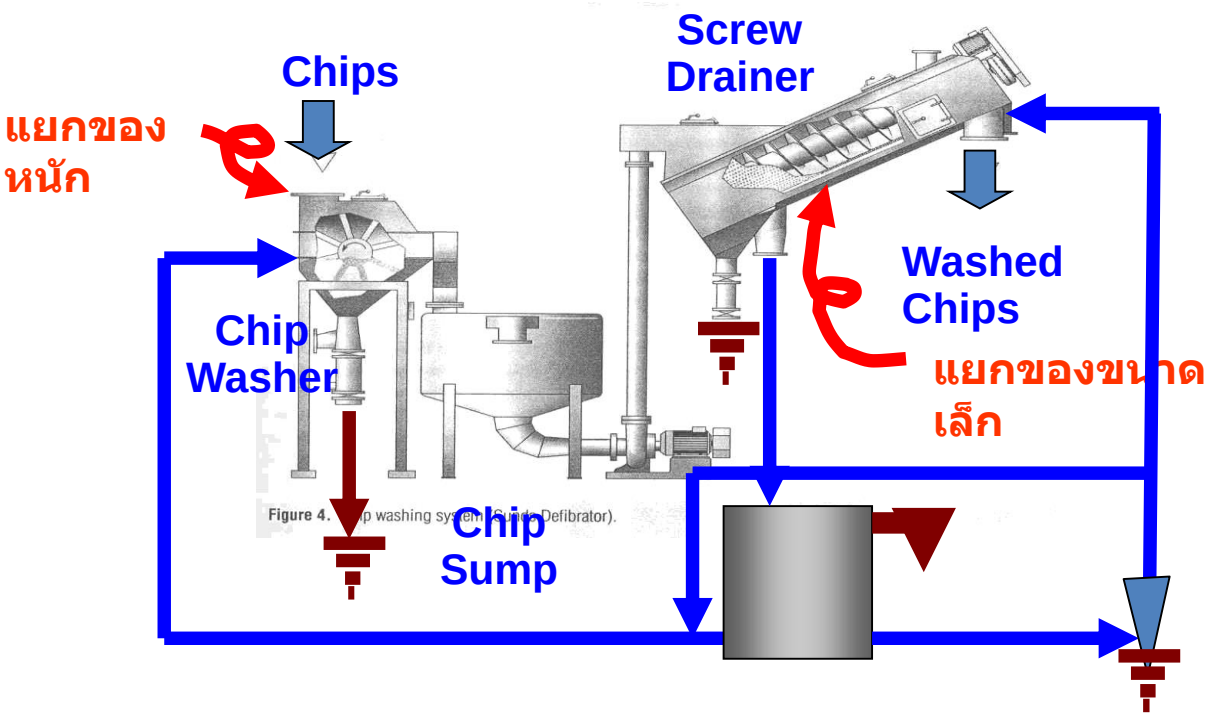


เหล็กเส้น

SCP – Chip Washing

ล้างวัตถุดิบ

1. ลด-แยกสิ่งเจือปน (ของหนัก, เศษ-กากยูคาฯ, ทราย)
2. ทำให้ความชื้นของ Chips สม่ำเสมอ
3. อุ่น Chips ให้ร้อน (อุณหภูมิ ~70-80 c)





กากยูคาลิปตัสที่แยกออก

SCP – Chip Washing

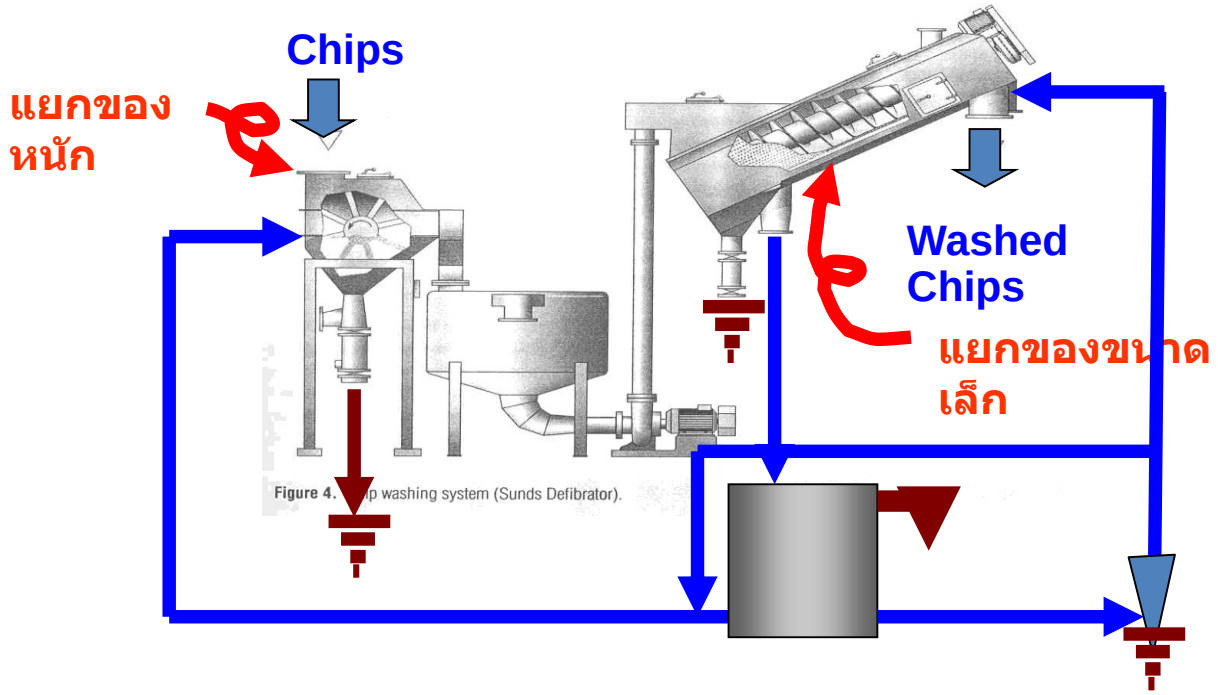
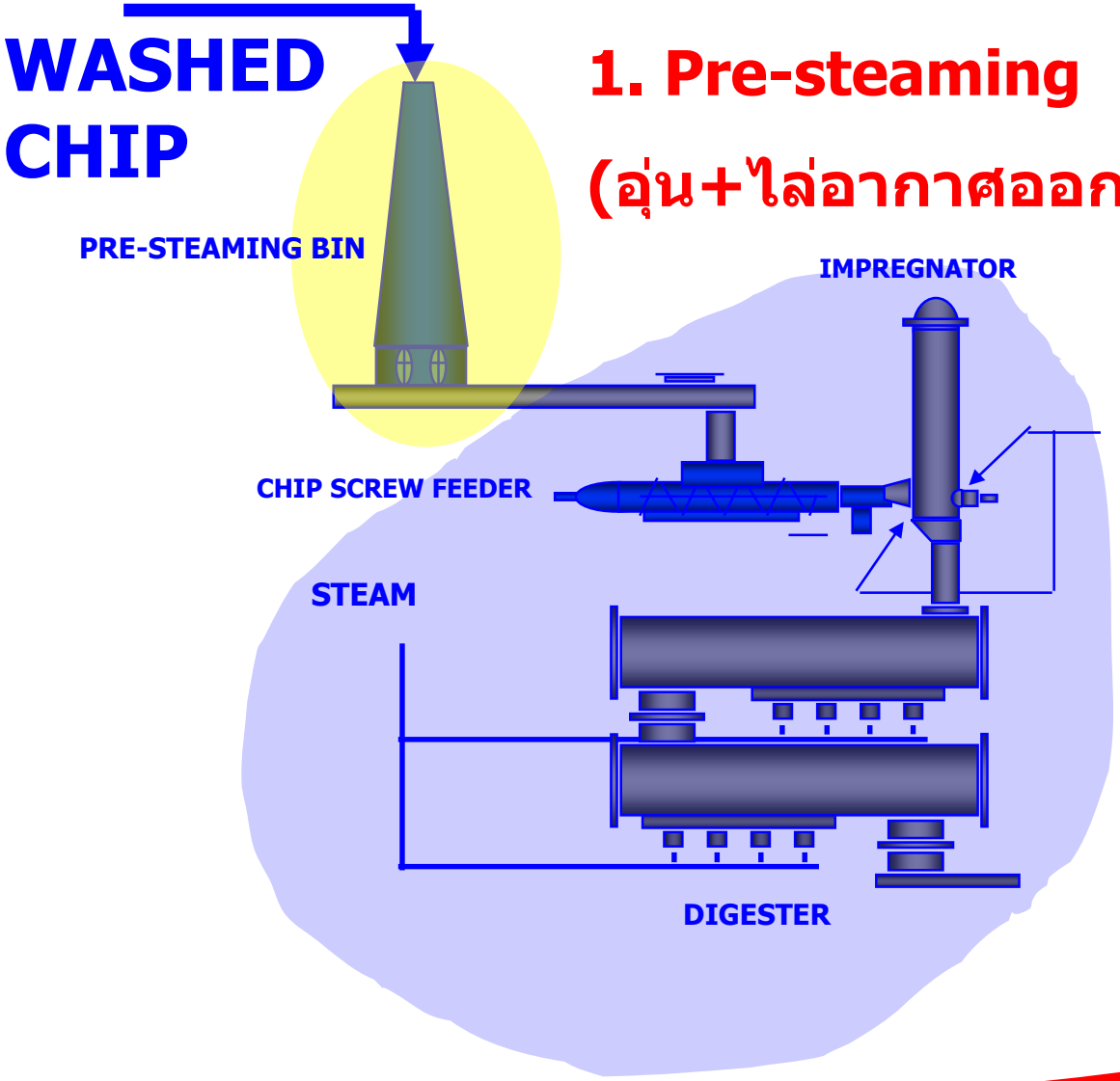


Figure 4. ip washing system (Sunds Defibrator).

การควบคุมเครื่องจักร

1. ประสิทธิภาพในการแยกสิ่งเจือปน
2. การแยกน้ำส่วนเกินออกจาก Chips

SCP – Partial Digestion



1. Pre-steaming

(อุ่น+ไล่อากาศออกจาก Chips ด้วยไอน้ำ)

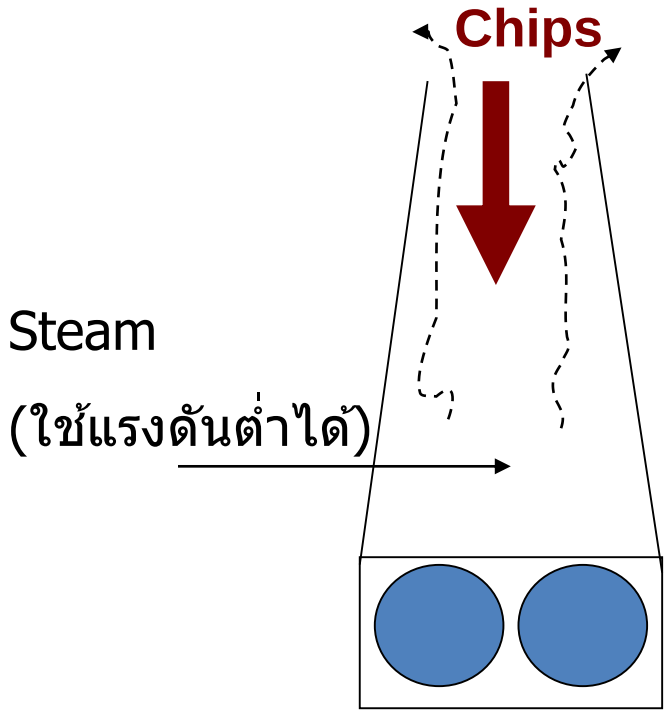
2. Partial Digestion

(ต้มย่อยด้วยสารเคมี)

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SCP – Partial Digestion

Pre-steaming (อุ่น + ไล่อากาศออกจาก Chips)





Presteaming Bin

Pre-Steaming Bin

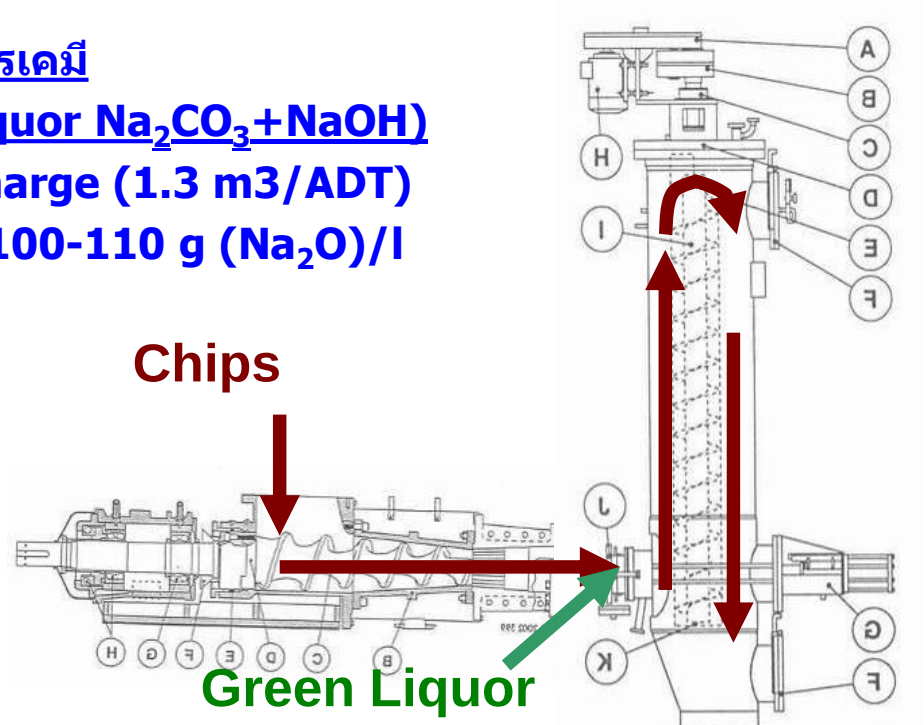
1. อุณหภูมิ (~90-100 c)
2. ระดับ (Level) ของ Chips (>50%)
3. ความเร็วรอบของ Discharge Screw

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SCP – Partial Digestion

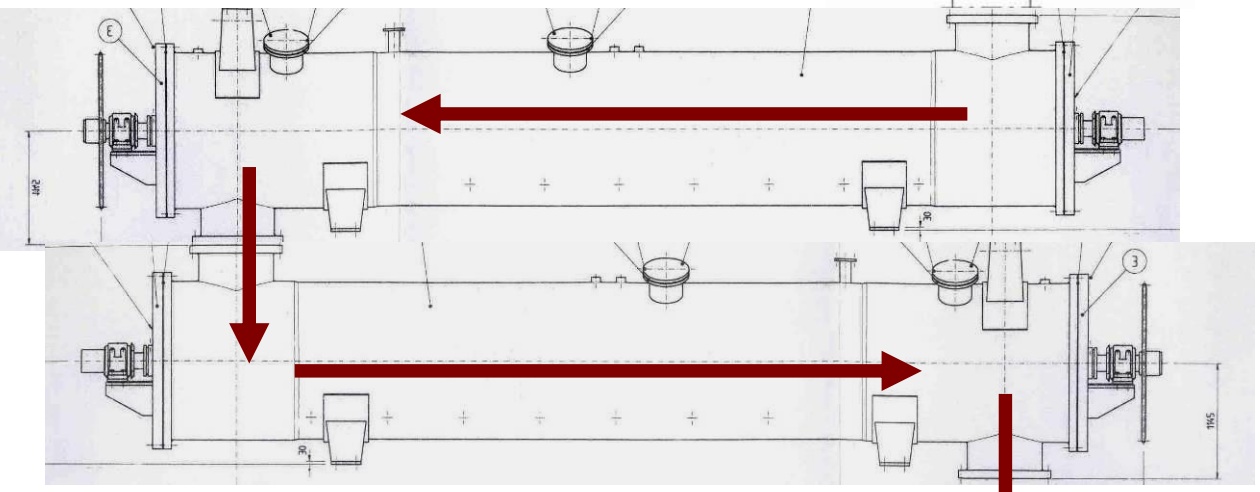
การป้อนสารเคมี
(Green Liquor Na_2CO_3+NaOH)

1. %Charge (1.3 m3/ADT)
2. TA (100-110 g (Na_2O)/l



Impregnator Condition

1. อุณหภูมิ (175c)
2. เวลา (~30 sec)
3. Liquor Level (<90%) (ระดับสารเคมีใน Impregnator)

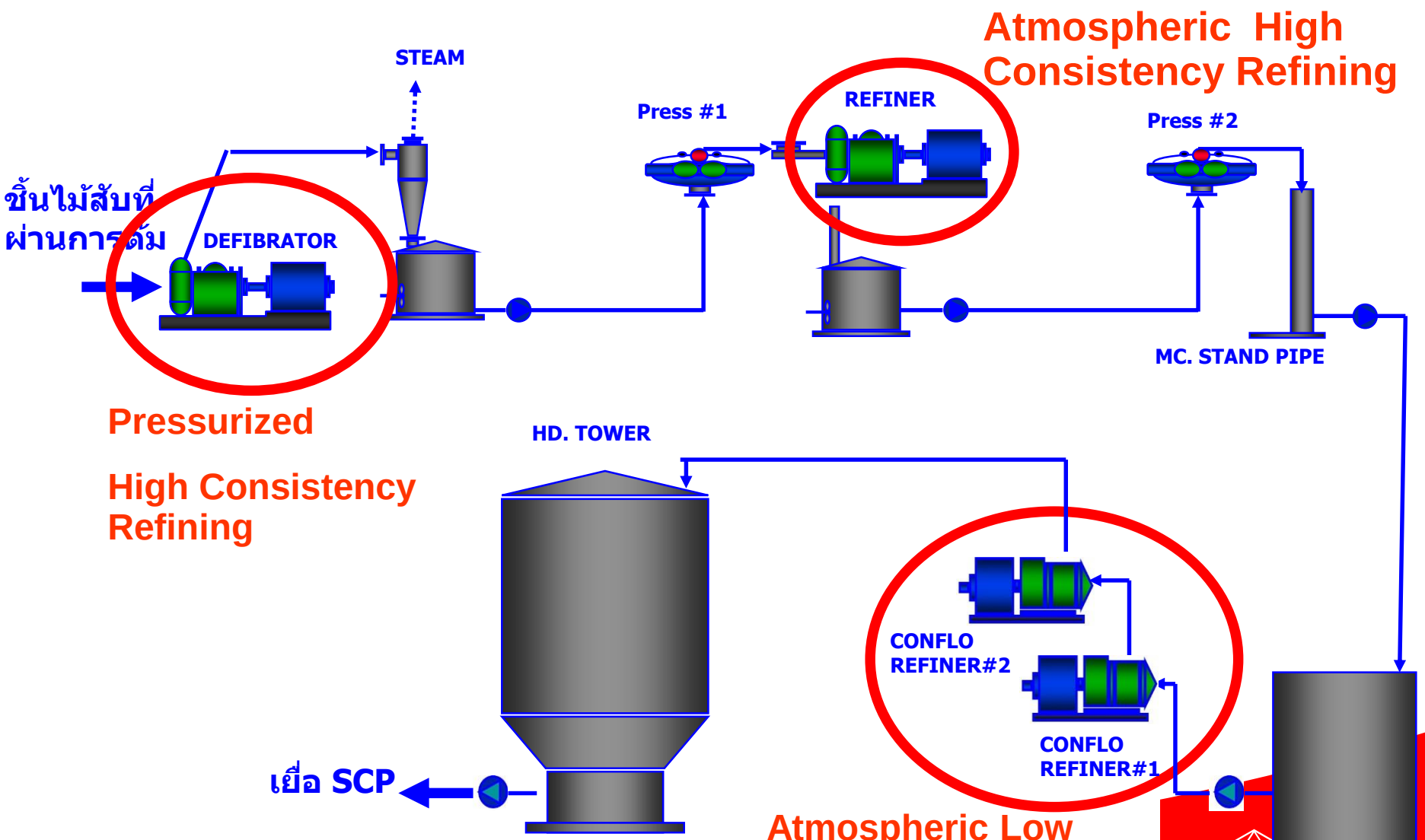


Digester Condition

1. อุณหภูมิ (175c)
2. เวลา (~25 min)
3. % Load Motor

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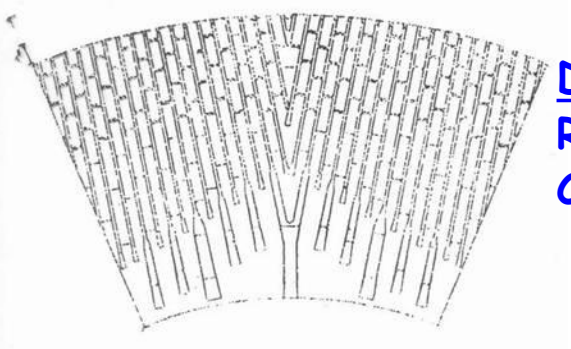
SCP Refining Process



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SCP Refining Process

Plate Pattern



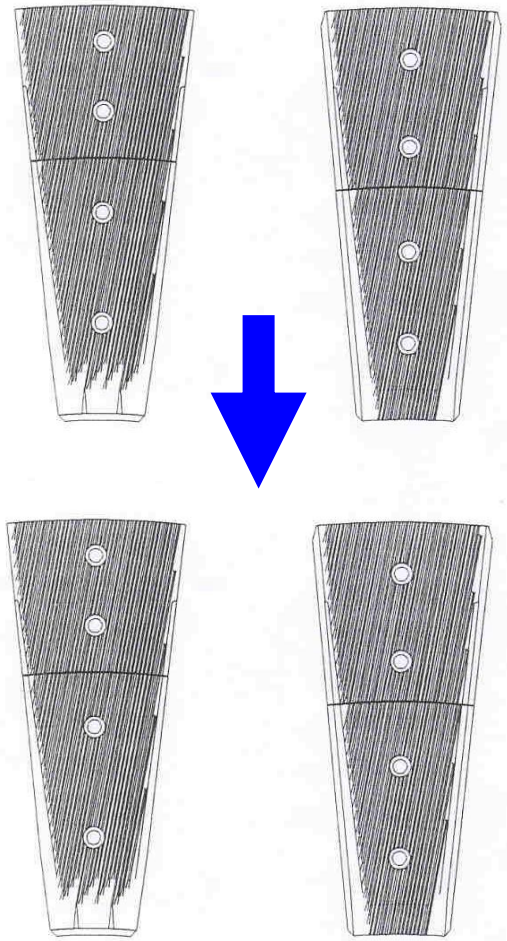
Defibrator
Refining Zone 2.5-8.5-10
Cutting Length ~10.5 km/rev



Refiner 11979s*

Fine Zone	2-3-6
Middle Zone	2-4-10

Conflo Refiner#1 JC03SF

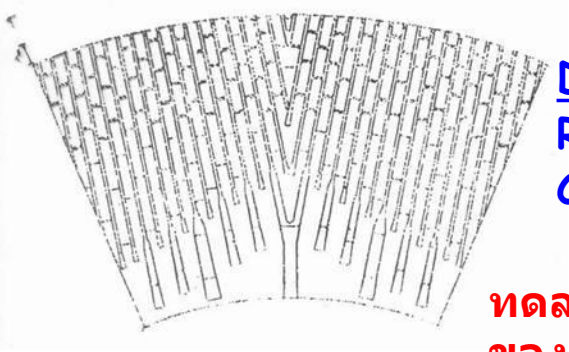


Conflo Refiner#2 JC03SF
Refining Zone 2-3-7
Bar Angle 10 mm
Cutting Length 45.7 km/rev

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SCP Refining Process

Plate Pattern



Defibrator
Refining Zone 2.5-8.5-10
Cutting Length ~10.5 km/rev

ทดลองใช้ New Patturn Design ของ Dura สามารถช่วยลด Load ที่ Swing ได้ (horizontal steaming Tube)



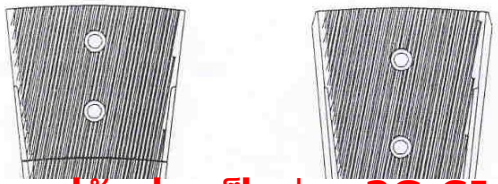
Refiner 11979s*

Fine Zone	2-3-6
Middle Zone	2-4-10

เปลี่ยนเป็นรุ่น 11979VT
ปรับ Groove width

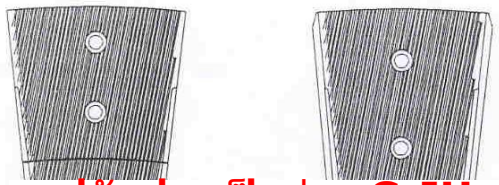
Fine zone	2-2.5-6
Middle zone	2-3.5-10

Conflo Refiner#1 JC03SF



ปรับปรุงเป็นรุ่น -2G-SJ
เพิ่มขอบของ Bar ให้สูงขึ้น





ปรับปรุงเป็นรุ่น -G-JH
เพิ่มความแข็งแรง



Conflo Refiner#2 JC03SF
Refining Zone 2-3-7
Bar Angle 10 mm
Cutting Length 45.7 km/rev

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SCP Refining Process

Refining Parameters

	Defibrator	Refiner	Conflo#1	Conflo#2
Type	Single Disc	Single Disc	Conical	Conical
	Pressurized	Atmospheric /High Consistency	Low Consistency	Low Consistency
Plate Pattern (Bar Width-Groove Width-Bar Height)	2-8.5-10	2-2.5-6 2-3.5-10	2-3-7	2-3-7
% Pulp Consistency	~20%	~30-35%	~5.0-5.5%	~5.0-5.5%
SEC (kWh/ADT)	50-100	150-250	80-120	80-120
Pulp Quality Control	%Shive <70%	Freeness 500-700 ml CSF (Shive <30%)	Freeness 500-530 ml CSF	Freeness 380-430 ml CSF

©

SCP Refining Process





1st Stage Refiner

Pressurized

High Consistency

2nd Stage Refiner

Atmospheric

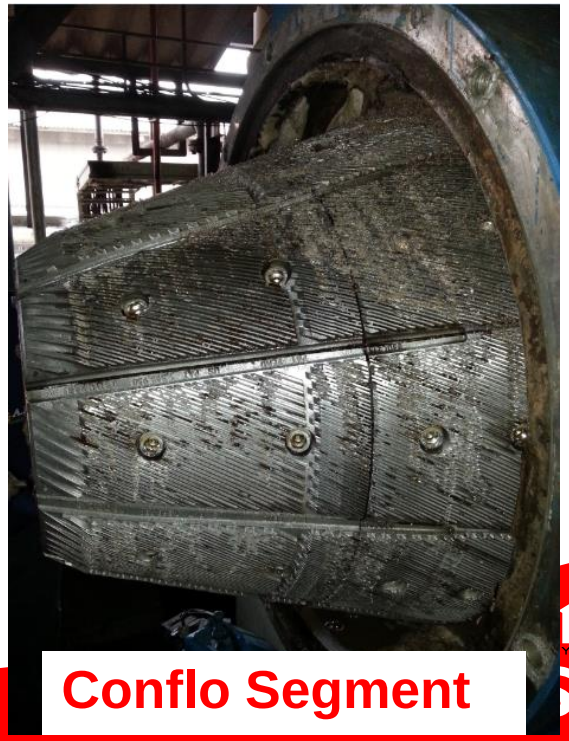
High Consistency

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SCP Refining Process







Conflo Segment

Atmospheric

Low Consistency Refiner

(Conflo Refiner; Conical Refiner)

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SCP
Pulp Characteristics

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Pulp Yield vs คุณสมบัติเยื่อ

Pulp Strength

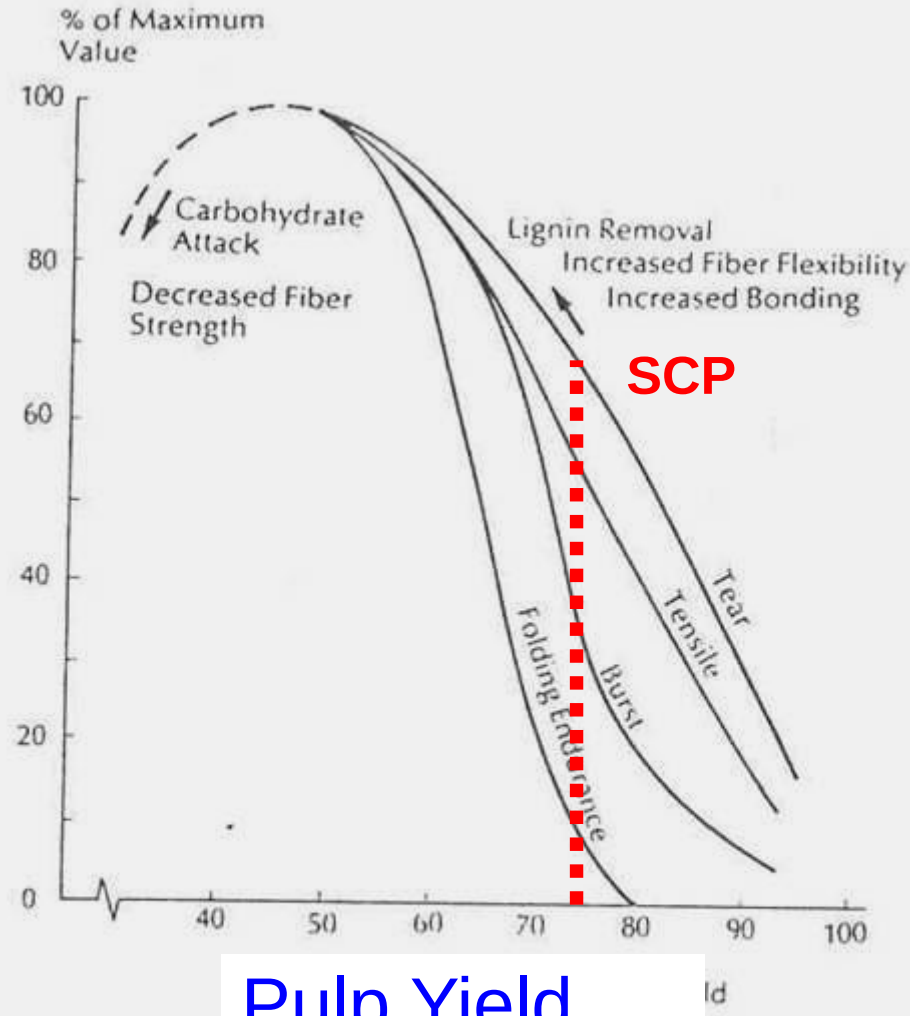
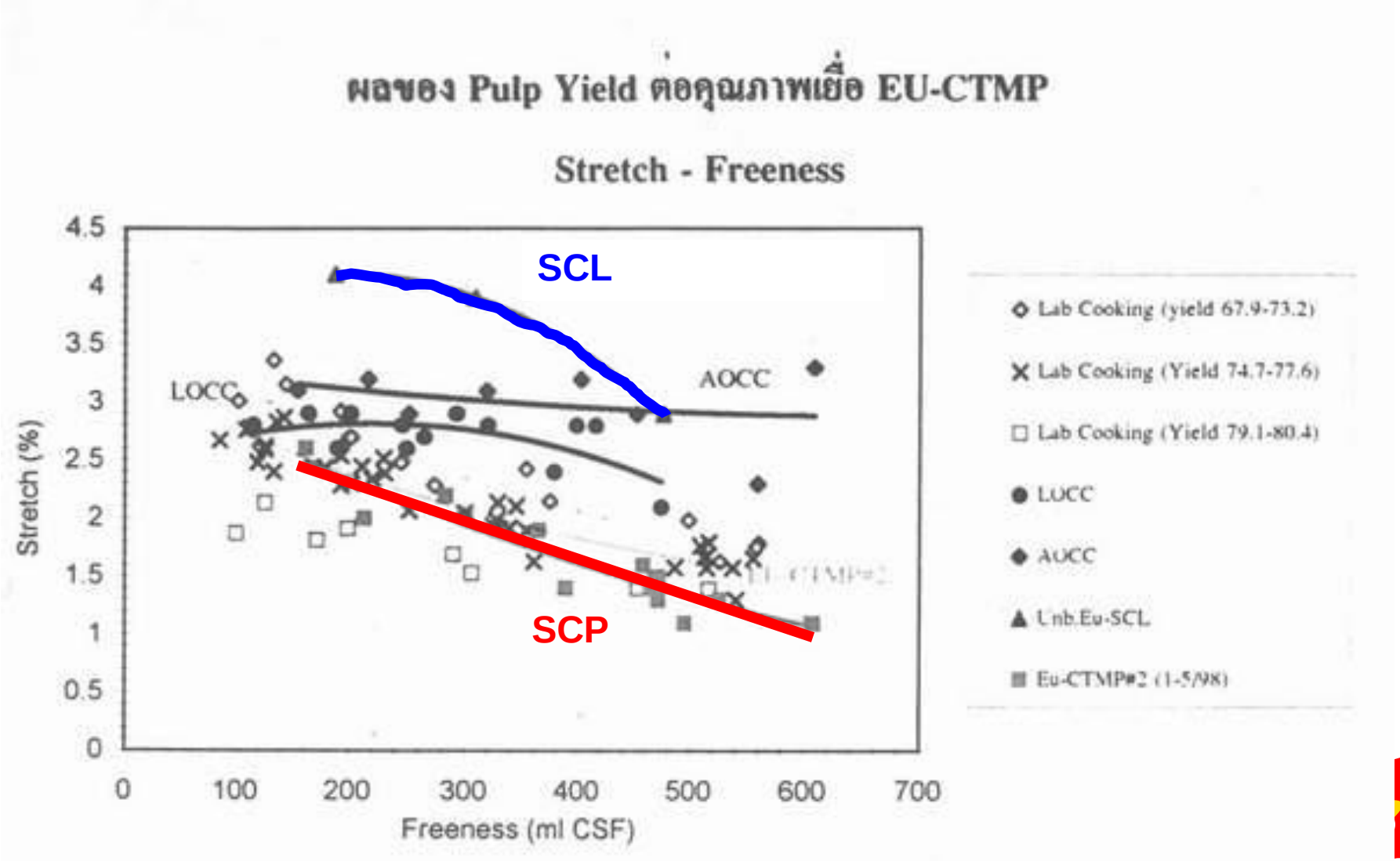


FIGURE 4-7. Effect of kraft pulp yield on selected strength parameters (at 500 CSF).

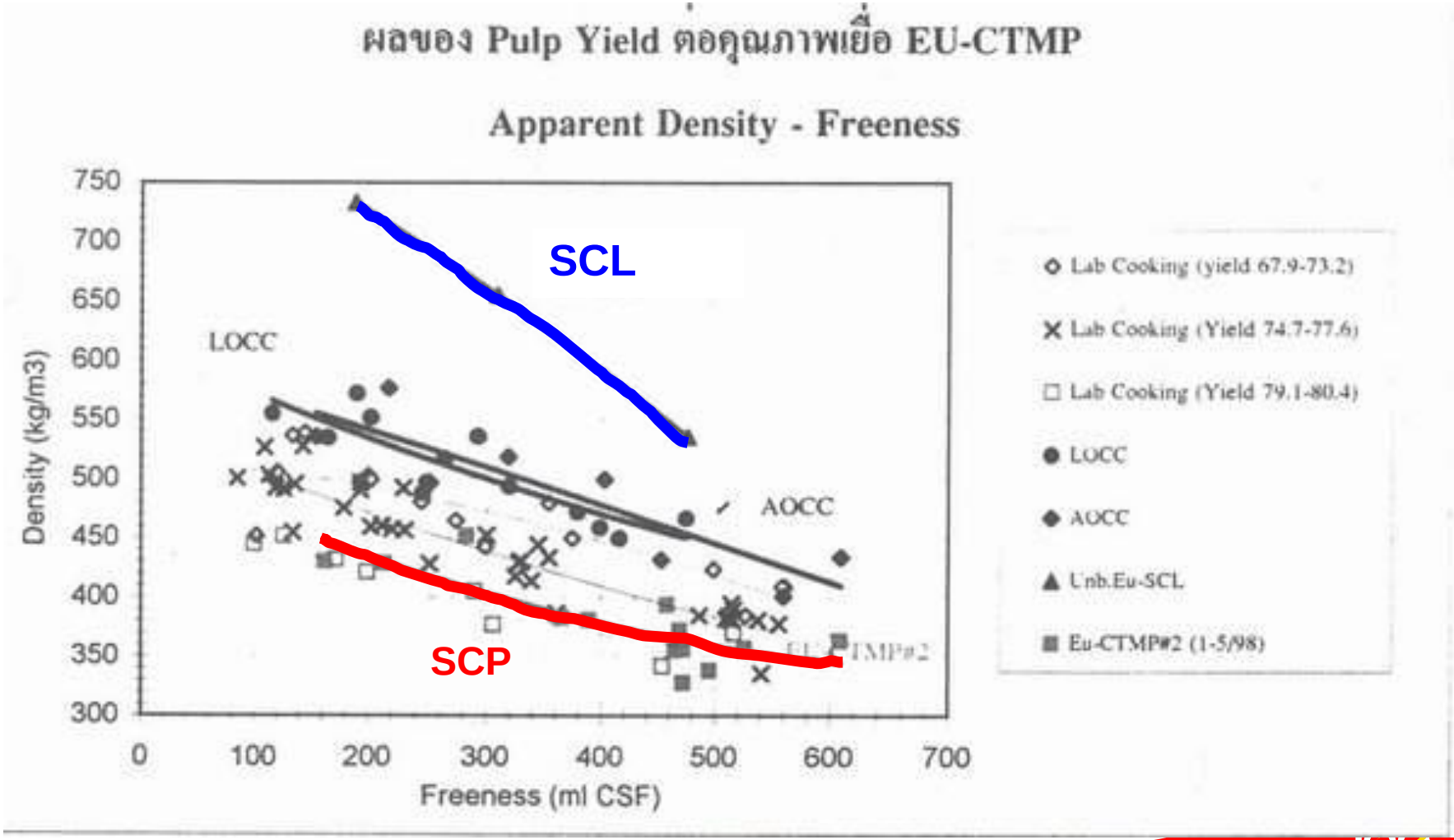
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คุณสมบัติเยื่อ SCP

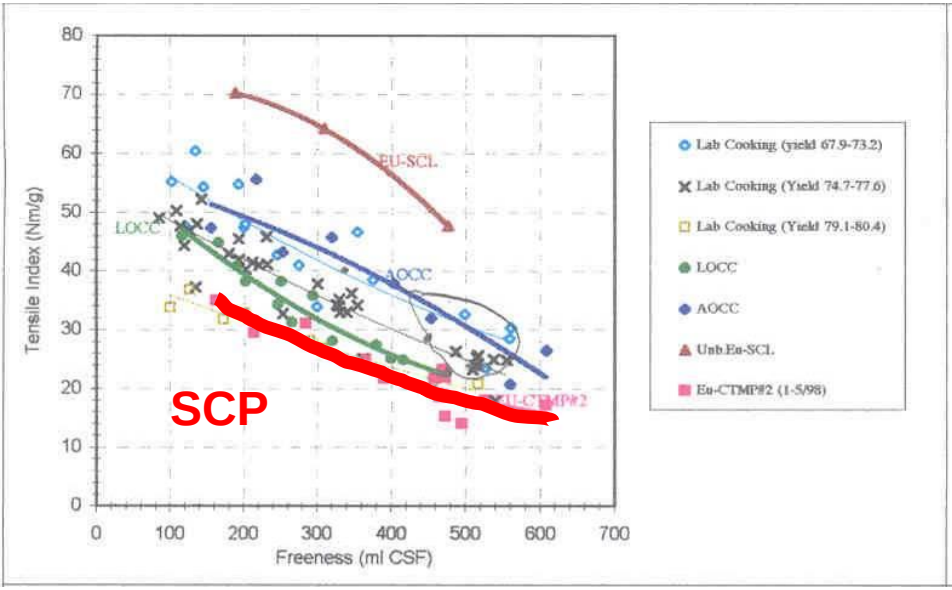


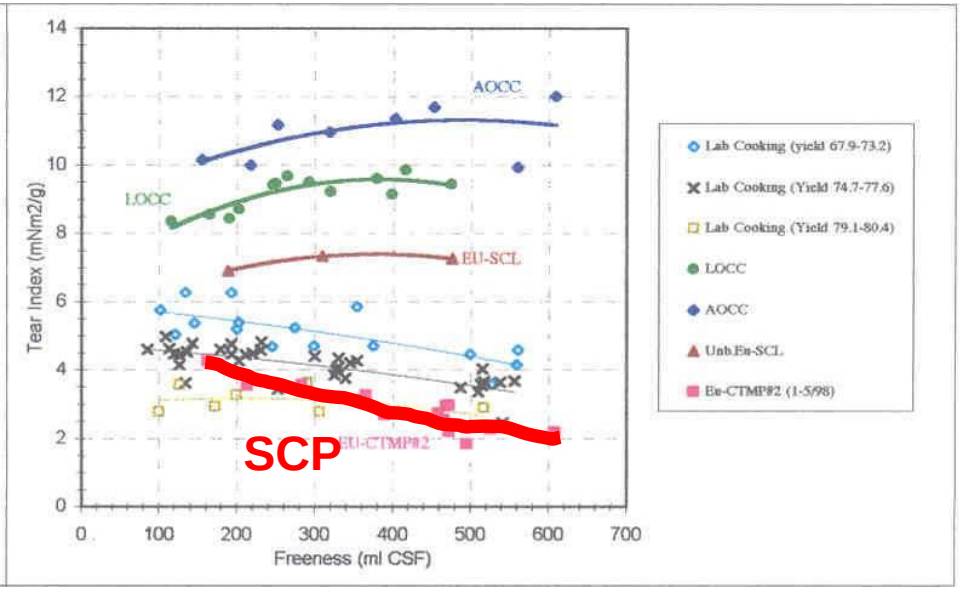
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คุณสมบัติเยื่อ SCP



คุณสมบัติเยื่อ SCP





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คุณสมบัติเยื่อ SCP vs AOCC vs LOCC

Pulp Strength Comparison Table at Freeness 300 ml.CSF

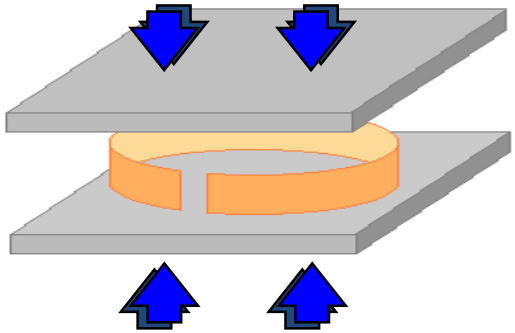
Item	Unit	SCP	AOCC	LOCC
Process Summary				
Process	-			
- Cooking	-	Semi-Cooking	n/a	n/a
- Bleaching	-	no use	n/a	n/a
- Kappa no.	-	no test	no test	no test
Raw Material	-	Mainly Eucalyptus Chip	Recycle Pulp	Recycle Pulp
Physical Property				
Brightness	%ISO	no test	no test	no test
Initial Freeness	ml. CSF	400.0	591.2	376.6
Burst Index	$kPam^2/g$	2.0	3.4	2.0
Tear Index	mNm^2/g	3.7	9.7	7.7
Tensile Index	Nm/g	29.7	50.3	34.4
CMT Index	N	264.0	271.0	228.9
RC Index	N	315.0	292.0	230.3

Remark : 1. Strength Property tested at 300 ml. CSF
2. Physical Property based on Year 2014

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คุณสมบัติเยื่อ SCP vs AOCC vs LOCC

Ring Crush Test (RCT) : The maximum compression force per unit width that a test piece of the paper or board can support until the onset of failure in a compression test
ISO 9895-1989 Unit : N

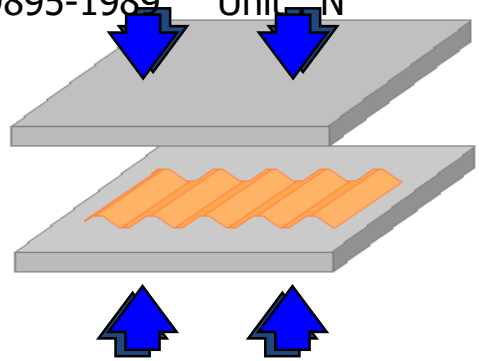


Testing Machine





Concora Medium Test (CMT) : The maximum compression force per unit width that a test piece of the paper or board can support until the onset of failure in a compression test
ISO 9895-1989 Unit : N



Testing Machine





คุณสมบัติเยื่อ SCP

การใช้งานเยื่อ SCP

- กระดาษลอนลูกฟูก (Corrugating Medium)
- ชั้นล่าง/ชั้นกลางของ Linerboard

เยื่อ SCP vs เยื่อเศษกระดาษ

- Stiffness / CMT / RCT สูงกว่า
- ต้นทุนการผลิตต่ำกว่า
- ความสะอาดของเยื่อ
- การ Drain น้ำบนลวดเดินแผ่นดีกว่า
- Tensile, Tear, Stretch ต่ำ
- เยื่อมีสีคล้ำ / มีฟอง / เสี้ยน
- บดเพิ่มคุณภาพยาก (ใช้พลังงานมากกว่า)

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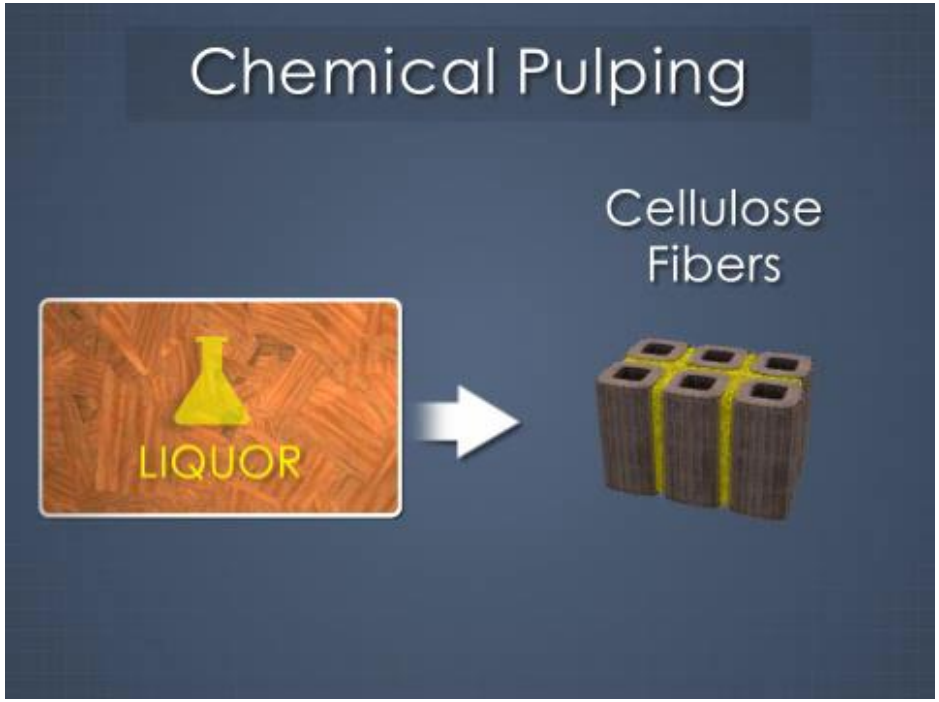
4. Overview of Pulping methodology



Credit: Cor

4. Overview of Pulping methodology





The two major chemical pulping processes are:

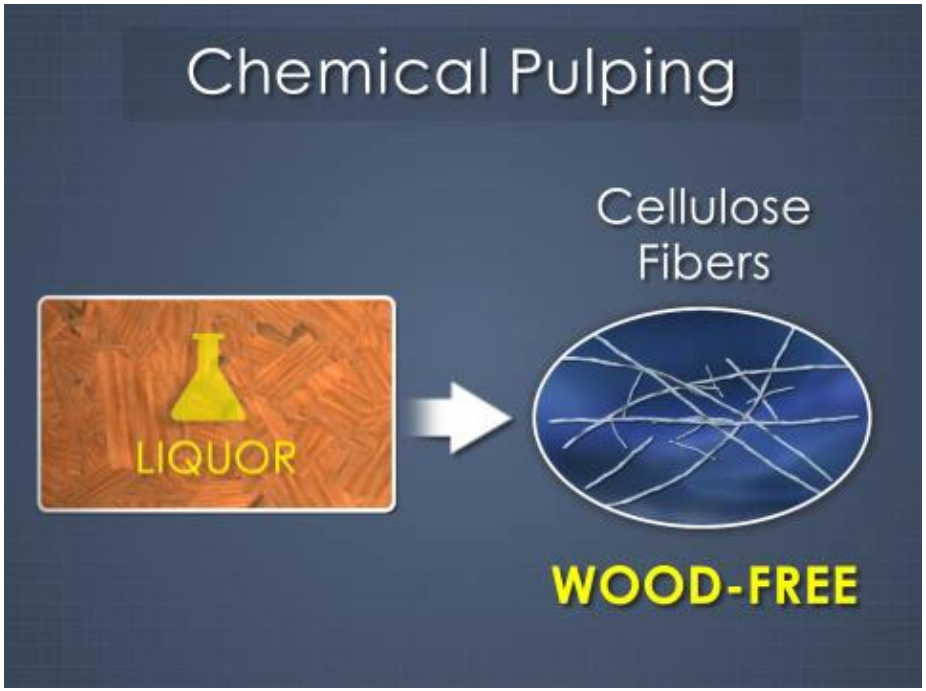
- Sulfite
- Kraft, or sulfate

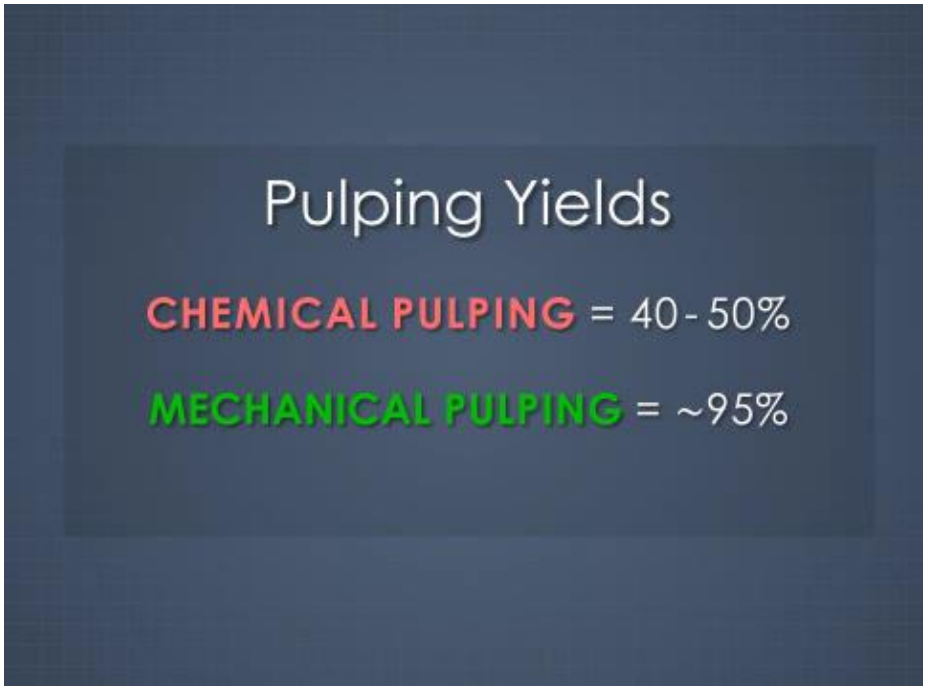
In both processes, wood chips are "cooked" in a chemical solution called "liquor" to remove the lignin and separate the cellulose fibers from each other.

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4. Overview of Pulping methodology





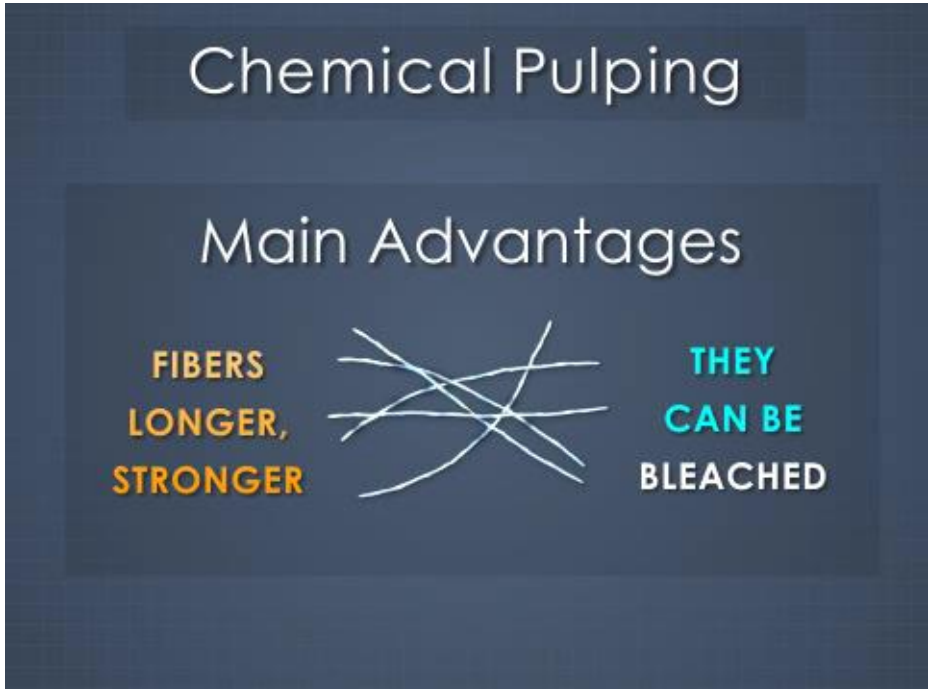
Because lignin and some other wood components are dissolved and removed, these pulps are referred to as "wood-free."

One major disadvantage of chemical pulping processes is that they have yields in the range of 40 to 50%, whereas mechanical pulping yields run around 95%.

Credit: Convergence Training

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4. Overview of Pulping methodology



The main advantages of chemical pulps are that the fibers are longer and stronger, and they can be bleached to a high brightness.

Credit: Convergence Training

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4. Overview of Pulping methodology

Cooking

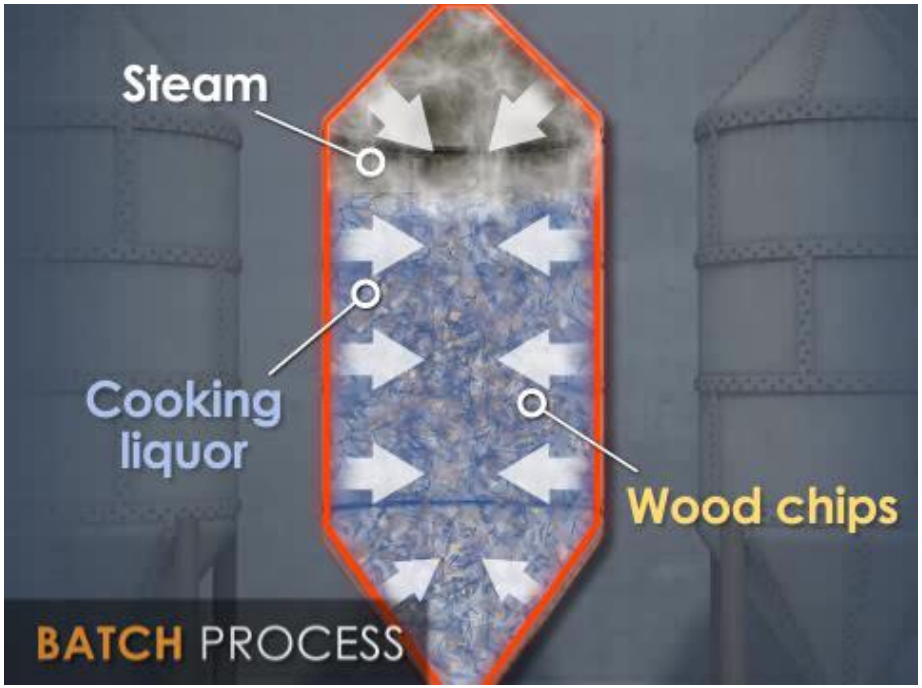




4. Overview of Pulping methodology



In chemical pulping, the cooking is accomplished in a "digester" (a large pressure cooker), in either a batch or a continuous process.

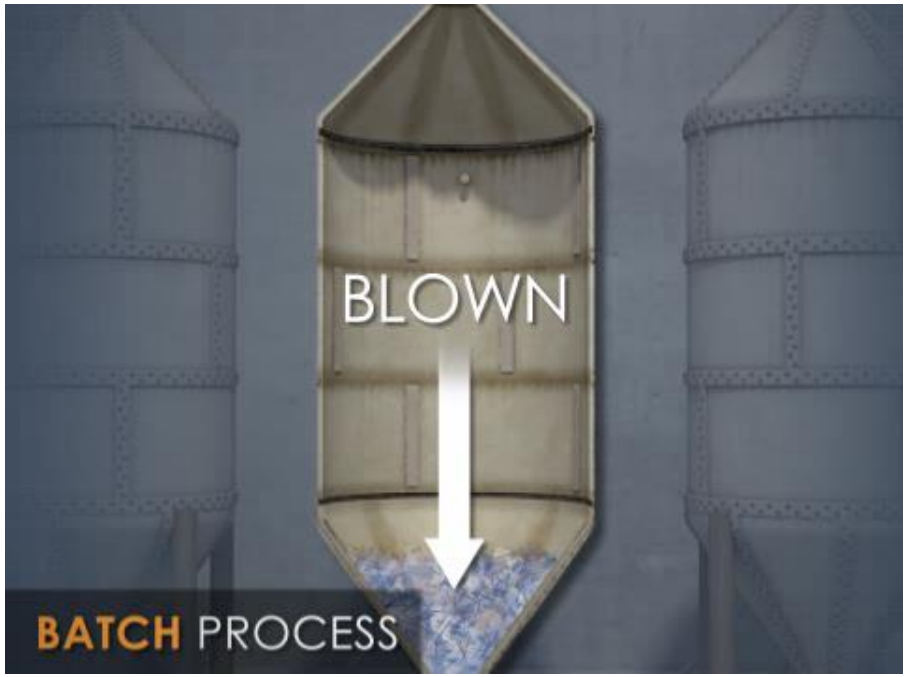


In the batch process, the digester is loaded with wood chips, filled with cooking liquor, and then heated and pressurized with steam.

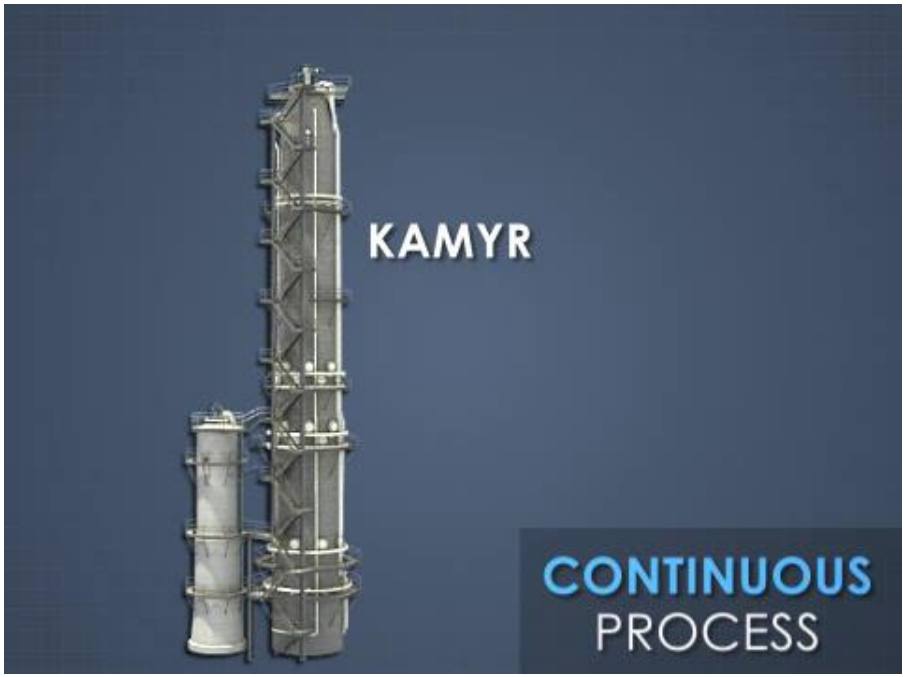
Credit: Convergence Training

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4. Overview of Pulping methodology



After several hours at these conditions, the liquor-fiber mixture is released or "blown" from the bottom of the digester.



In a continuous digester (Kamyr being the most common name),

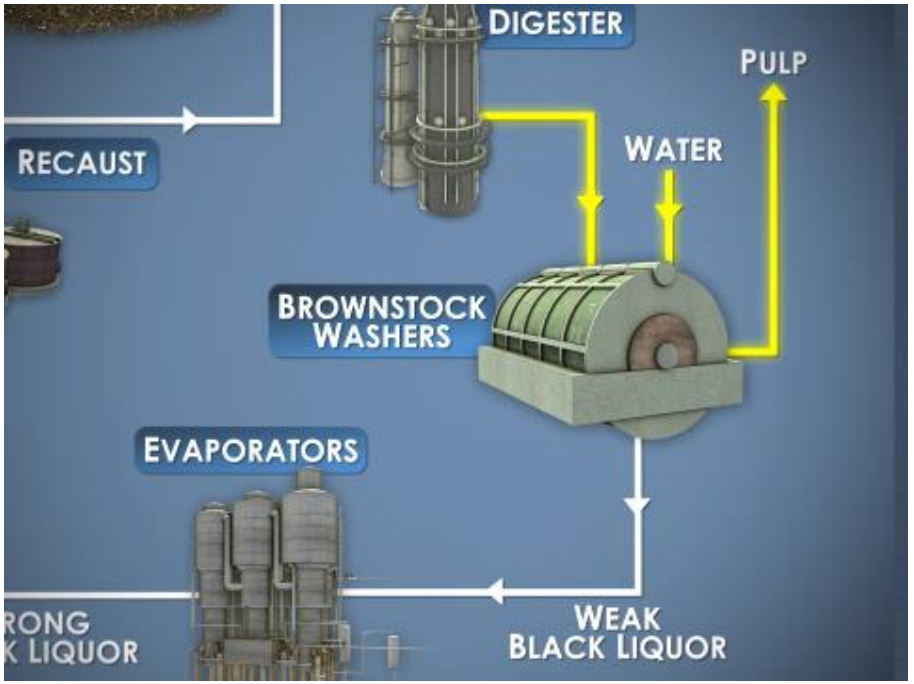
Credit: Convergence Training

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4. Overview of Pulping methodology



the wood chips and heated cooking liquor are continuously fed into the top, the cooking occurs as they travel through the digester, and the liquor-fiber mixture is continuously blown from the bottom.



After cooking, the used pulping chemicals and dissolved wood components are washed from the pulp using water. The pulp is directed to the next process step, while the wash water is pumped to the recovery area of the mill

Credit: Convergence Training

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4. Overview of Pulping methodology

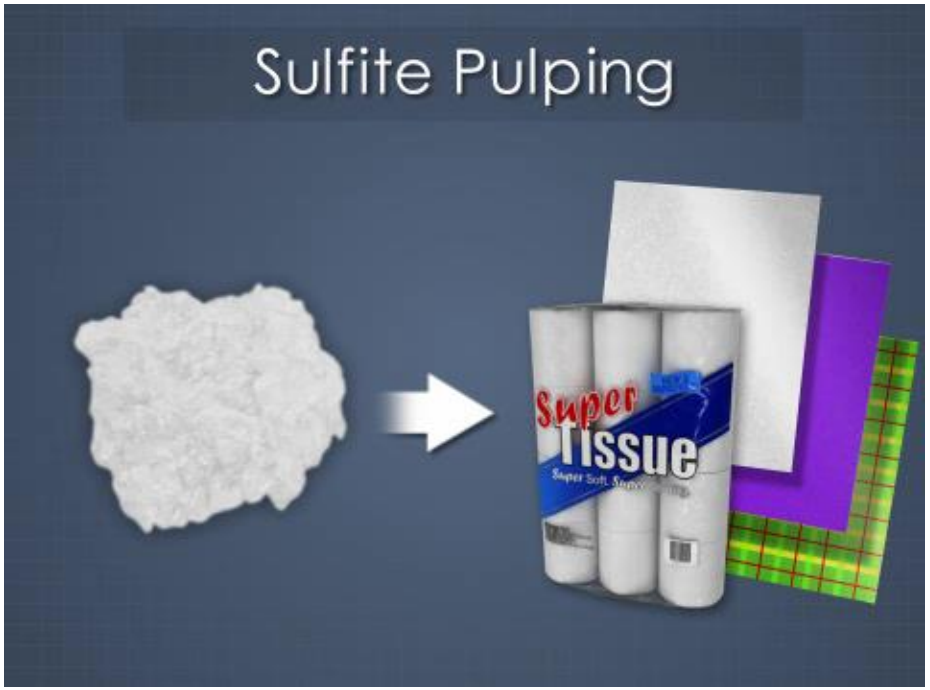


Credit: Co

4. Overview of Pulping methodology



The sulfite process uses an acidic cooking solution. Sulfite pulps are easily bleached to remove all of the remaining lignin,

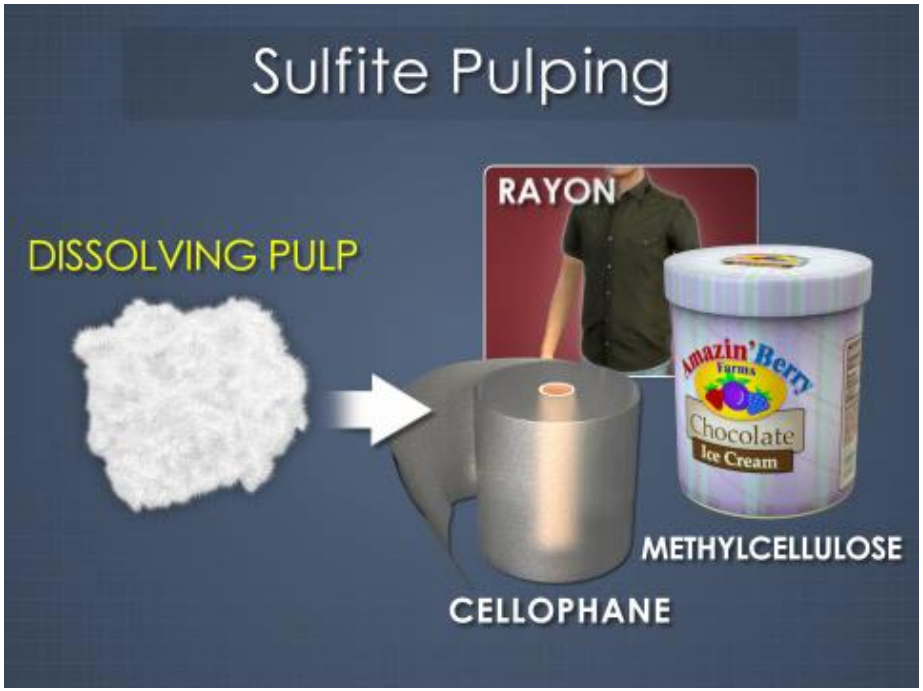


They are currently used to produce tissue, fine paper, and specialty grades.

Credit: Convergence Training

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4. Overview of Pulping methodology



One significant specialty grade of highly-bleached sulfite pulp is called "dissolving pulp," which can be used to produce rayon, cellophane, and methylcellulose (a thickener and emulsifier that is used in food and cosmetic products).



One drawback to the sulfite process is that it is sensitive to some wood species; it cannot handle resinous softwoods, dark-colored hardwoods, or bark-contaminated wood.

Credit: Convergence Training

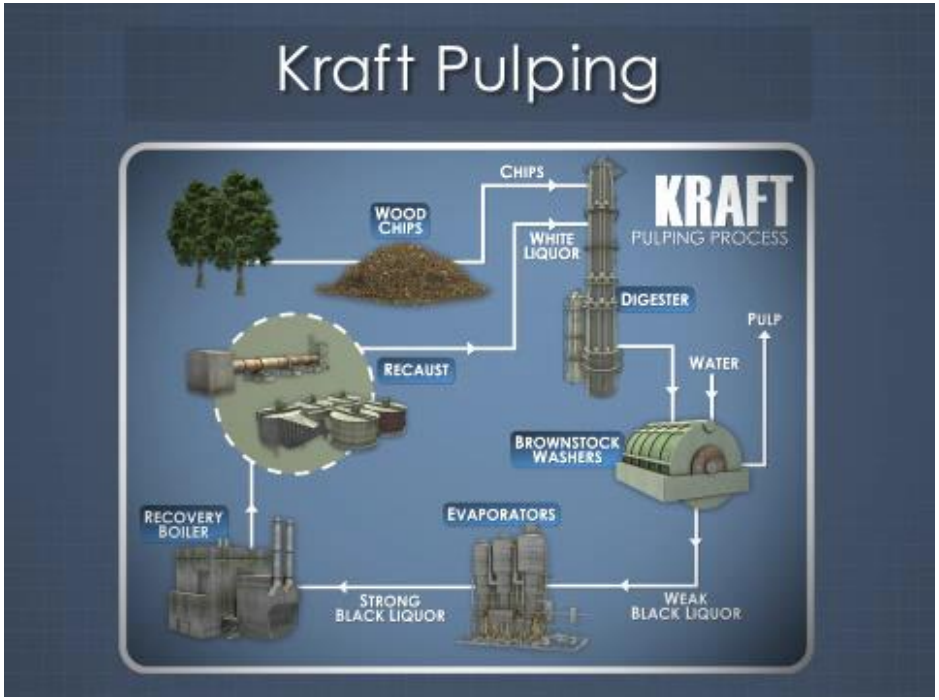
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4. Overview of Pulping methodology



Credit: Co

4. Overview of Pulping methodology





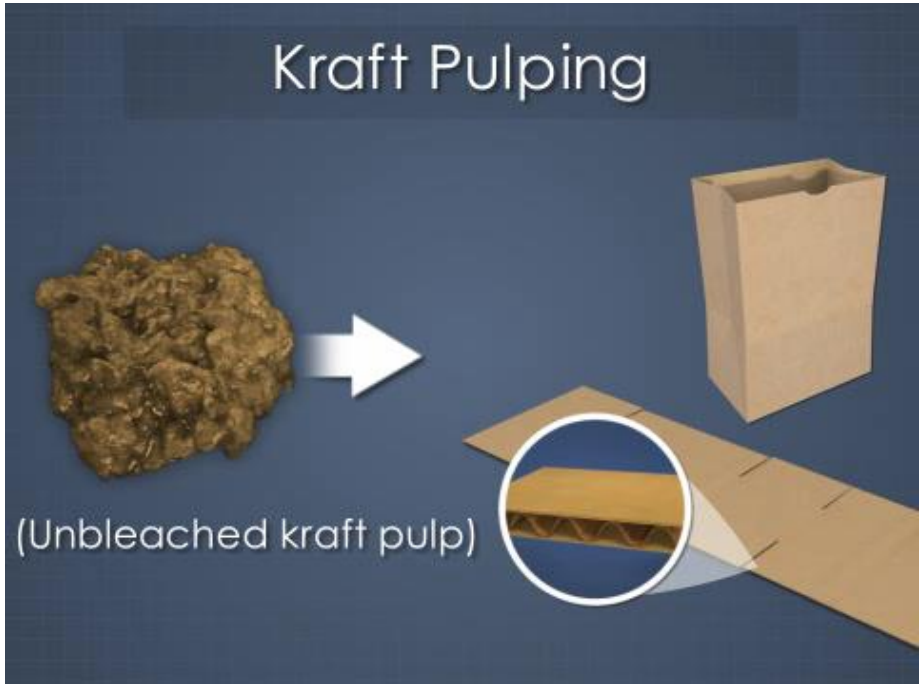
The kraft (or sulfate) process is by far the dominant chemical pulping process. It uses an alkaline cooking solution and has several advantages over the sulfite process, including:

- Handles a wide variety of wood species
- Tolerates bark-contaminated wood chips
- Produces the highest strength pulp
- Has a proven, efficient chemical recovery process

Credit: Convergence Training

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4. Overview of Pulping methodology





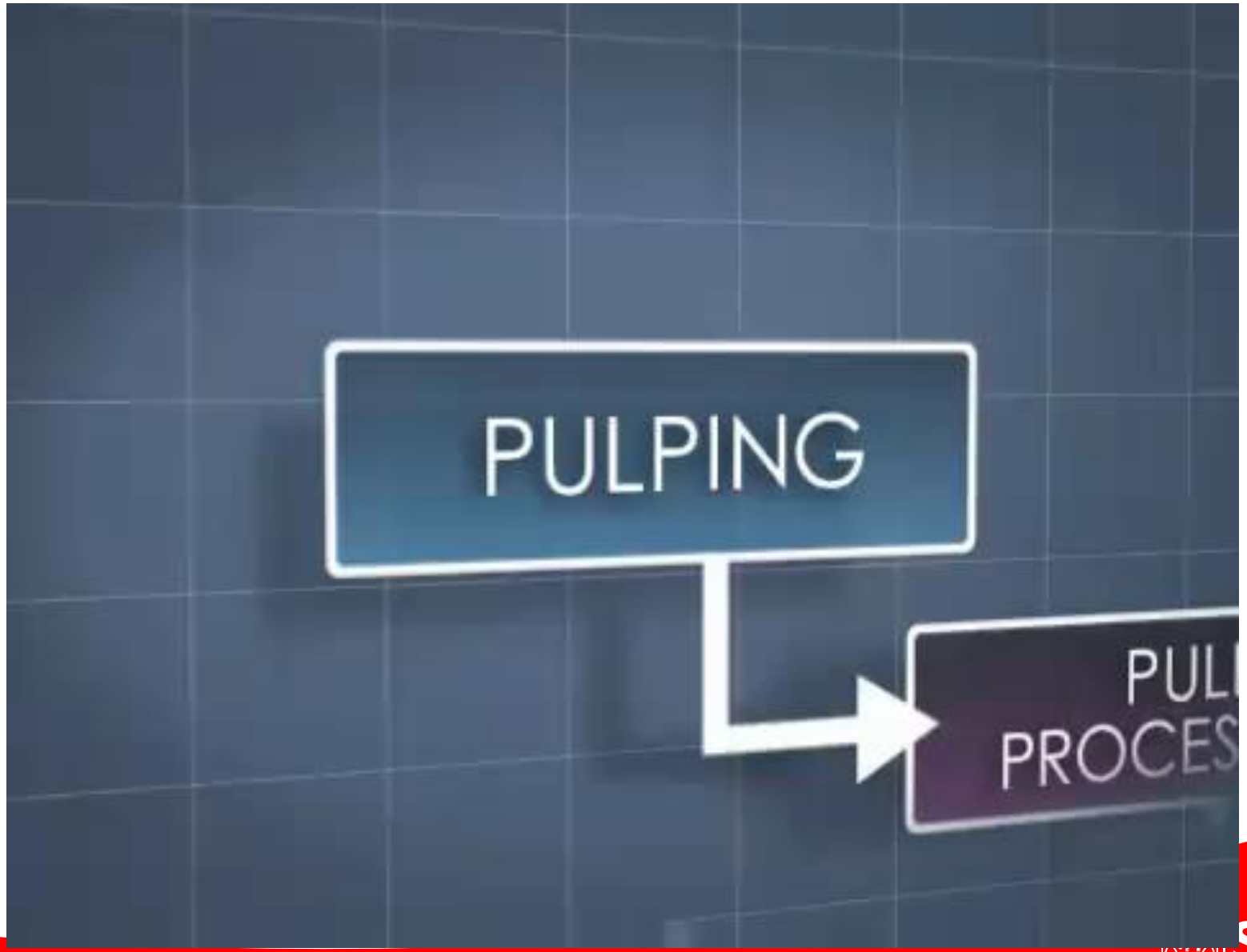
Unbleached kraft pulp is dark brown in color and is very strong. It is used to produce packaging materials, including corrugated board and paper bags.

Kraft pulp that has been bleached white is commonly used to produce white papers that require strength, including tissue, printing, and packaging grades.

Credit: Convergence Training

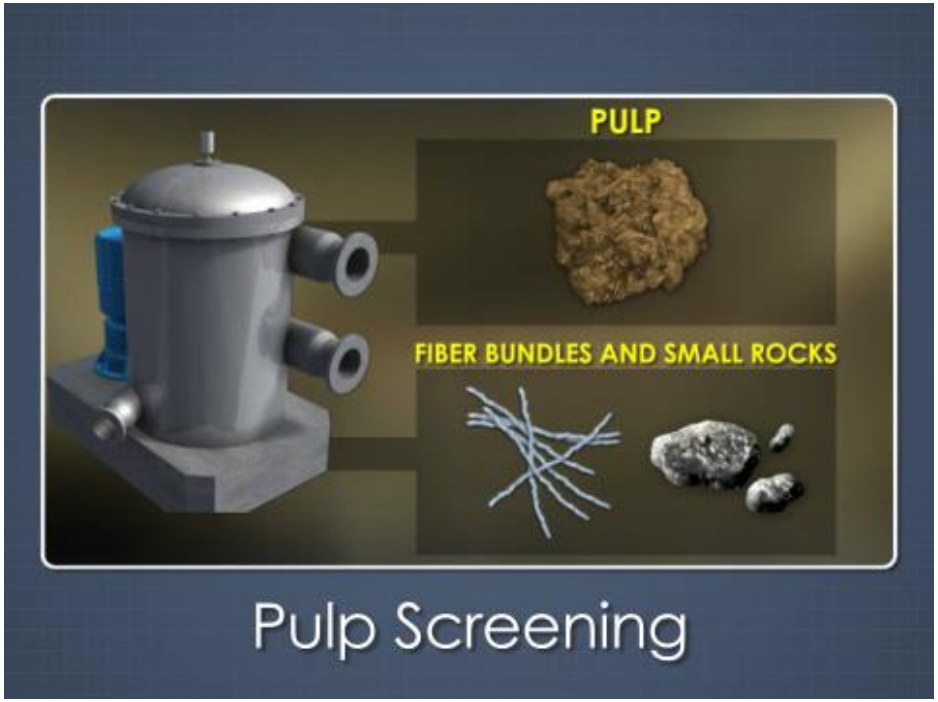
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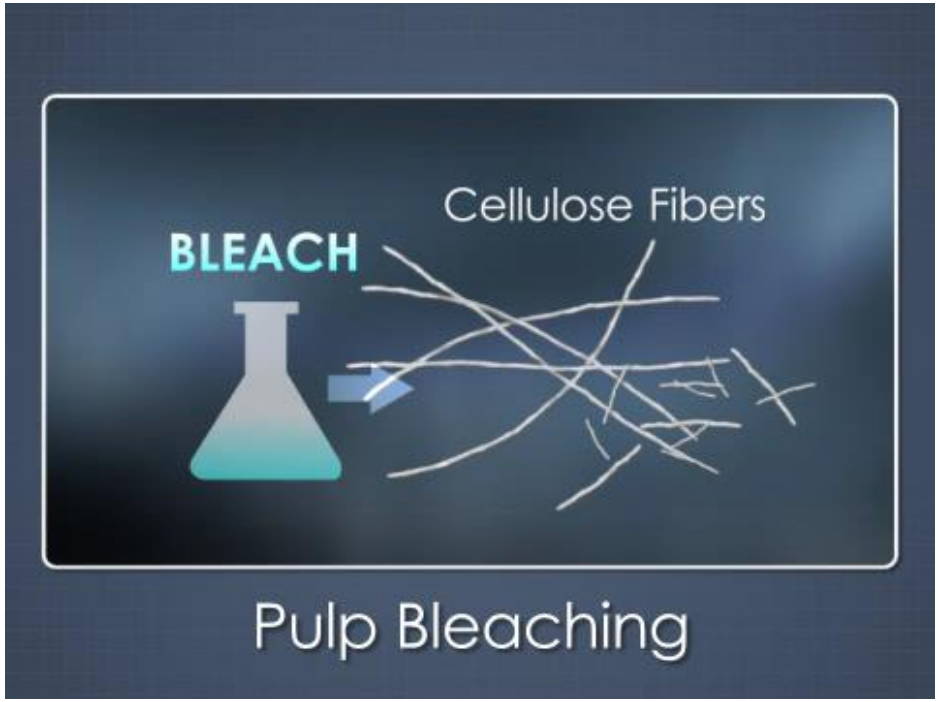
4. Overview of Pulping methodology



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4. Overview of Pulping methodology





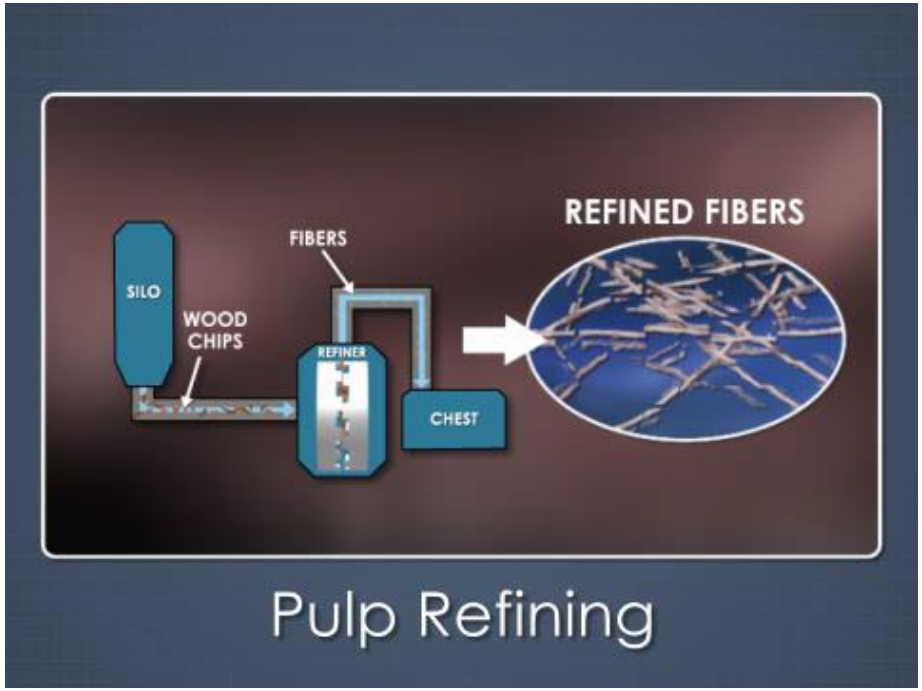
After pulping, the cellulose fibers liberated from the wood may undergo one or more additional processes to prepare them for papermaking. These include:

- Pulp screening to remove contaminants and impurities, such as fiber bundles and small rocks
- Pulp bleaching with chemicals to increase brightness (this will be covered in greater detail on the next page)

Credit: Convergence Training

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4. Overview of Pulping methodology





- Pulp refining to make the fibers more pliable and increase the surface area available for bonding; these actions improve the strength characteristics of the paper produced from the fibers

The desired characteristics of the paper being produced dictate the type and amount of processing performed on the pulp fibers.

Credit: Convergence Training

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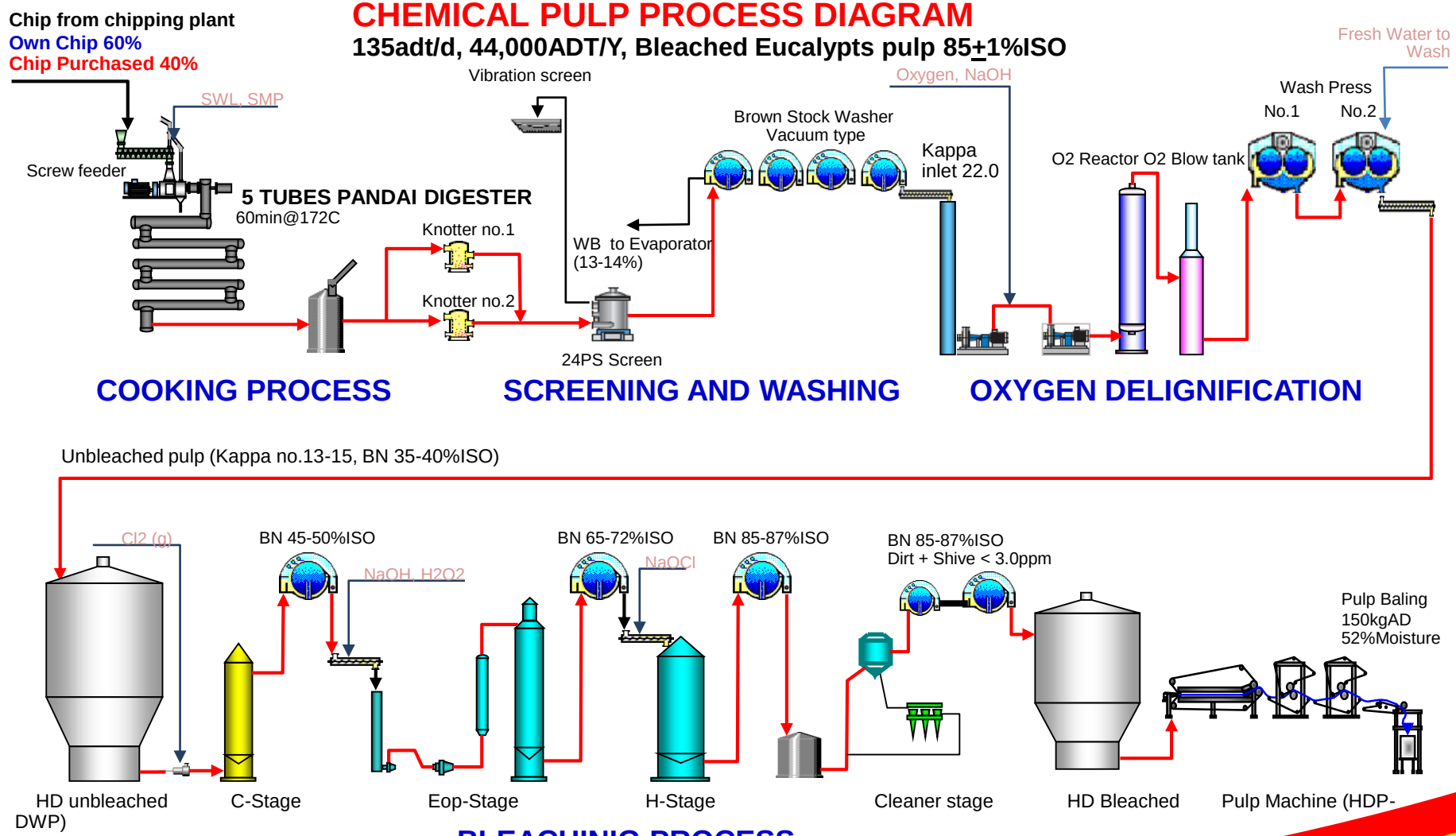
SCG - CP Mill

(Chemical Pulp Mill)

Soda Process

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4. Overview of Pulping methodology



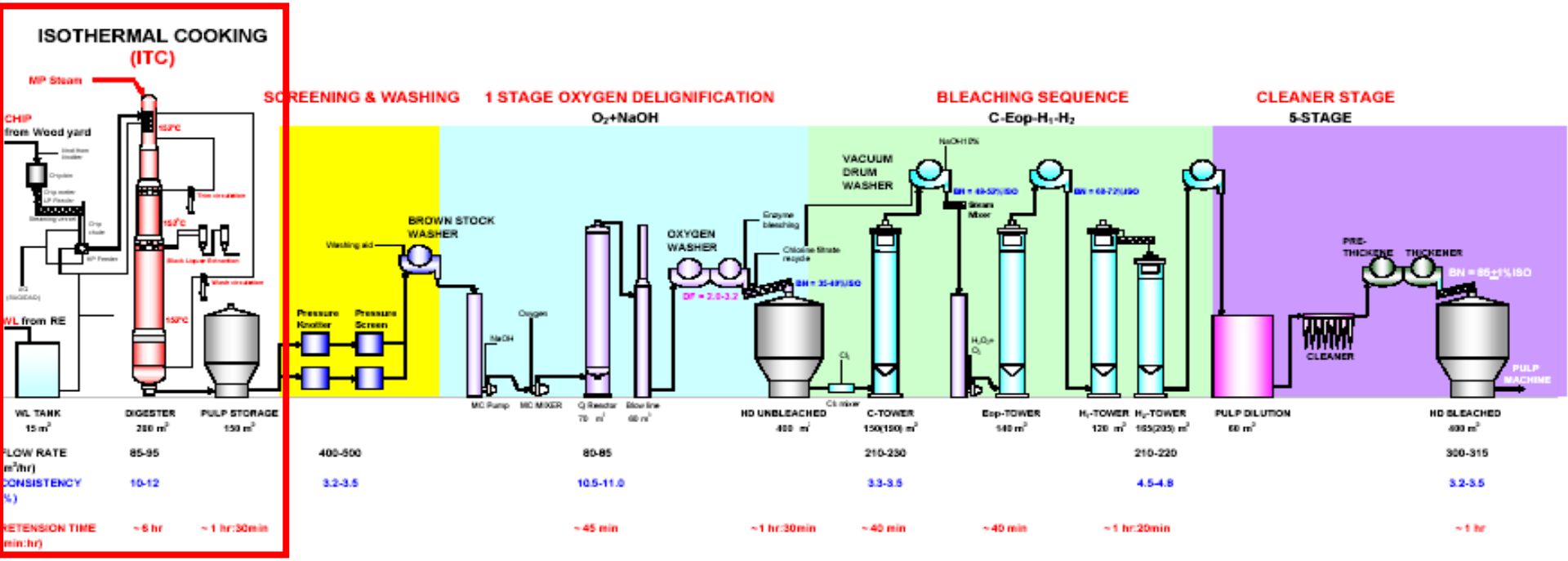
Credit: Weerachai Seachao

4. Overview of Pulping methodology

SCL Wangsala Pulp Mill

CHEMICAL PULP PROCESS DIAGRAM

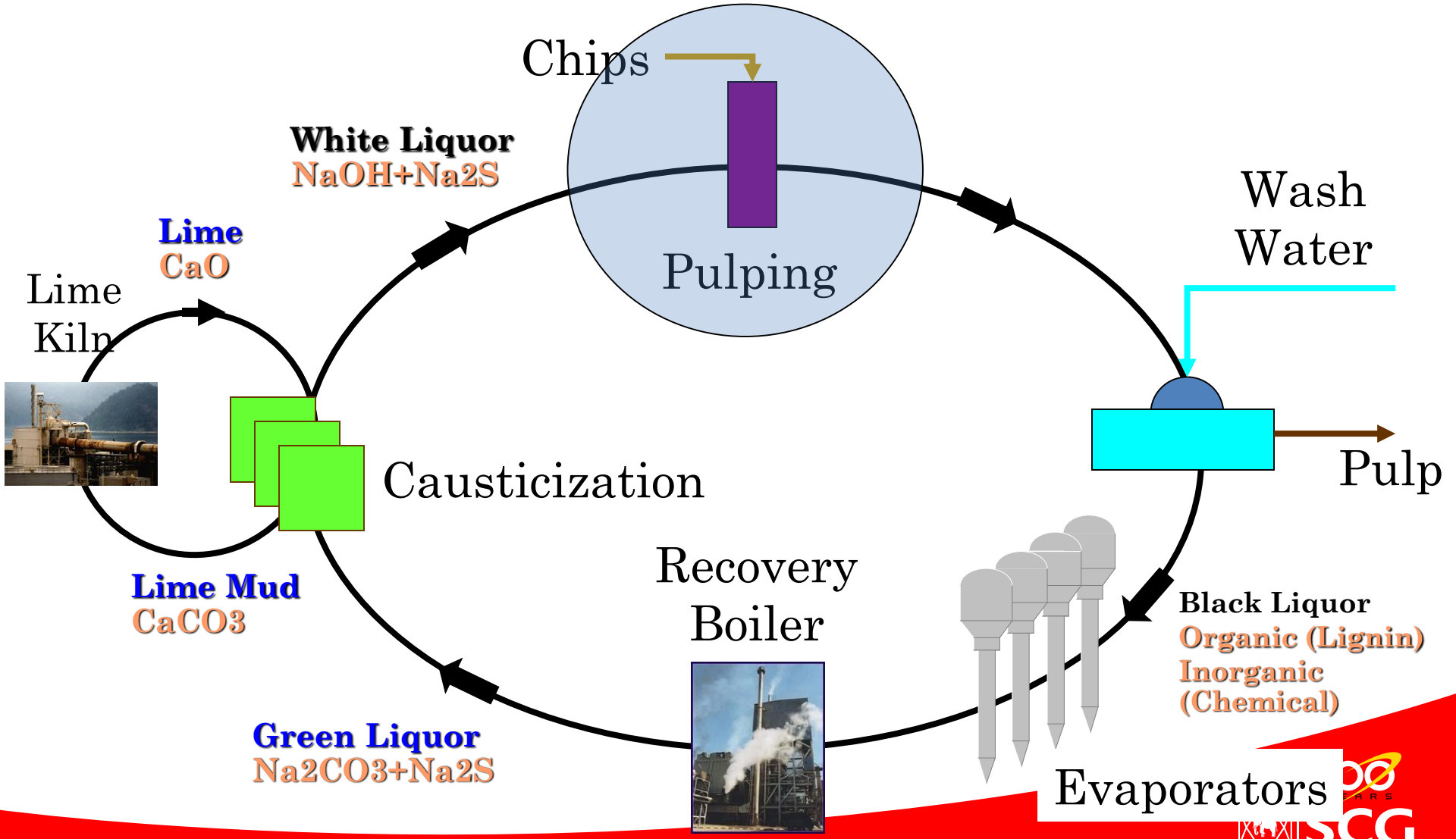
250 adt/d, 65,000ADT/Y, Bleached Eucalypts pulp 85±1%ISO



Credit: Weerachai Seachao

Chemical pulping

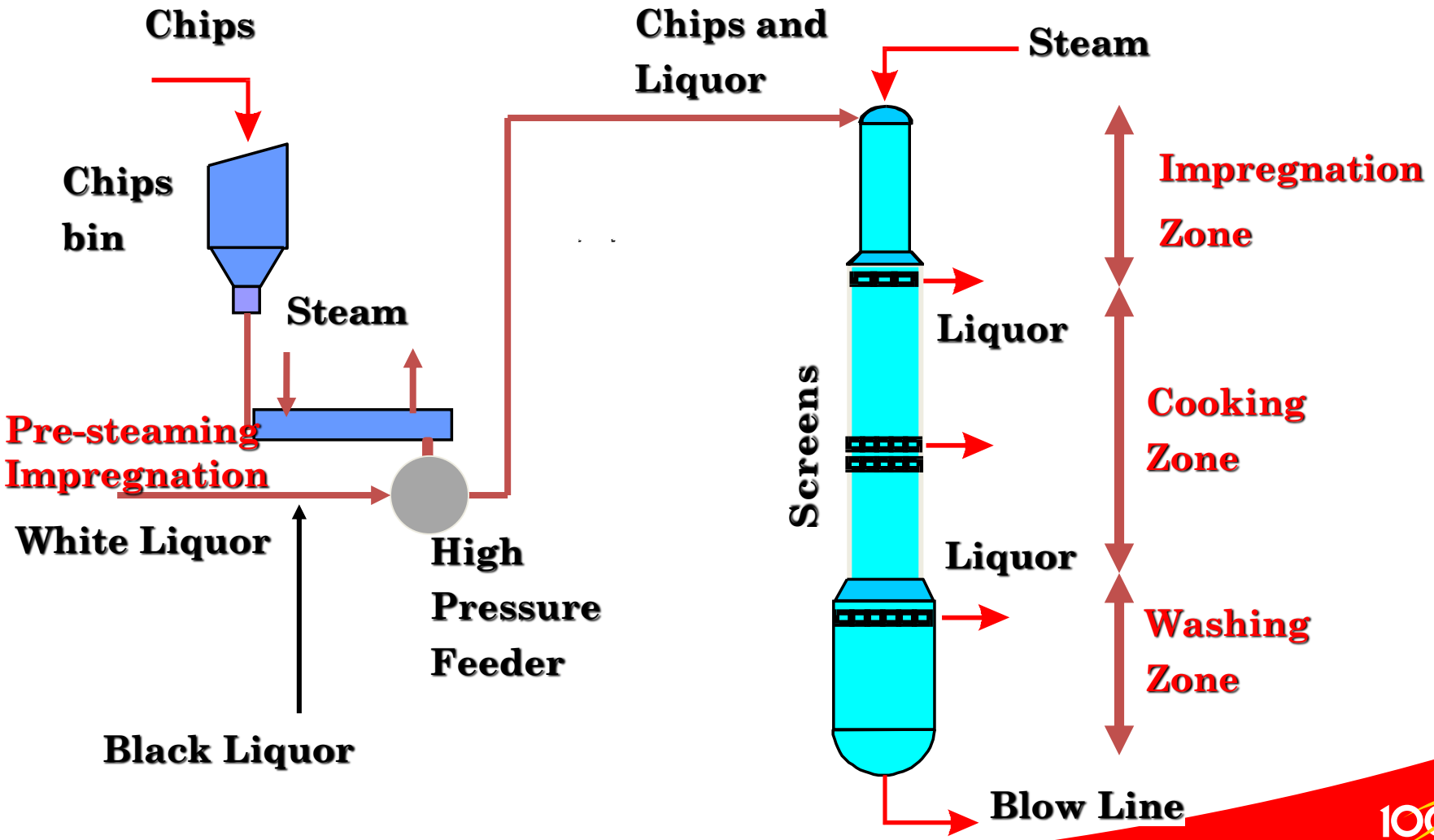
Simplify process diagram of kraft process



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Continuous digester

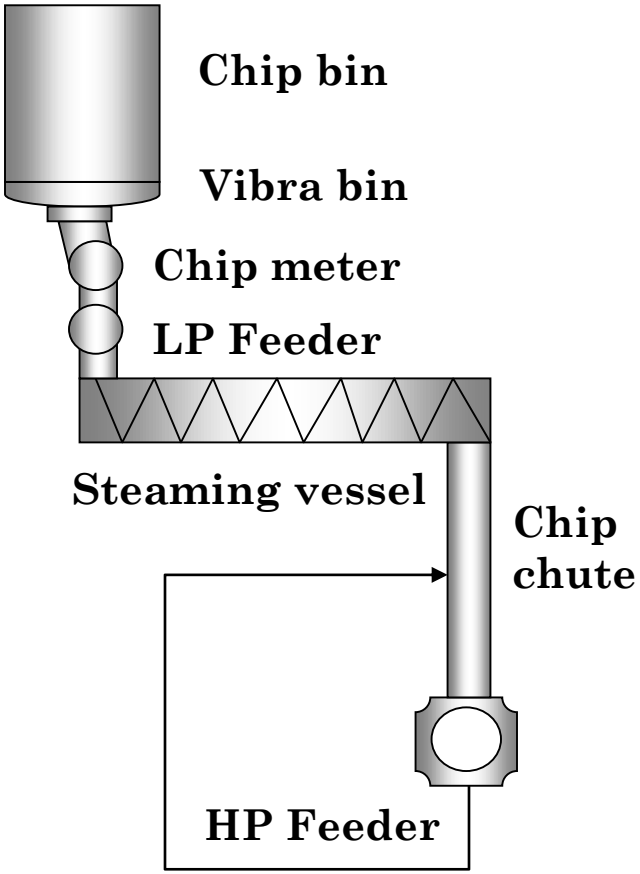
Unit operation and machinery



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Continuous digester

Chip pre-steaming and feed system

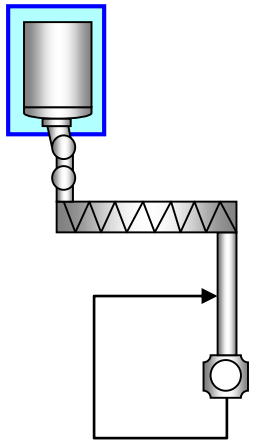


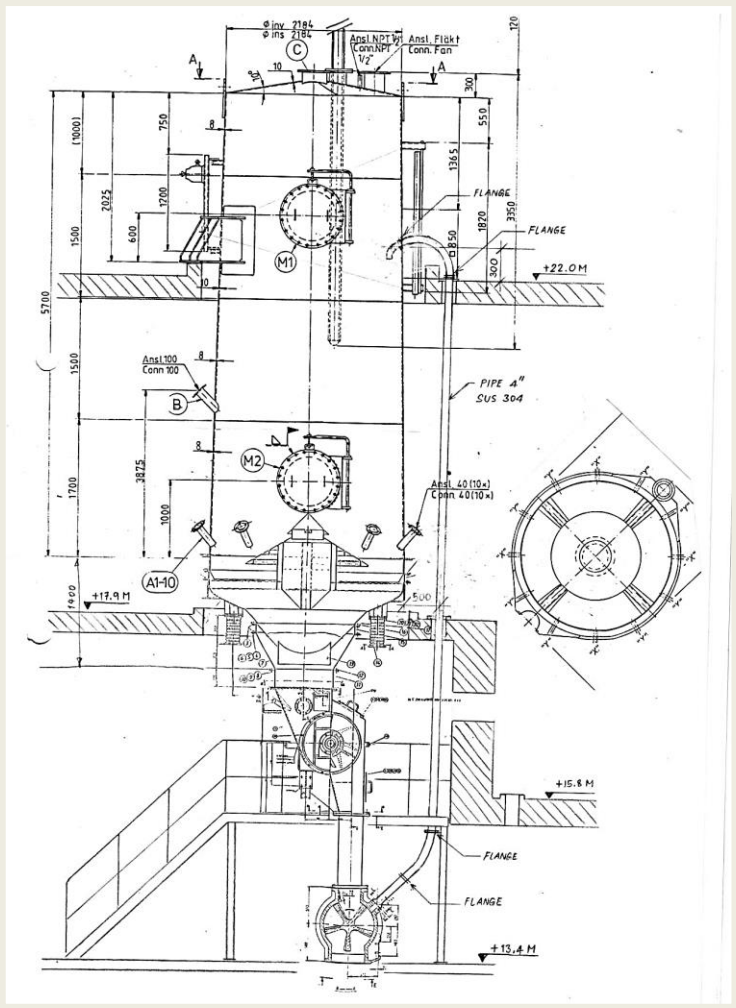
1. Chip bin (Vibra bin) (C23)
2. Chip meter (C24)
3. Low pressure feeder (LP feeder) (C25)
4. Steaming vessel (C26)
5. Chip chute (C29)
6. High pressure feeder (HP feeder) (C30)
7. Chip chute circulation pump (C3)
8. Sand separator (C31)
9. Inline drainer (C32)
10. Level tank (C34)
11. Top circulation pump (C4)

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Continuous digester

Chip bin VS Vibra bin





Vibra bin : **ตำแหน่งและหน้าที่**

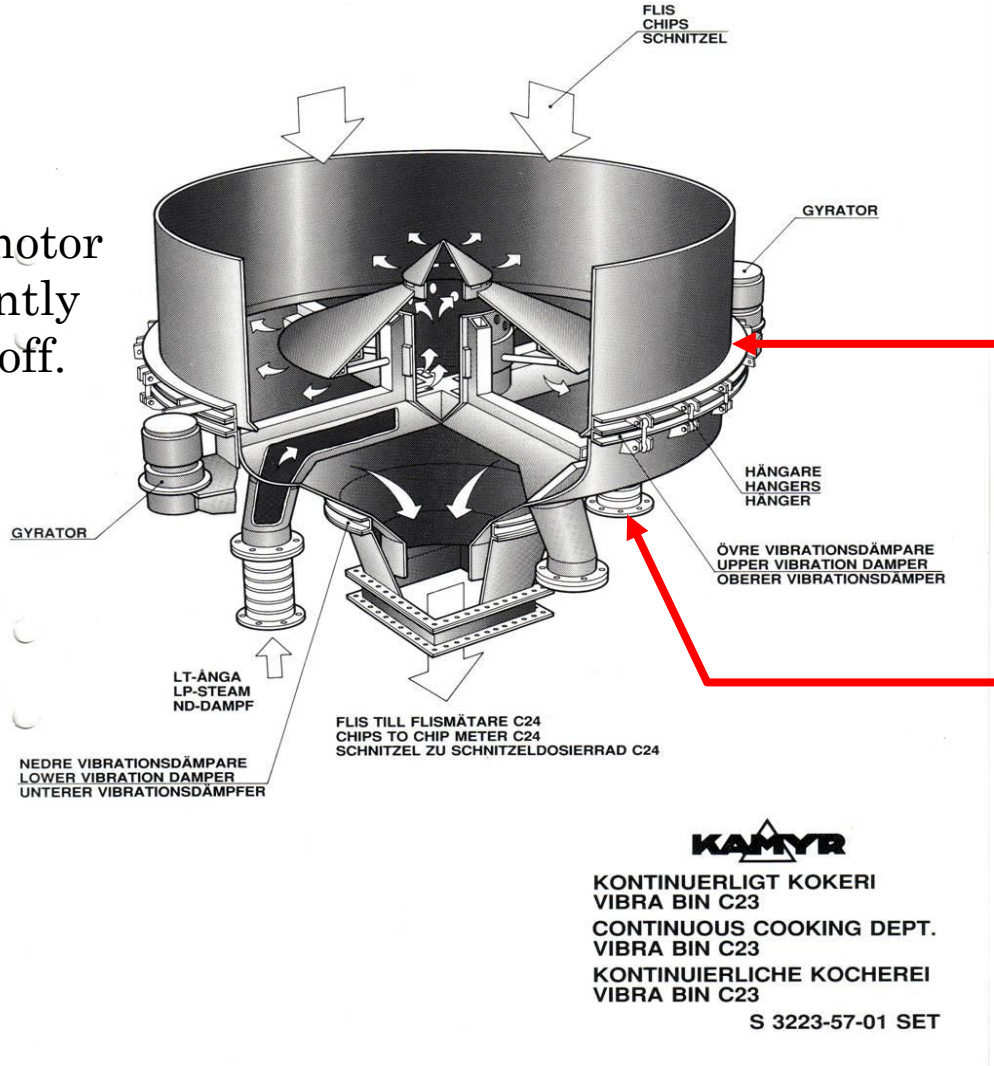
Vibra bin อยู่ระหว่าง Chip bin และตัวควบคุมปริมาณการป้อนของชิ้นไม้สับ (Chip meter) มีหน้าที่ป้อนชิ้นไม้ออก Chip bin ลงสู่ Chip Meter โดยอาศัยการสั่น จากMotor Vibration ขนาด 5 แรงม้า เพื่อให้ชิ้นไม้สับไหลออกจาก chip bin

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Continuous digester

Chip bin VS Vibra bin

- Motor No.1 runs continuously while motor no.2 runs intermittently with 20s on and 20s off.

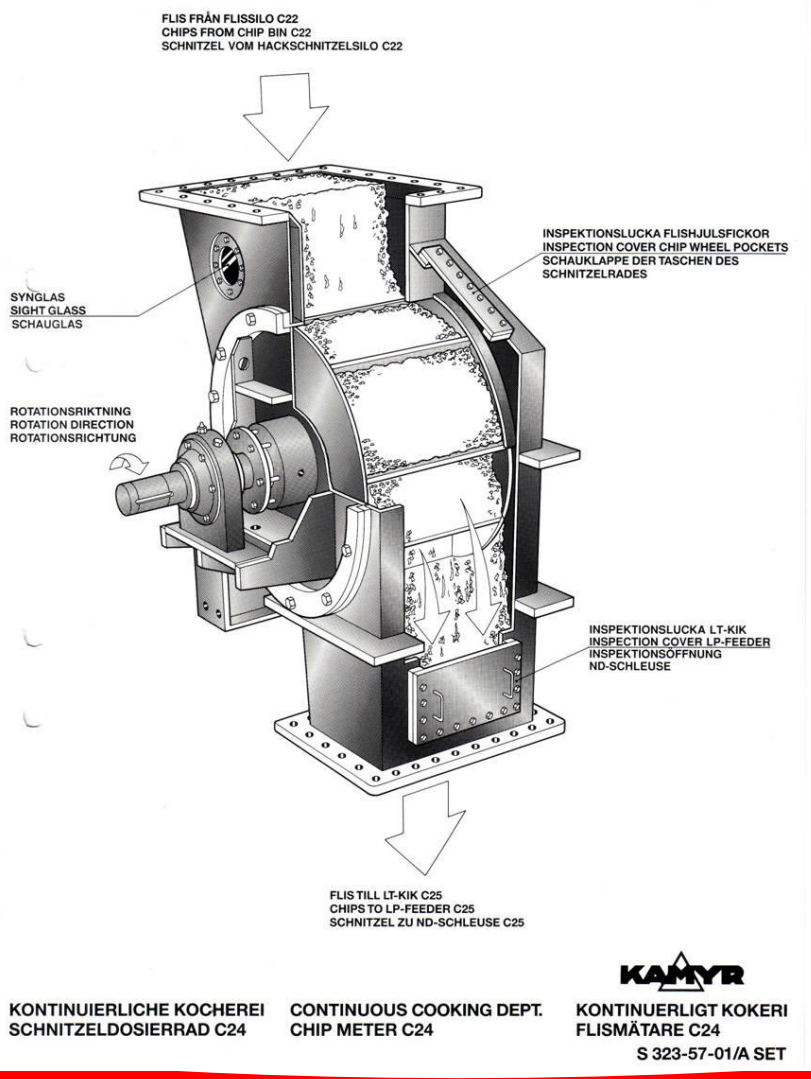


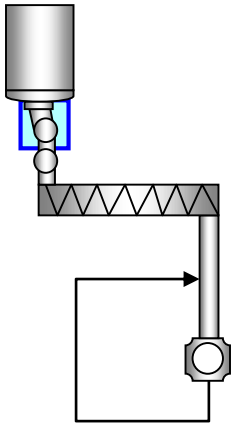
LP Steam

Flash steam from flash cyclone no.2

Continuous digester

Chip meter (C24)





- **The rotor in chip meter** is equipped with 7 pockets, which feed a pre-determined quantity of chip forward per revolution. The rotor rotates clockwise.

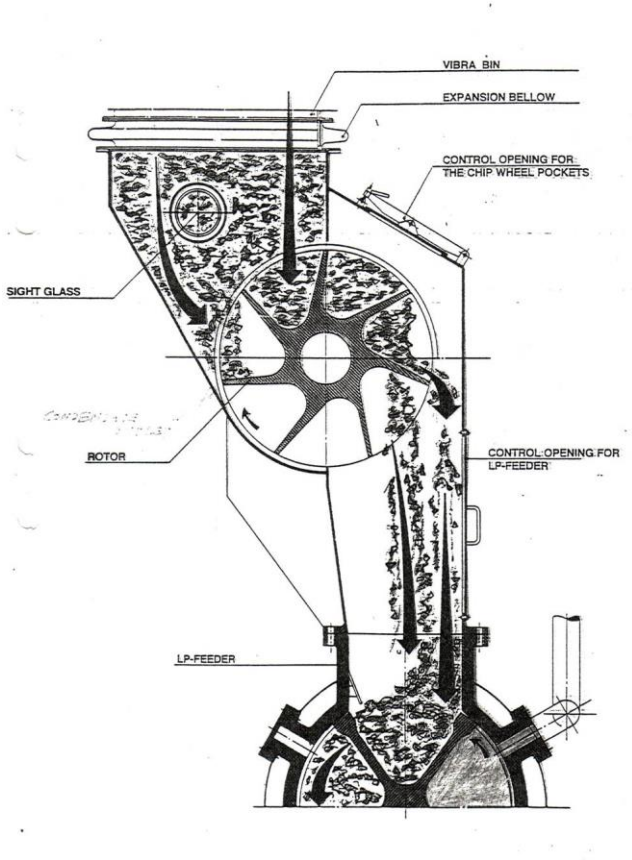
- ***Chip meter*** *is equipped With 2 sight glasses to check that the level is correct and steady.*

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Continuous digester

Chip meter (C24)

Chip meter ติดตั้งอยู่ระหว่าง Chip bin และ Low pressure Feeder มีหน้าที่ในการควบคุมปริมาณการป้อนชิ้นไม้เข้าสู่ cooking plant ซึ่งก็หมายถึงกำลังการผลิต (capacity) จะถูกกำหนดการโดยตัว chip meter นี้



โครงสร้างของ chip meter

ตัวป้อนนี้ประกอบด้วยตัว housing และเพลาหมุนเป็นลักษณะวงล้อ (Wheel) ซึ่งประกอบด้วยช่องป้อน (pocket) ซึ่งปริมาตรแน่นอน วงล้อหมุนนี้สามารถปรับรอบการหมุนได้ตั้งแต่ 7 ถึง 21 rpm ซึ่งจะเป็นตัวควบคุมปริมาณ chip ที่ป้อนเข้า Digester

การทำงาน

ประสิทธิภาพทางด้านปริมาตร หรือความจุของแต่ละ packet นี้จะขึ้นอยู่กับ*ความเร็วรอบ ของ wheel และคุณภาพของ chip*



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Continuous digester

Chip meter (C24)

Production calculation

Total volume of chip meter pocket = 0.145 m3

So,

	1 wheel	= 0.145	m3
If,	chip bulk density	= b	t/m3

So,

	1 wheel	= 0.145 x b	ton (BDT)
If,	chip meter run	= n	rpm (round/min)

So,

Total BDT CHIP to Digester = 0.145 x b x n x 60 BDT/hr

Example : Speed chip meter 15.7 rpm, chip bulk density 170kg/m3

Solution

Chip flow to digester = 0.145 x 15.7 x 60

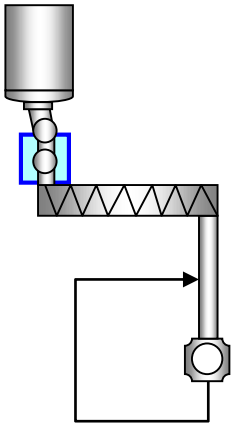
= 136.58 m3/hr

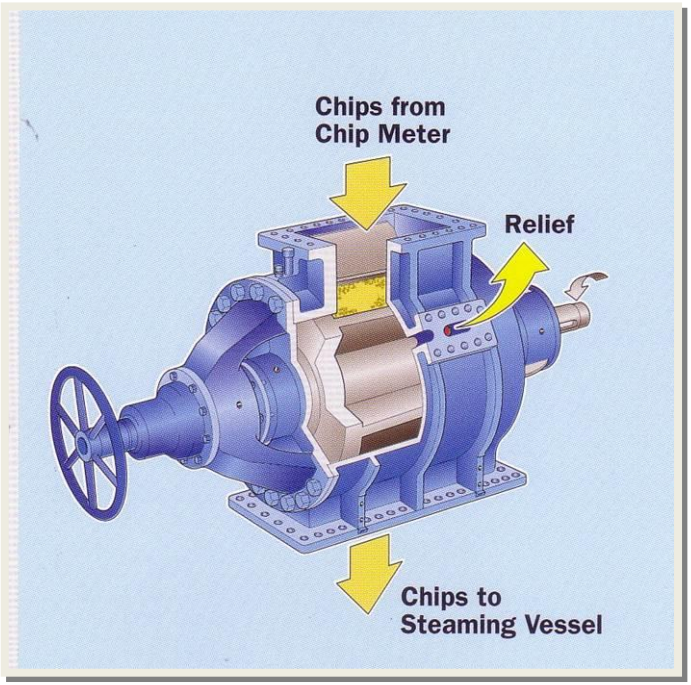
= 23.22 bdt/hr

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Continuous digester

Low Pressure feeder (C25)



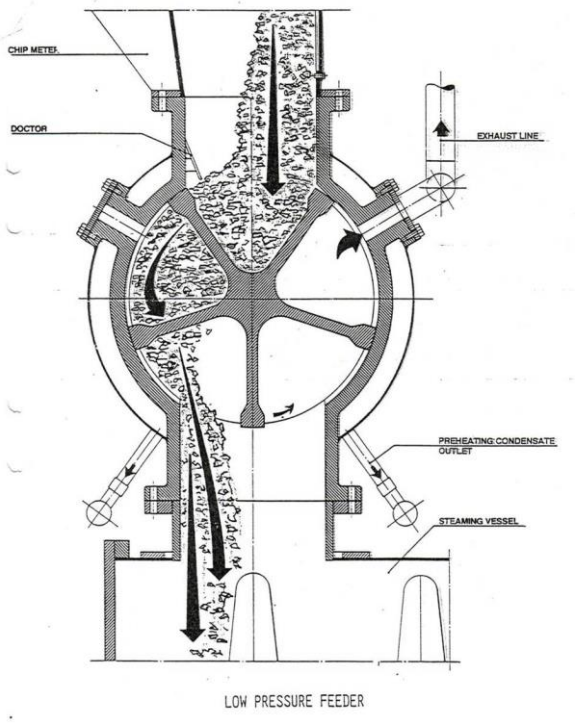


LP feeder ติดตั้งอยู่ระหว่าง chip meter และ Steaming vessel ทำหน้าที่เป็นตัวกั้นระหว่าง pressure ที่เป็นความดันบรรยากาศใน Chip Meter และความดันซึ่งสูงกว่าความดันบรรยากาศ (แต่ยังเป็น low pressure) ในระบบของ cooking plant

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Continuous digester

Low Pressure feeder (C25)



LOW PRESSURE FEEDER

โครงสร้าง

LP-feeder ประกอบด้วยตัว housing และตัวหมุน rotor มีการป้อน Chip เข้าทางด้านบนและออกทางด้านล่าง และจะมีช่องเปิดเพื่อระบายความดันภายใน pocket (decompression connection) ที่ค้างมาจาก Steaming vessel ตัว doctor (แผ่นรองรับการไหลของ chip) จะติดไว้ที่ทางท่อ inlet เพื่อให้ chip ไหลสู่ pocket ได้สะดวกขึ้น

ตัว rotor จะประกอบด้วย pocket 5 ช่อง แยกกันโดยมีผนังกั้นแนวยาว ***จุดที่จำเป็นอย่างยิ่งคือ ระยะห่าง clearance ระหว่าง rotor และ housing ควรมีระยะห่างน้อยที่สุด เพื่อลดอัตราการรั่วไหลของความดันจากภายใน Steaming Vessel*** ด้วยเหตุนี้ตัว housing และ rotor จะถูกออกแบบให้เป็นลักษณะของ taper ตามแนวแกน เพื่อที่จะสามารถปรับระยะห่างภายในนี้ได้โดยการปรับตัว rotor ตามแนวแกนได้เมื่อระยะห่างเพิ่มขึ้น เนื่องจากการสึกกร่อนจากการหมุน

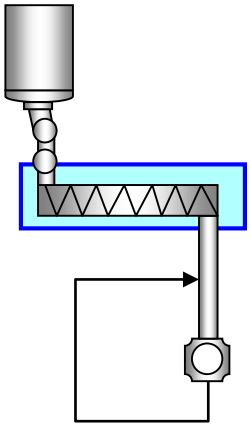
การทำงาน

chip จะตกลงจาก chip meter เข้าทางด้านบนของ LP-feeder ลงสู่ Pocket ซึ่ง rotor จะหมุนด้วยความเร็วรอบประมาณ 22 rpm. เมื่อตัว rotor หมุนไปครึ่งรอบ Chipใน pocket นั้น ก็จะตกลงสู่ทางเข้า Steaming vessel ส่วน pocket ตัวที่ว่างที่ส่ง Chip เข้า steaming vessel แล้วจะมีความดันสูงอยู่ เมื่อหมุนกลับมาถึงช่องระบายความดัน (decompression line) ความดันก็จะถูกระบายสู่ Chip bin เพื่อลดความดันเป็นความดันบรรยากาศ

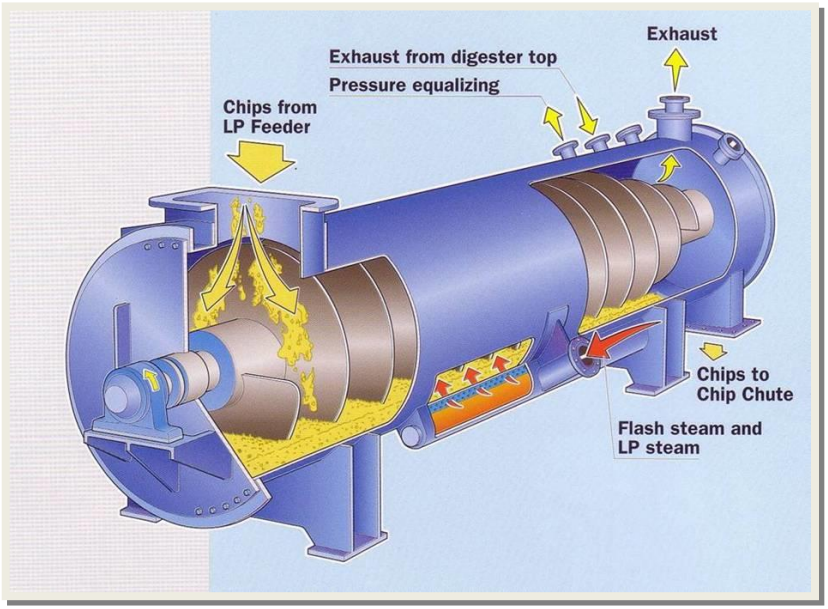
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Continuous digester

Steaming vessel (C26)



Steaming vessel ติดตั้งอยู่ระหว่าง LP-feeder และ HP-feeder chip จะถูกให้ความร้อนจากไอน้ำให้อุณหภูมิสูงขึ้น เพื่อไล่อากาศภายในชิ้นไม้ออก ซึ่งในขั้นตอนนี้เป็นขั้นตอนที่เตรียมเพื่อให้น้ำยาต้มเยื่อ (Liquor) ซึมเข้าภายใน เนื้อไม้ง่ายขึ้นในช่วงของการ impregnation



โครงสร้าง

ตัว Steaming vessel ประกอบด้วย body ซึ่งเป็นลักษณะทรงกระบอกแนวนอนภายในมี screw conveyor ซึ่งถูกยึดด้วย bearing เป็นตัวนำพา chip ที่ผ่านเข้าทางด้านบนที่ปลายข้างหนึ่งของ vessel

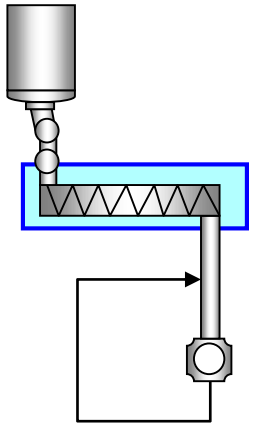
การทำงาน

chip จาก LP-feeder จะถูกป้อนผ่านโดยตัว screw conveyor โดยที่ chip จะอยู่ภายในตัว Steaming vessel เป็นเวลาประมาณ 1.5-2 นาที ไอน้ำที่ป้อนเข้าจะเข้าทางด้านล่างผ่านตะแกรงกระจายไอน้ำเพื่อให้อัตราการถ่ายเทความร้อนดีที่สุด รอบการหมุนของ screw จะ set ไว้ประมาณ 6 หรือ 8 rpm. ขึ้นอยู่กับปริมาณที่ต้องการผลิต

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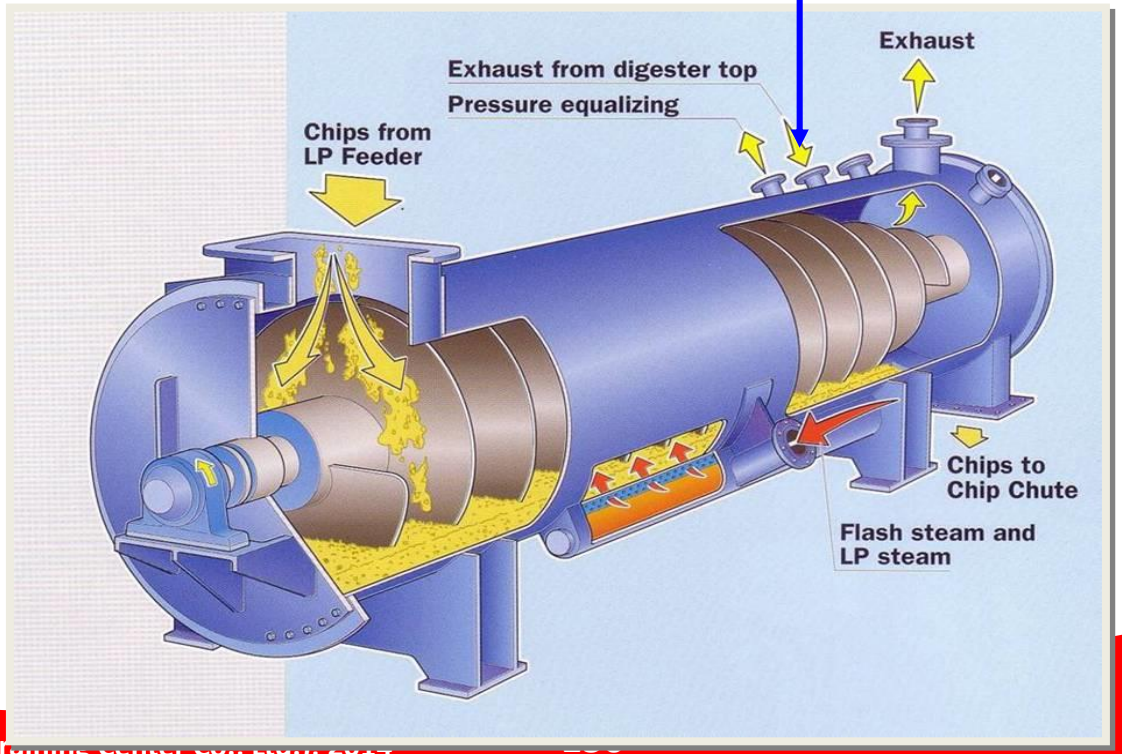
Continuous digester

Steaming vessel (C26)



Flash steam from flash cyclone no.1

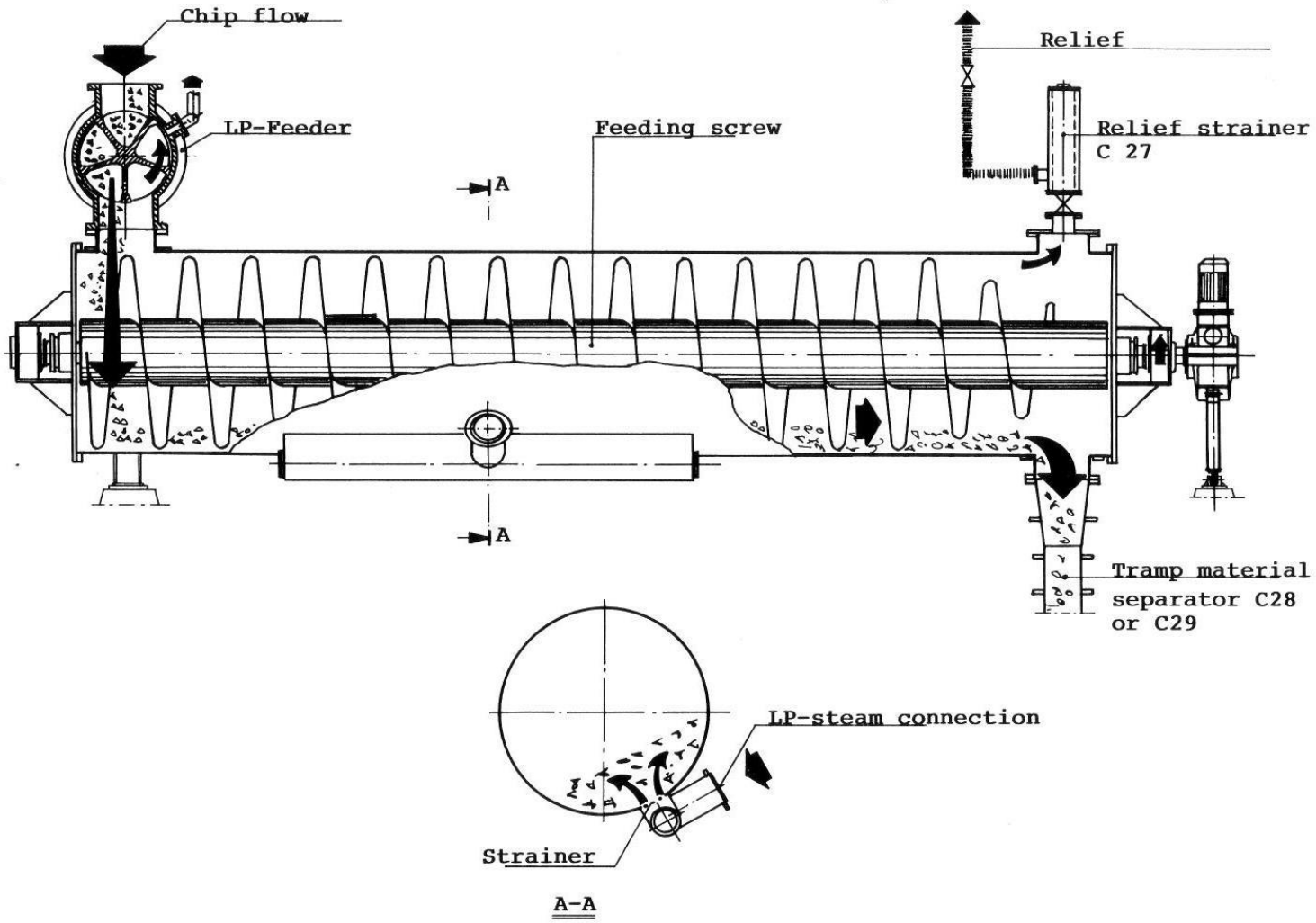
- **The steaming vessel** is equipped with a conveying screw which transports the chips through the vessel. This takes about 2 minutes. During this period the chips are steamed at a *pressure of 1.0-1.3 bar.*



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Continuous digester

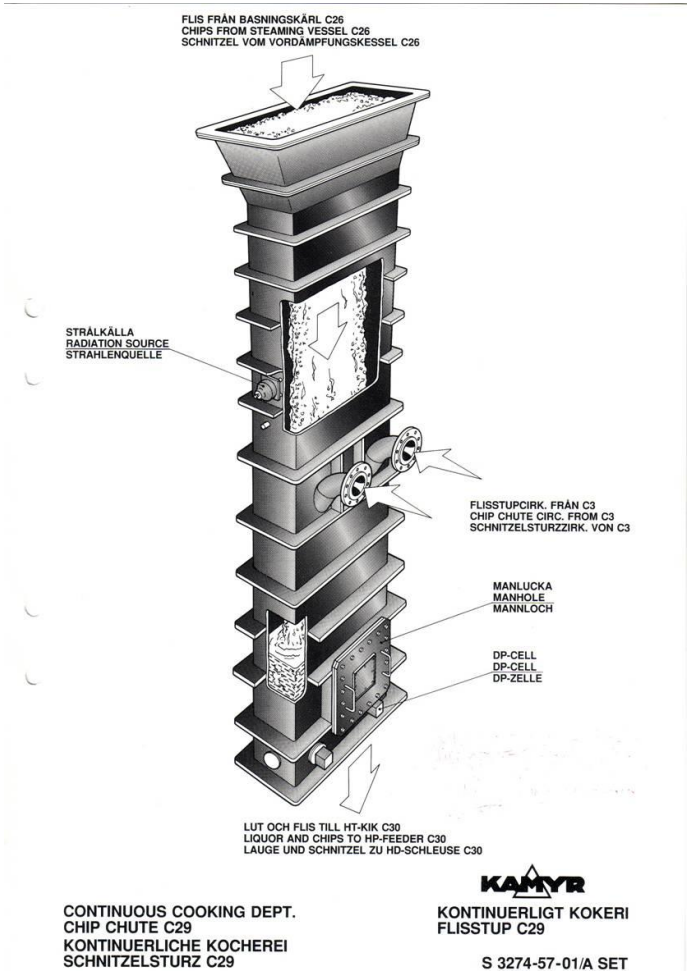
Steaming vessel (C26)

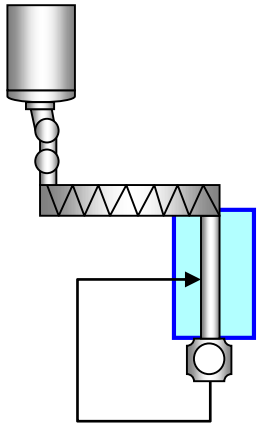


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Continuous digester

Chip chute (C29)





- The chips from steaming vessel fall into ***chip chute***. It is important that the liquor level in the chip chute is correct. Too low level may cause shocks in the top circulation piping and too high level may cause a decrease in HP-feeder efficiency.

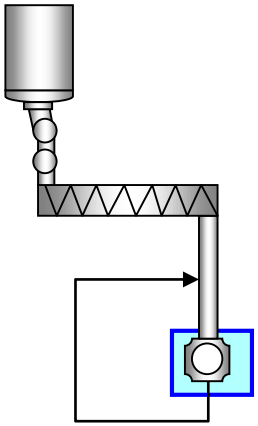
- If the level in the *chip chute is too high* alarm The screw in steaming vessel stops and chip feeding also stops.

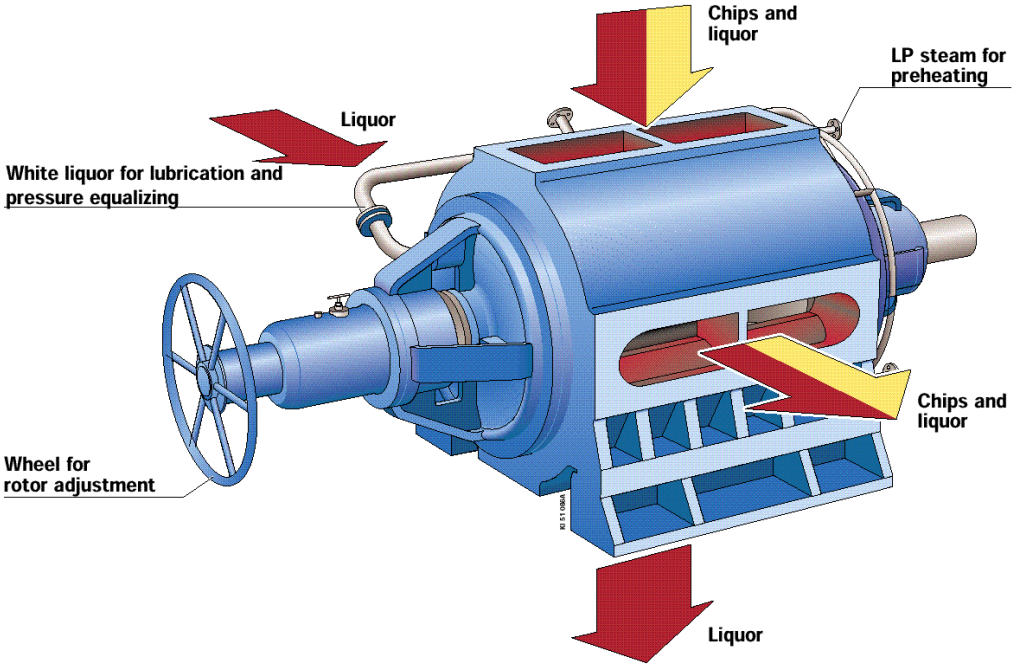
- If the *level is too low* HP-feeder stops.

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Continuous digester

High Pressure feeder (C30)





- **The HP feeder** is equipped has a rotor with 4 through going pockets, each of which feeds forwards twice per revolution. The wells are places around the rotor with an angle of 45° between the openings.

- *The HP feeder must never be run dry* and is therefore interlocking to stop if the level in the chip chute is low.

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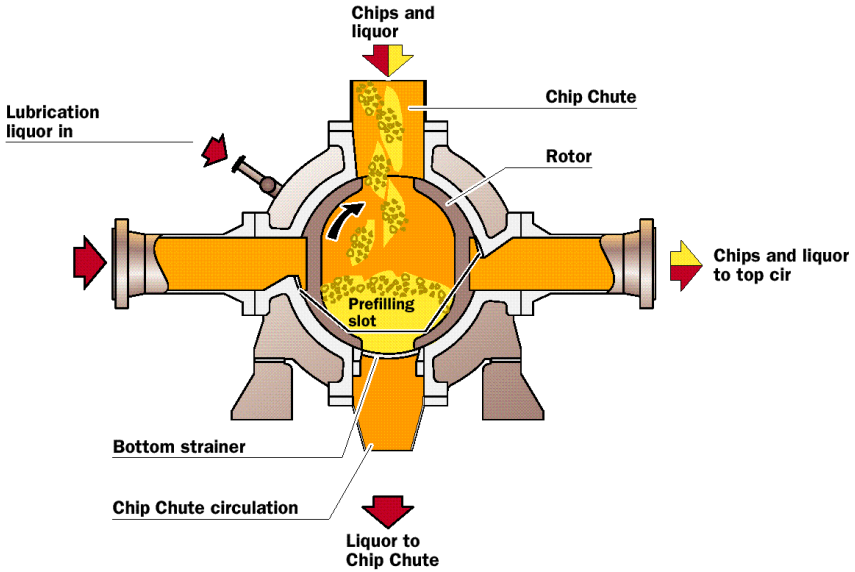
Continuous digester

High Pressure feeder (C30)

เมื่อ chip ผ่านตัว HP-feeder นี้ ระบบจะถูกเปลี่ยนจากระบบความดันต่ำไปสู่ความดันสูงซึ่งใกล้เคียงกับความดันใน digester น้ำยาต้มเยื่อที่ไหลเวียนใน chip chute (chip chute circulation liquor) จะถูกแยกออกจาก chip จะปั้มให้ไหลเวียนกลับไปยัง Chip chute ในขณะเดียวกัน chip ก็จะถูกน้ำยาต้มเยื่อความดันสูง (Liquor) ใน Top Circulation ผลักดันให้ขึ้นไปสู่ยอดหอ digester

โครงสร้าง

HP-feeder ใช้หลักการของการหมุนของตัว plug valve ซึ่งประกอบด้วย Packet 2 ชุด แต่ละชุดจะวางในแนวตั้งฉากซึ่งกันและกัน โดยออกแบบให้ pocket ด้านหนึ่ง (แนวตั้ง)อยู่ตรงกับแนวท่อด้าน Low pressure (chip chute circulation) ในขณะที่ pocket อีกด้านหนึ่ง (แนวนอน) จะอยู่ตรงกับแนวท่อด้าน High pressure (Top circulation) pocket 2 ชุดนี้ถูกออกแบบให้วางทำมุม 45° ต่อกัน เมื่อปากทางเข้าออกชุดหนึ่งเริ่มปิด อีกชุดหนึ่งจะเริ่มเปิดขณะที่หมุนไป ทำให้ช่องเปิดของแต่ละ Line เปิด 100 % ทุกขณะที่ rotor หมุนไป *เราสามารถปรับระยะ clearance ระหว่าง rotor และ housing ได้เช่นเดียวกับ LP-feeder เนื่องจากตัวโครงสร้างเป็น taper สามารถควบคุมได้ โดยการปรับตัว rotor เข้าไป*

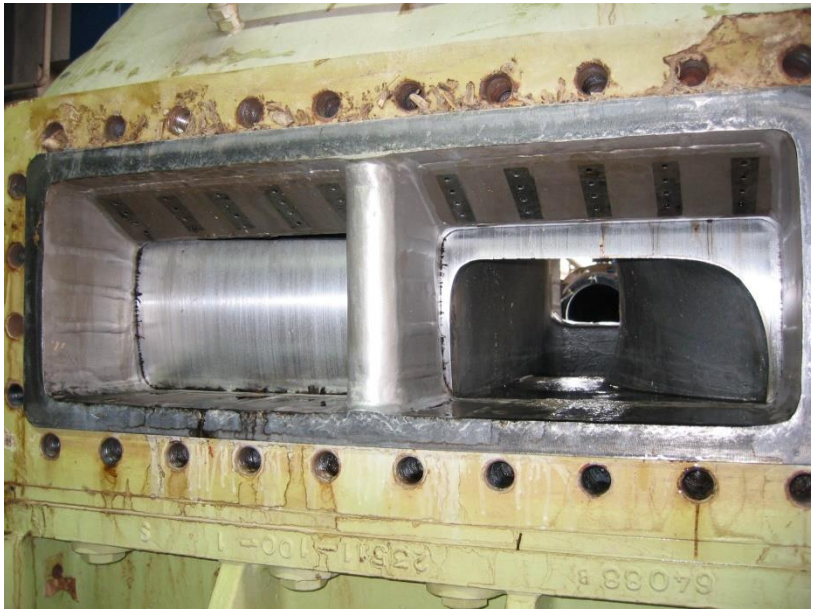


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Continuous digester
High Pressure feeder (C30)

<u>การทำงาน</u>

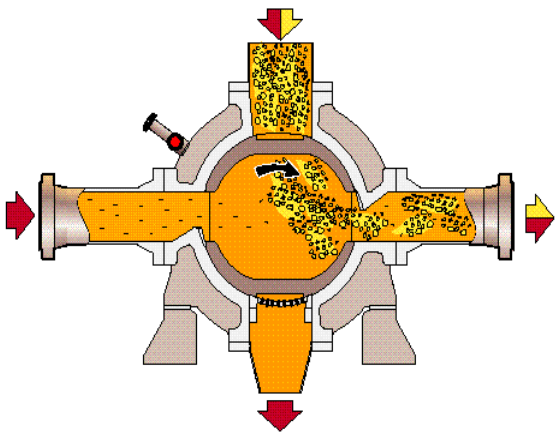
chip ที่อยู่ร่วมกับ Liquor จาก chip chute circulation จะผ่าน chip chute ลงมาเข้า HP-feeder ทางด้านบนและเข้าสู่ pocket ตัวที่อยู่ในตำแหน่งแนวตั้ง Liquor จะผ่านลงสู่ท่อในวงจรหมุนเวียน chip chute circulation ส่วน chip จะยังคงค้างใน pocket โดยไม่ผ่าน screen ที่อยู่ด้านล่างของทางออก เมื่อ pocket นี้หมุนไป 1/4 รอบจนอยู่ในแนวนอนตรงกับ Top circulation Liquor ใน Line นี้ ซึ่งมีความดันสูงจะดัน chip ป้อนขึ้นสู่ Digester



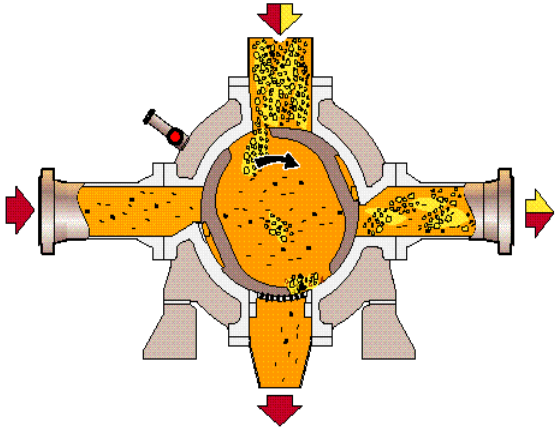
Continuous digester

High Pressure feeder (C30)

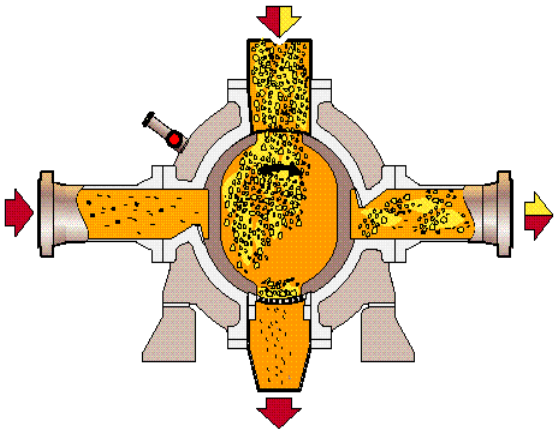
Conventional filling sequence



Chip to Top Circulation



Chip become to Chip chute Circulation

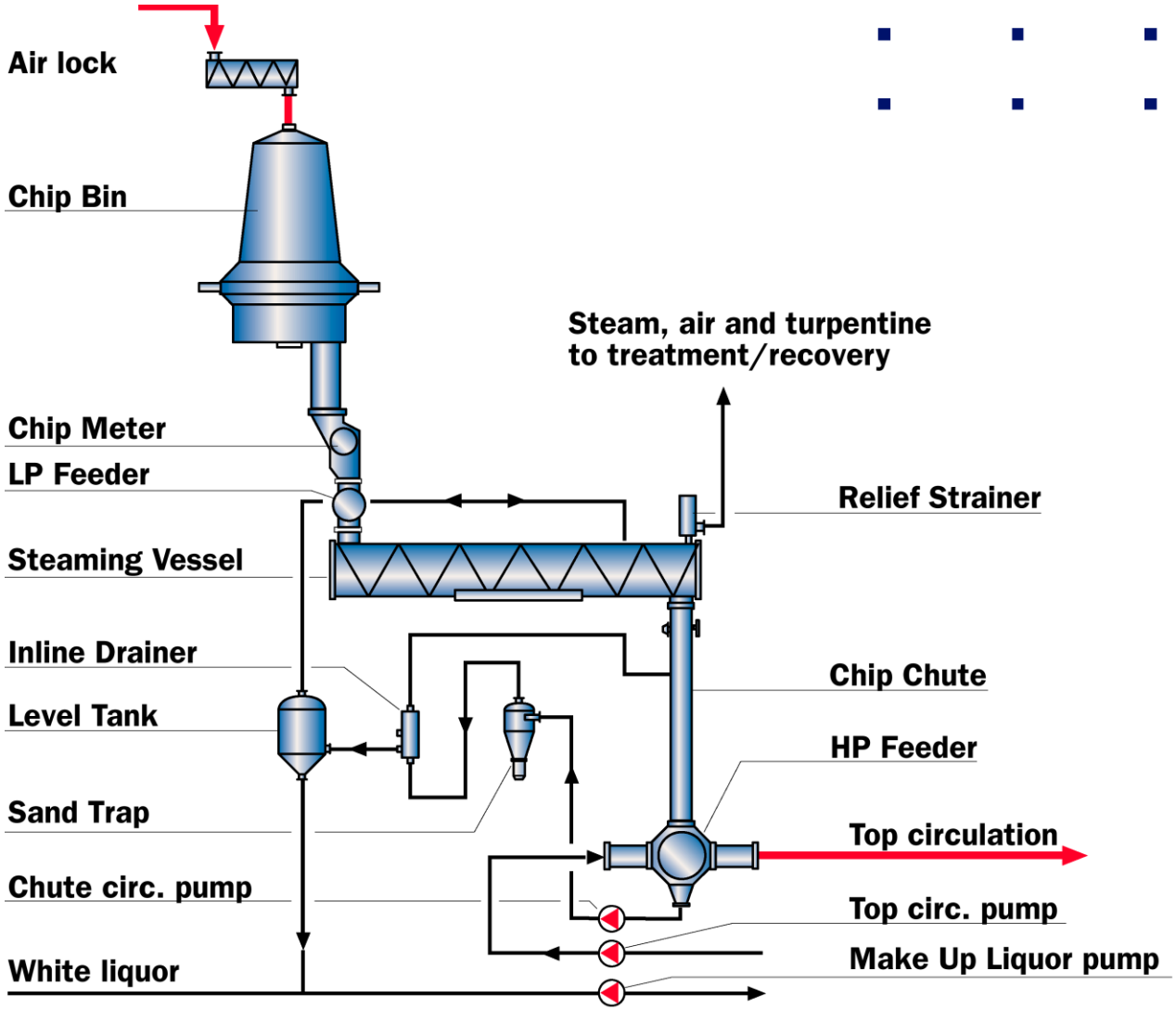


Chip to Chip cute Circulation

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Chip chute circulation (C3)



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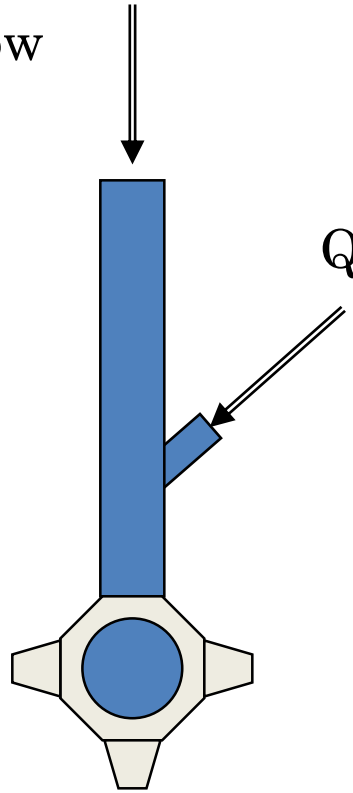
Continuous digester

Chip chute circulation (C3)

Chip chute filling degree

Q_F = Chip Flow

"Capacity"



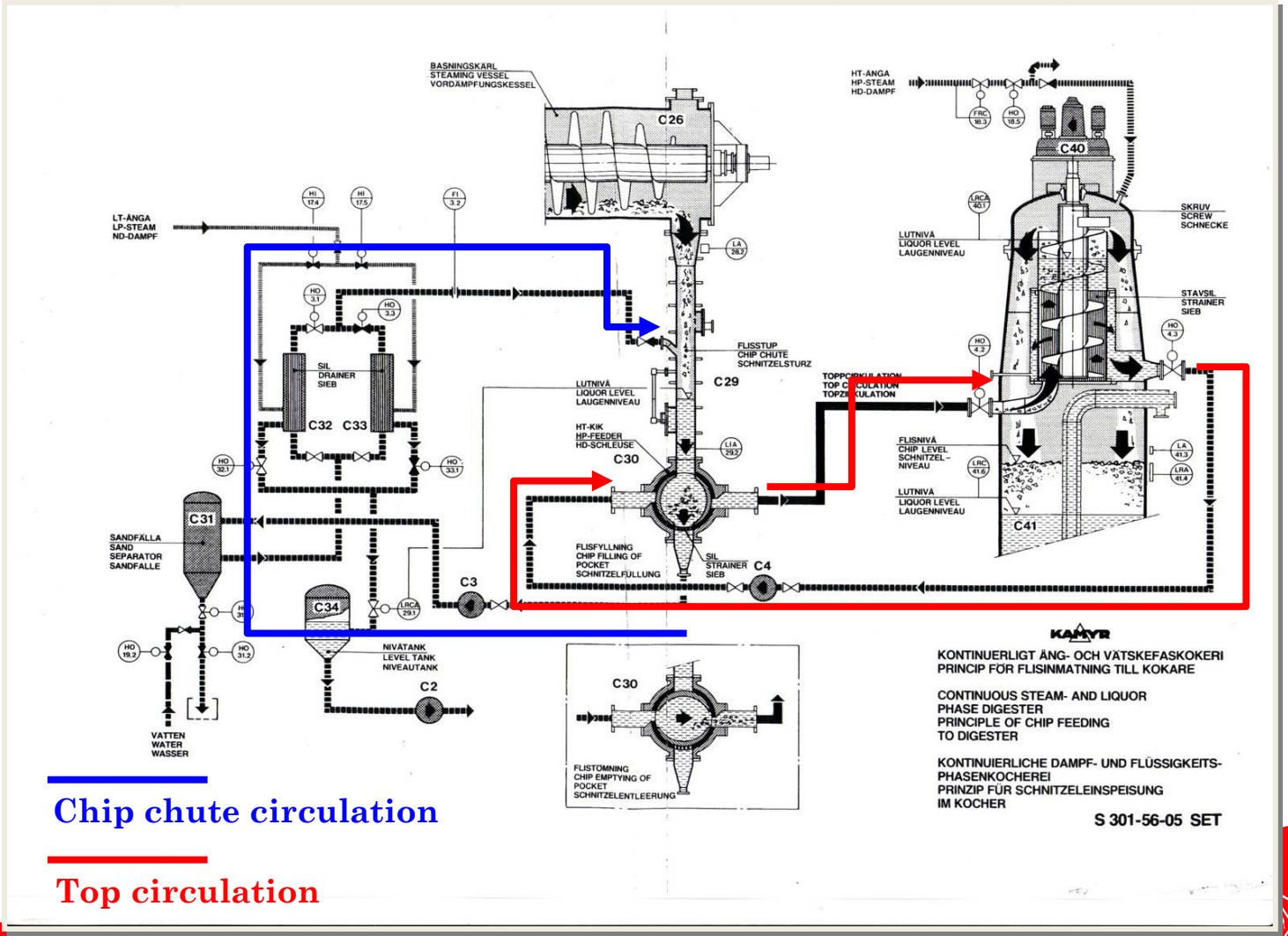
Q = Chip Chute Circulation Return Flow

$$\text{Filling Degree} = \frac{Q_F}{\frac{Q_F}{3} + Q}$$

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Chip chute circulation (C3)



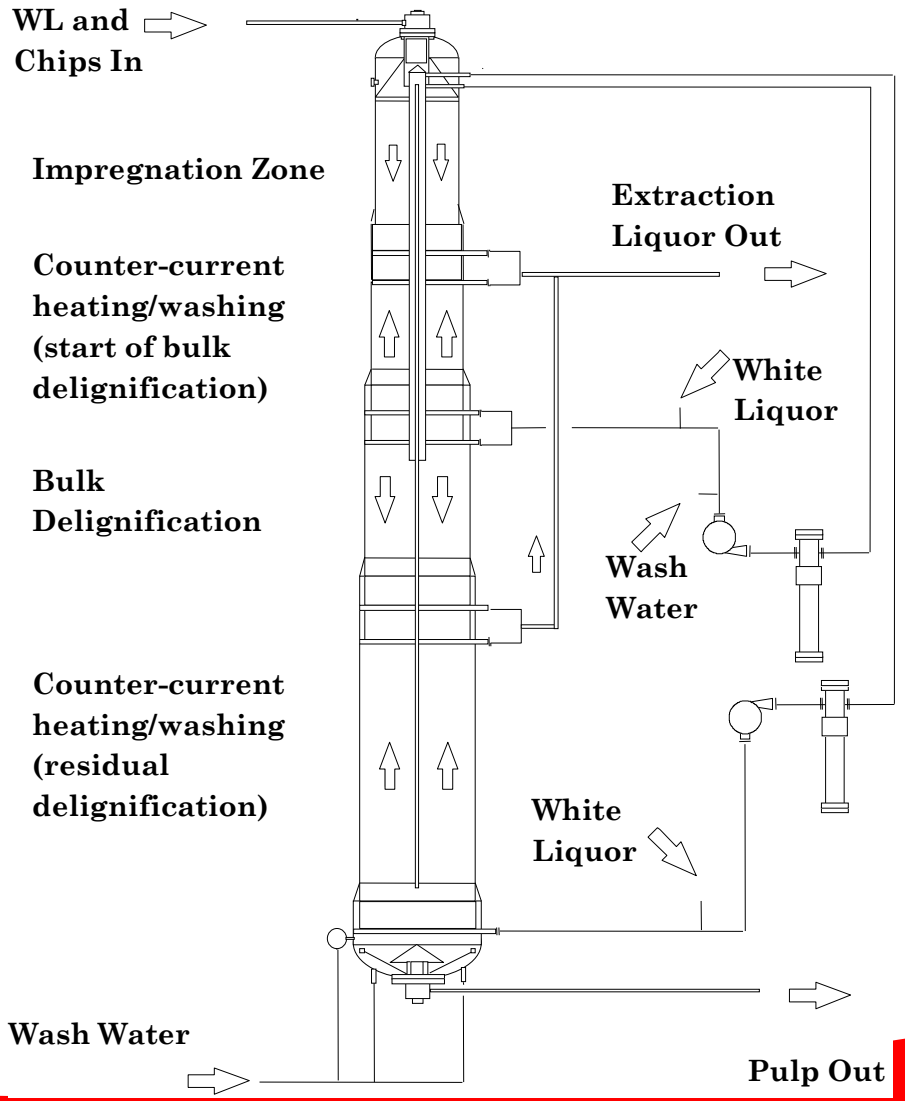
Chip chute circulation

Top circulation

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Cooking part

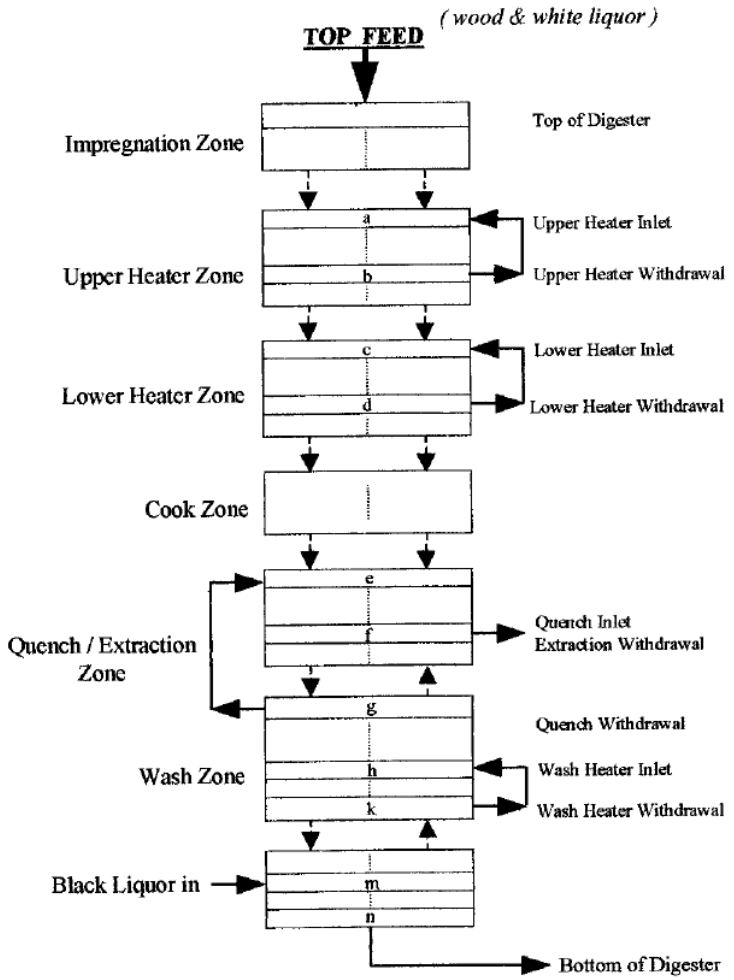


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Cooking part

- Controlled Variables
  - Outlet kappa
  - EA in various sections of digester
  - Chip level
- Manipulated Variables
  - Lower heater temperature
  - White liquor charge
  - Chip flow rate



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Cooking part

1. Top separator (C40)
2. Cooking liquor in Digester
3. Hi-Heat washing
4. The cooking zones and circulation
5. The trim circulation (C6)
6. Liquor extraction
7. The wash circulation (C8)
8. Outlet device (C42)
9. The flash cyclone
10. The condensate system

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Kraft Pulping

Basic concept for lower Kappa No.

1. Alkali concentration (NaOH and Na_2S) should be low in the beginning of cook and stay consistent during it. High alkali concentration in the beginning increases cellulose degradation and consumes alkali unnecessarily.

2. Hydrogen sulfide concentration (HS-, forms when Na2S dissolves and reacts with wood must be as high as possible in the beginning of the cook. Hydrogen sulfide increases lignin dissolval rate and it doesn't degradate cellulose in the beginning of cooking like sodium hydroxide does.

3. Dissolved lignin concentration should be as low as possible, especially in the end of cooking. Dissolved lignin slows delignification reactions. Moreover, it may repecipitate on fiber surfaces, especially when alkali concentration reduces too much.

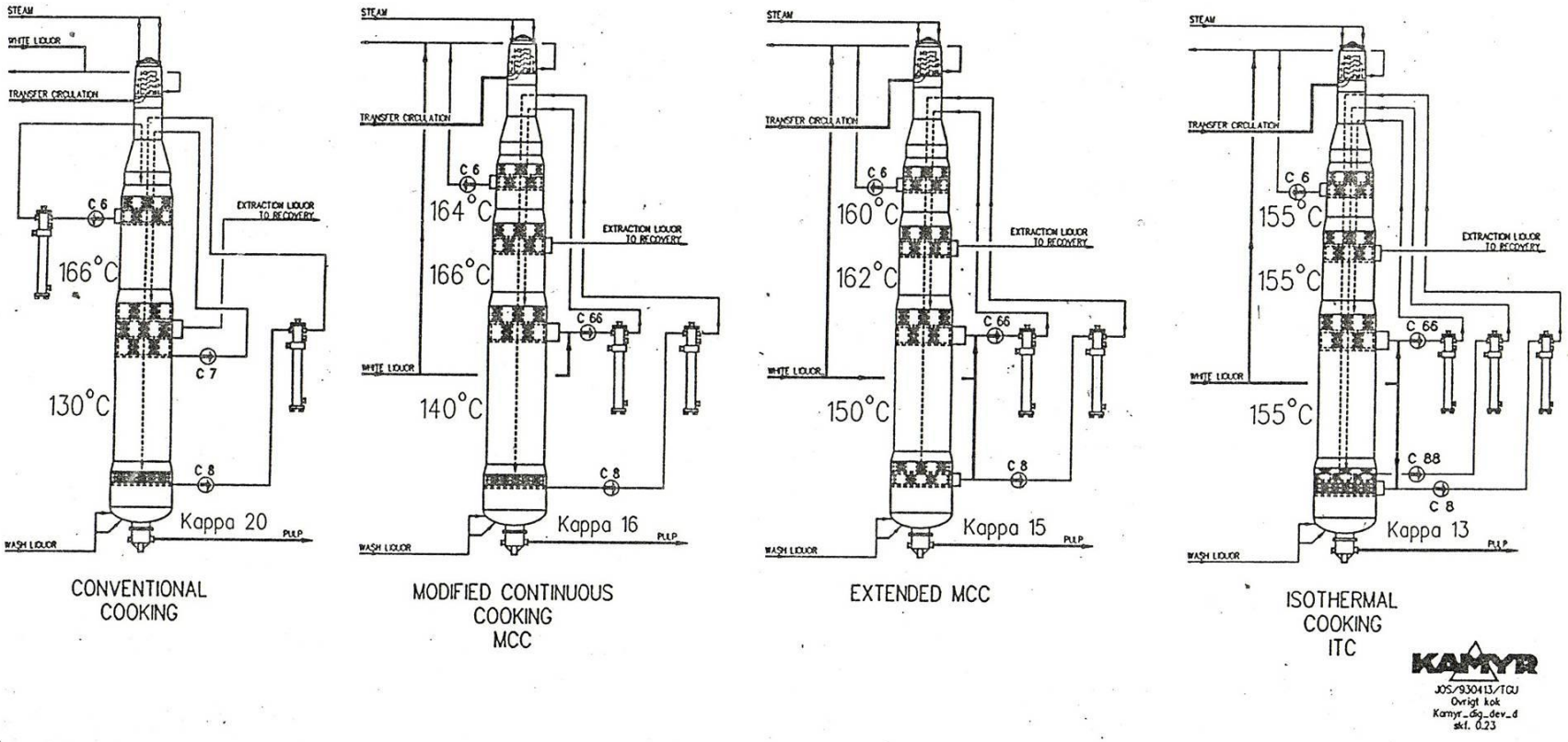
4. Temperature should be low, especially in the beginning of cooking. Low temperature decreases cellulose degradation. This degradation is especially strong in the beginning of cooking.

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Continuous digester

Continuous cooking development

HARDWOOD



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Continuous digester

Alkali profile

Conventional and MCC Cooking

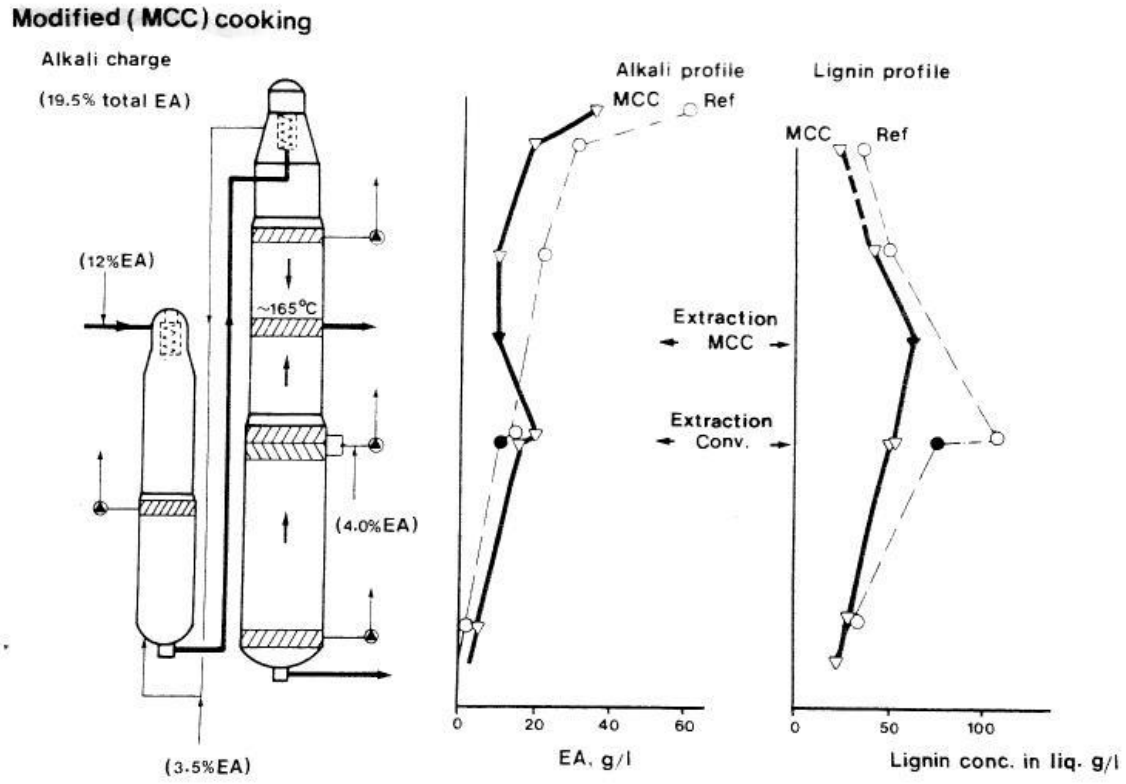


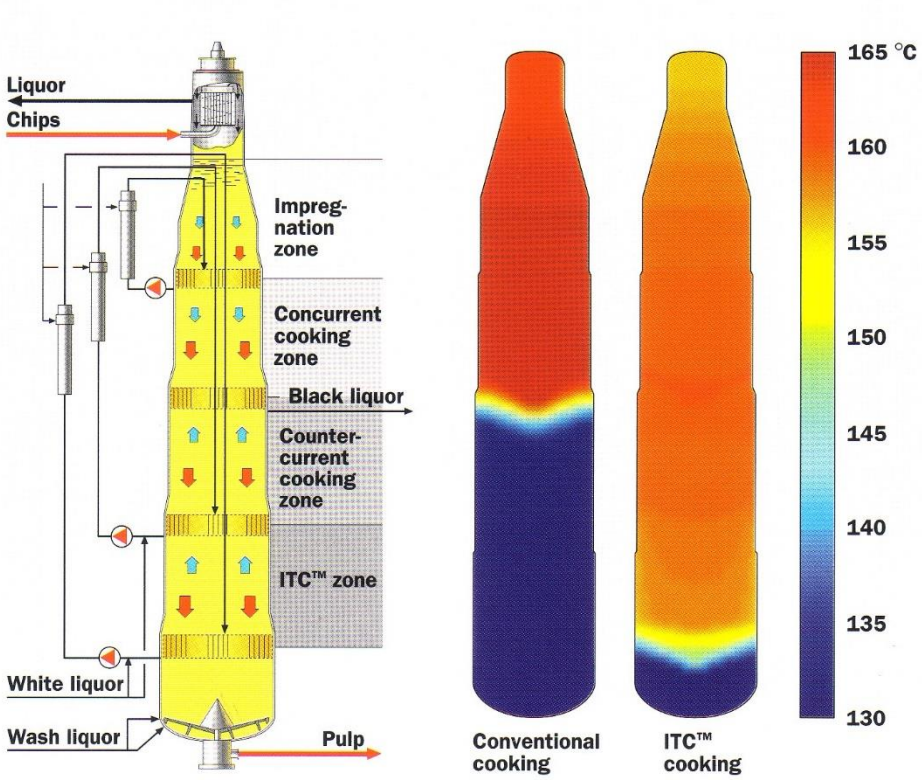
Fig. 19 Alkali and lignin profiles in the MCC-cook compared with a conventional cook

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Continuous digester

Conventional & ITC Cooking

HARDWOOD

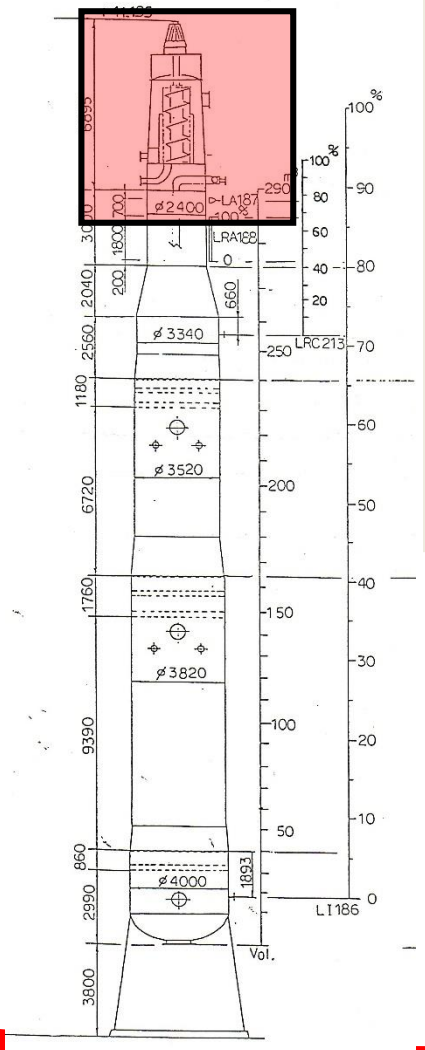


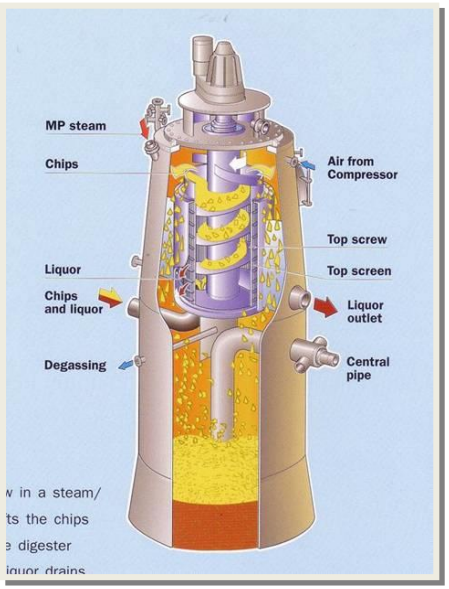
Concurrent cooking in the Upper part of the digester is followed by a *countercurrent zone* below The extraction screen. Cooking combined with displacement washing continues in the *ITC zone*. Additional white liquor is charged and heated in the circulation loops for final selective cooking to a low Kappa numbers.

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Continuous digester

Top Separator (C40)





- The Chips are transported by the top circulation to ***TOP SEPARATOR*** (C40) in the digester. This unit is comprised of a screw which rotates inside a strainer surrounded by a cylindrical inner shell.

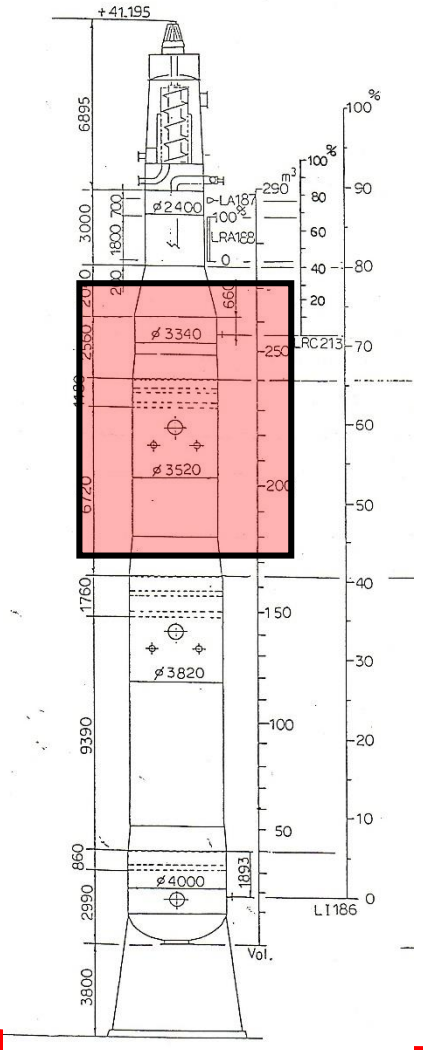
- Between the screw and the strainer there is only a small clearance to keep the strainer free from pin chips and other small projectiles which may plug the screen plate.

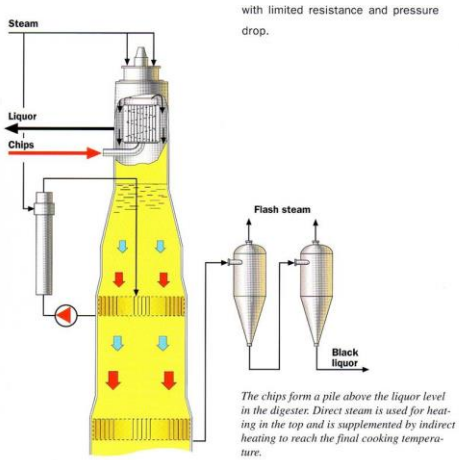
- There is,however, a larger space between the strainer and the outer cylinder where the liquor, which has been separated from the chips, collects prior to being returned to the HP feeder by pump C4

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Cooking zone





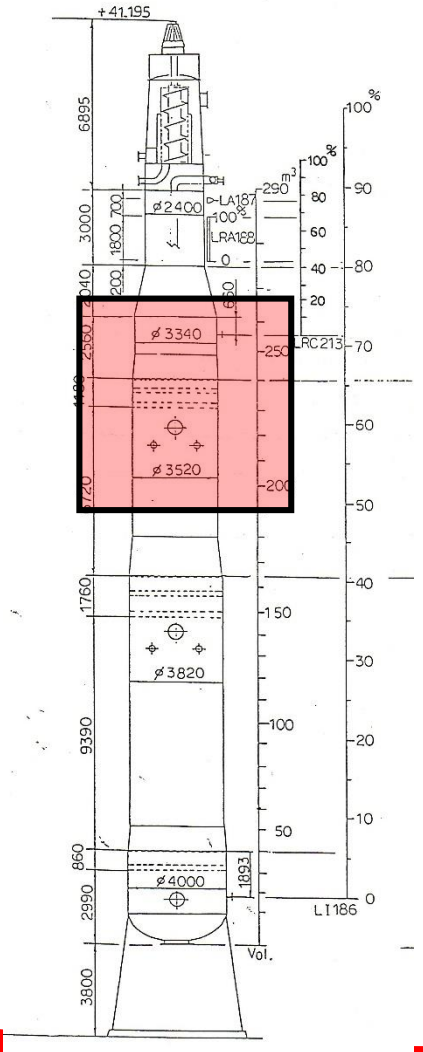
- In a single vessel digester the chips and some of the transport liquor drop into the impregnation zone of the digester vessel. *Gravity moves the column of wood chips downwards.*

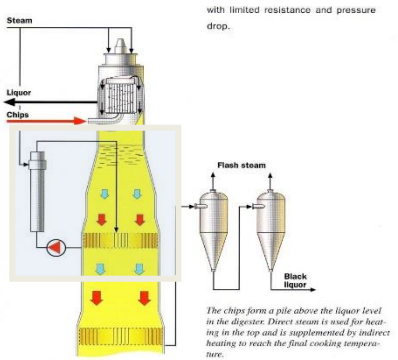
- Circulation of liquor within the digester begins with extraction through a peripheral screen. The liquor is heated externally in a heat exchanger and returned to the digester center through a *central pipe.*

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Continuous digester

Trim circulation (C6)





- **The Purpose of the TRIM circulation** is to adjust the temperature and alkali charge.

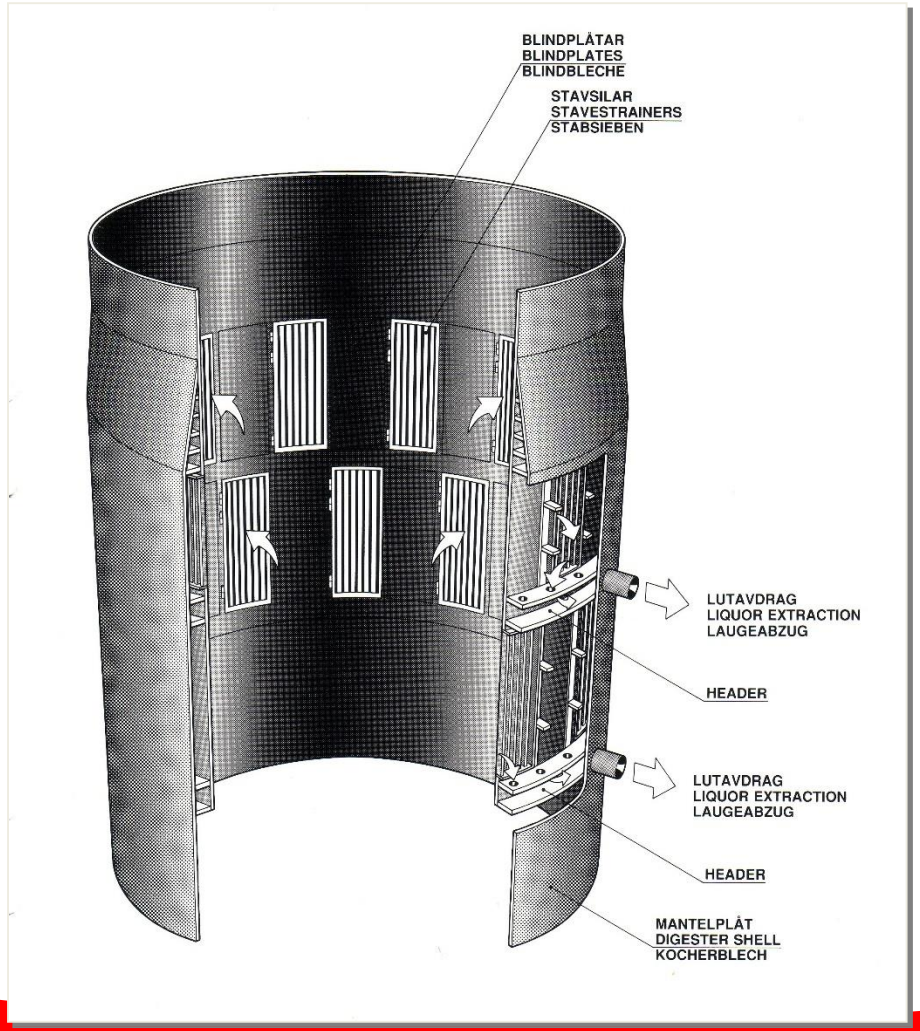
- **The trim cooking circulation** strainers consist of two rings stave type screens with separate headers placed underneath. Sets of time controlled switching valve,***KS185 in the trim circulation***,continuously switch strainers,***at interval of about 90 seconds***, so that liquor is extracted from first the upper ring and then the lower ring. ***This procedure prevents the strainers becoming plugged.***

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Continuous digester

Trim circulation (C6)

TRIM STRAINER



- **The C6 circulation liquor** is pumped, by pump C6, through the trim circulation and returned to the chip column in the digester, via the central pipe witch discharges just above the trim circulation strainers.

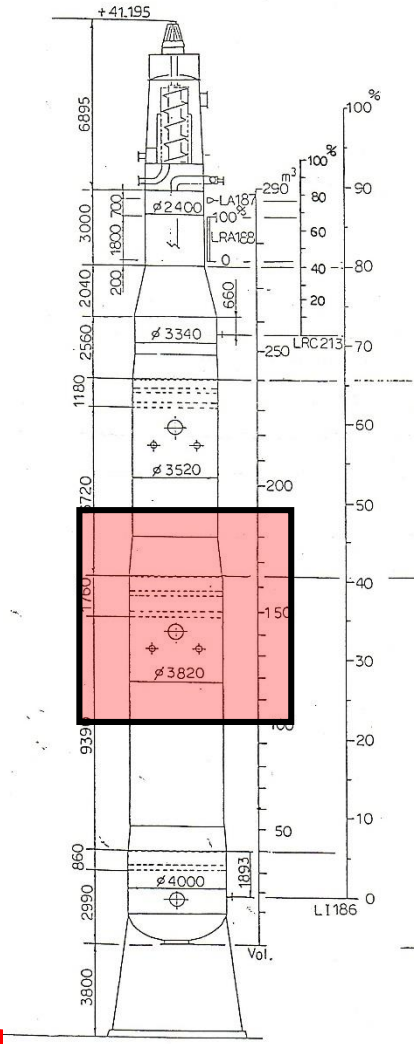
- **The differential pressure** ,d/p, across the upper strainer ring in the trim circulation is measured by ***PDIA184*** and lower strainer ring is measured by ***PDIA170***.

- **The flow** in the trim circulation at normal production,given in lts/min, is ***approximately 3 times the daily pulp production, given in ADT.***

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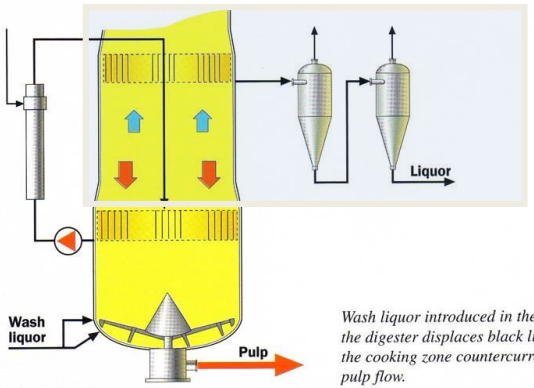
Continuous digester

Liquor Extraction



- The extraction strainers consists of two rings of stave type strainers with headers placed underneath.
  - ***60 % of total liquor*** through upper ring,***adjust with HIC195***
  - ***40% of total liquor*** through lower ring, ***adjust with HIC197***

- The differential pressure ,d/p
  - Upper ring : ***PDIA196***
  - Lower ring : ***PDIA217***

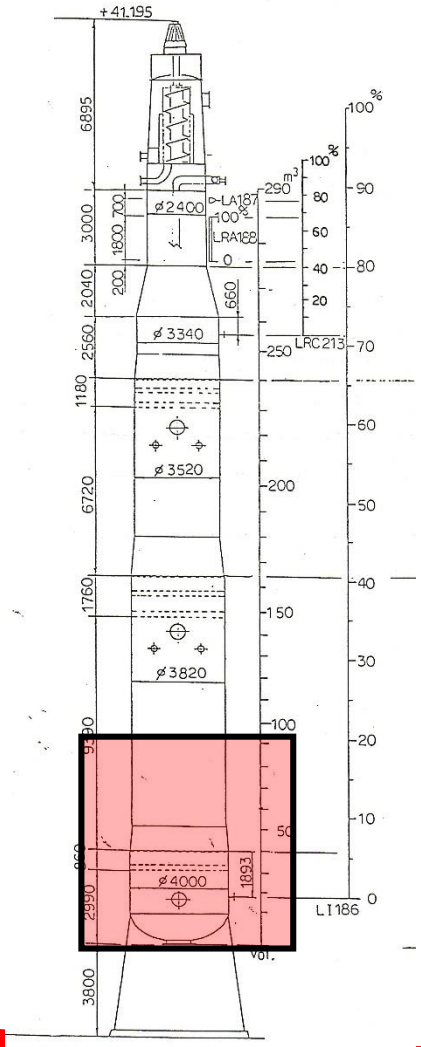


Wash liquor introduced in the bottom of the digester displaces black liquor from the cooking zone countercurrent to the pulp flow.

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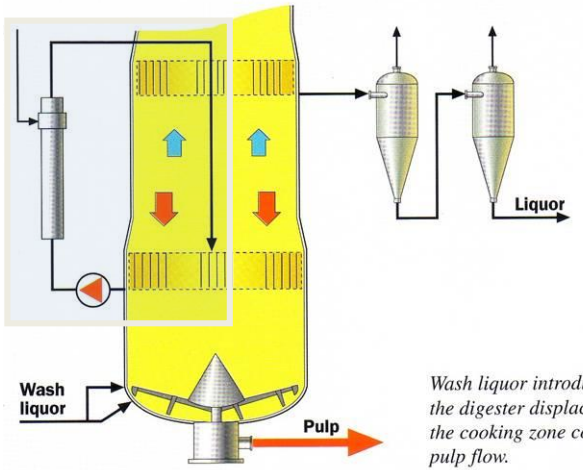
Continuous digester

Wash circulation (C8)



- The wash circulation ,C8, liquor is circulated by pump C8 through heater C48 where it is heated. It is returned to the digester via the *central pipe*. The amount of LP steam added, is controlled by ***TRC212***

- **The flow** in the wash circulation at normal production,given in lts/min, is ***approximately 2 times the daily pulp production, given in ADT***. The wash circulation flow is set and shown on FI203.

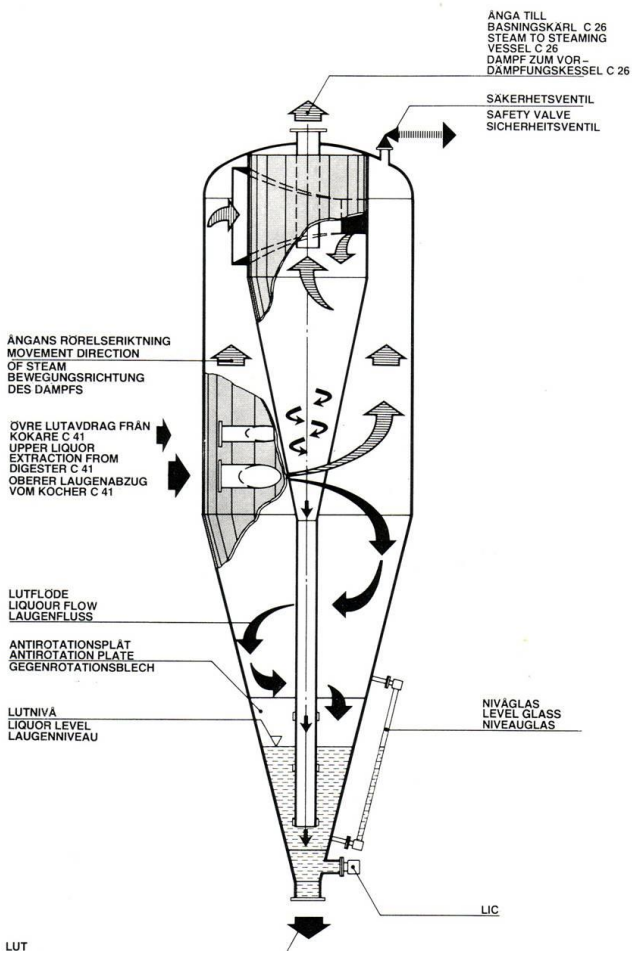


Wash liquor introduced in the bottom of the digester displaces black liquor from the cooking zone countercurrent to the pulp flow.

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Continuous digester

Flash cyclone (C50, C51)



- The temperature of the extraction liquor, which is about 150-160°C, is lowered in two stages in two cyclones C50 and C51.

- Level control
  - *First flash LIC218*
  - *Second flash LIC237*

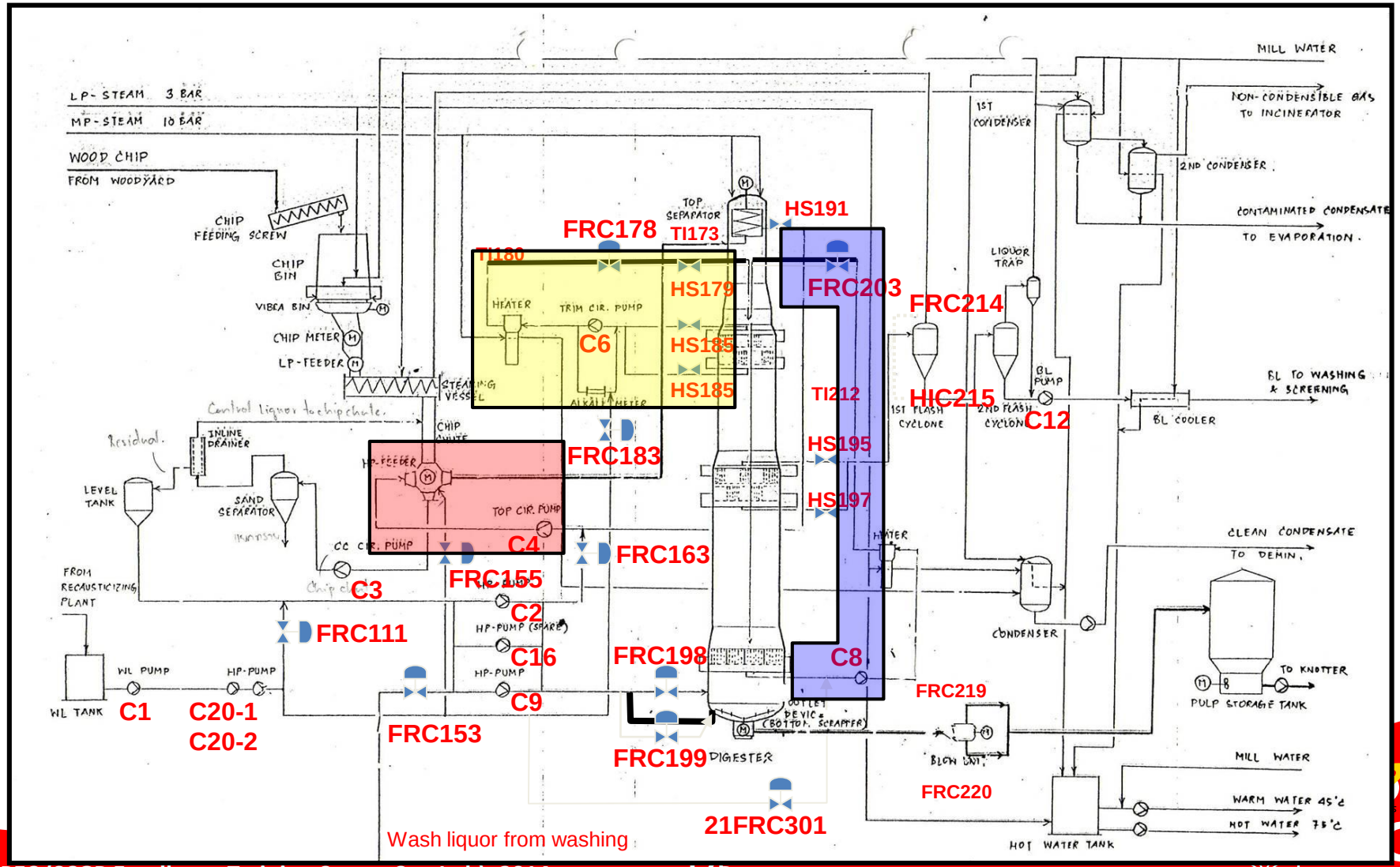
- Pressure range
  - *First flash 1.0-1.3 bar*
  - *Second flash 0.3-0.5 bar*

- Steam used :
  - *First flash to steaming vessel*
  - *Second flash to vibra bin or condenser system*

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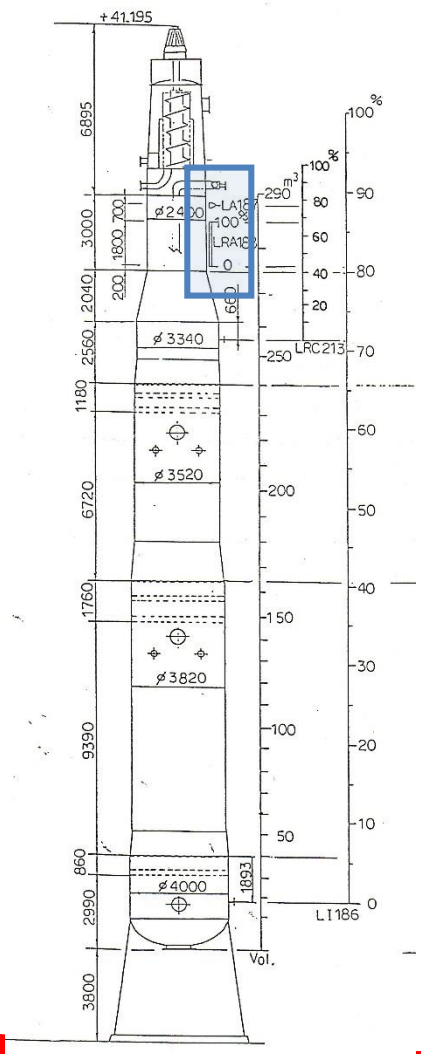
Continuous digester

Liquor circulation

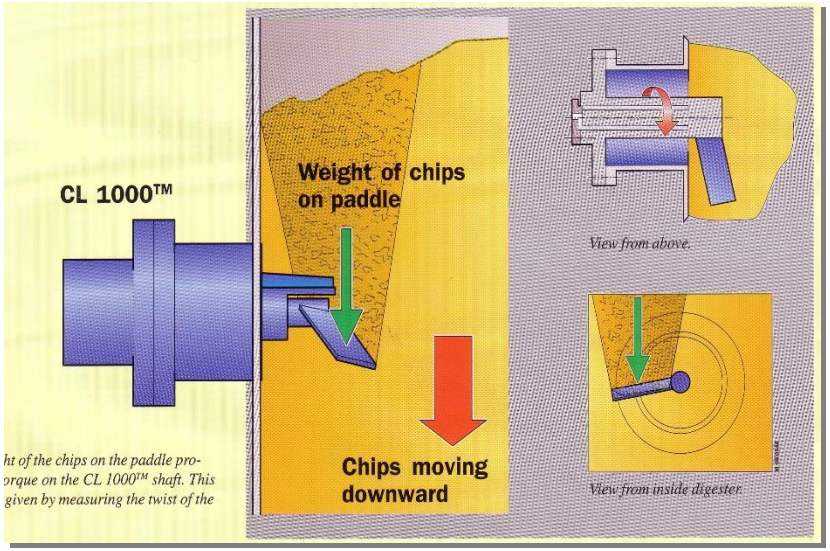


Continuous digester

Chip level control



- **The chip level** in digester is recorded on ***LRA188***. Under normal conditions it is maintained in the middle of the top cylindrical section approx. 2 meters below the top separator.

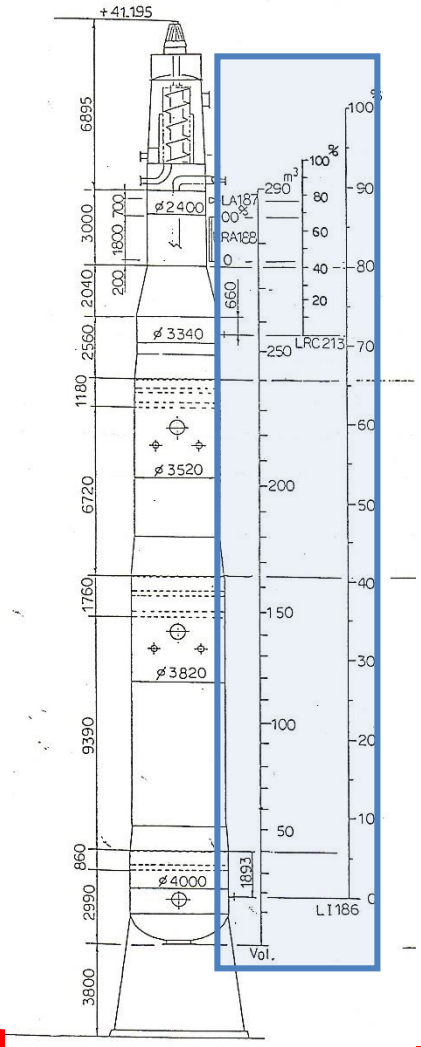


- The chip level is also recorded by *two mechanical chip level indicators*, ***LR189 and LR190***, which are mounted on the wall of top section of vessel.

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Continuous digester

Liquor level, Pressure, Temperature control



- **The liquor level** in maintained ***about 1 m below the chip level*** and is adjusted with ***LRC213***, which in turn controls flow controller ***FRC214,HIC215*** for the flow of extraction liquor to flash cyclone.

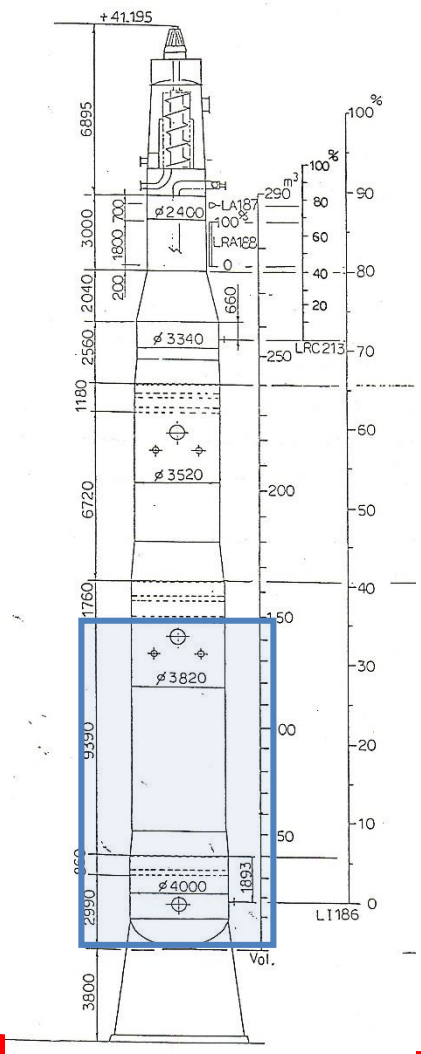
- At the top of the digester there is a relief line to the steaming vessel incorporating **pressure controller** ***PRCA192***, which open if the digester pressure is too high. If the pressure is too low compressed air will be added from compressor C58, ***PIC193***

- **The temperature** at the top of the digester is controlled by ***TRC173,*** which in turn controls the flow of medium pressure steam,***FRC172***, to the top of the digester.

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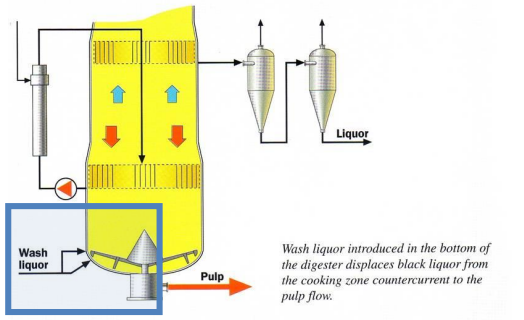
Continuous digester

Hi-Heat Washing



- The liquor in the pulp is displaced with weaker wash liquor. The wash liquor is pumped ,by pump C9, into the digester via nozzles at the bottom. The flow of wash liquor to the nozzles is controlled by ***FRC198*** and ***FRC199***

- From the bottom of the digester, the displacement liquor flows upwards and displaces the cooking liquor in the pulp.



Wash liquor introduced in the bottom of the digester displaces black liquor from the cooking zone countercurrent to the pulp flow.

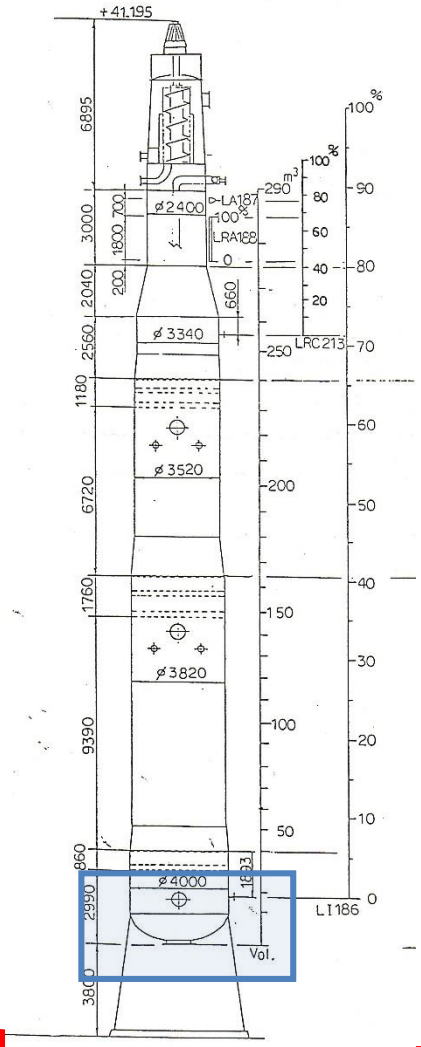
- The cooking liquor is extracted through the extraction strainers to the flash system and on to the evaporator plant.

- Optimize temperature of blow pulp = 75-85°C (not effect to pulp strength)

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Continuous digester

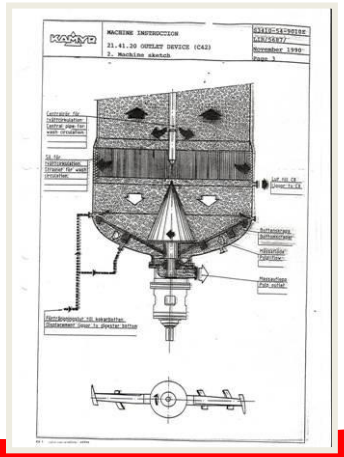
Outlet device (C42)

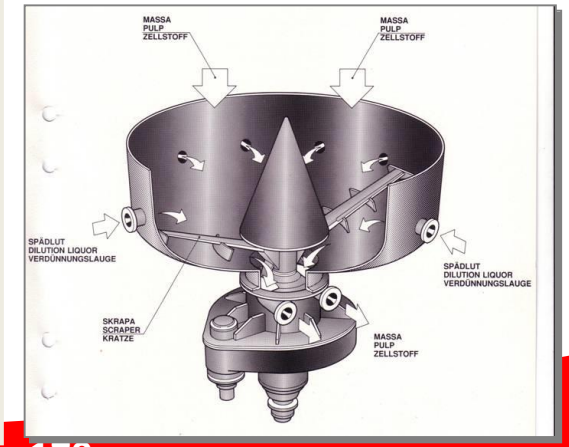


- Rotating outlet device, C42, is mounted in the bottom of the digester. **To obtain a uniform pulp quality**. The speed of the scrapper is adjustable from ***3 – 7.5 rpm.***

- The pressure difference over outlet device (C42) between the bottom of the digester and the outlet, ***shown on PDIA207***, is a very useful parameter when judging the discharge consistency.

- The speed of the scrapper effects both the concentration of the pulp being fed out and the chip level in the digester

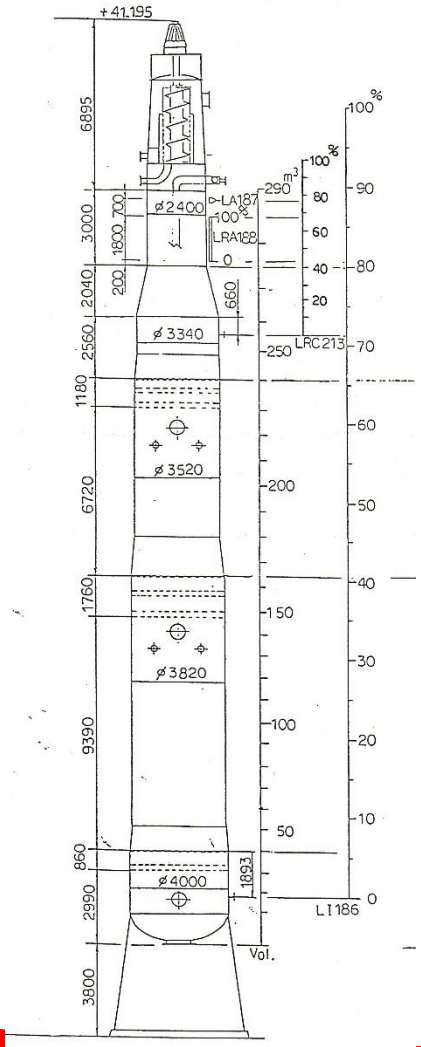




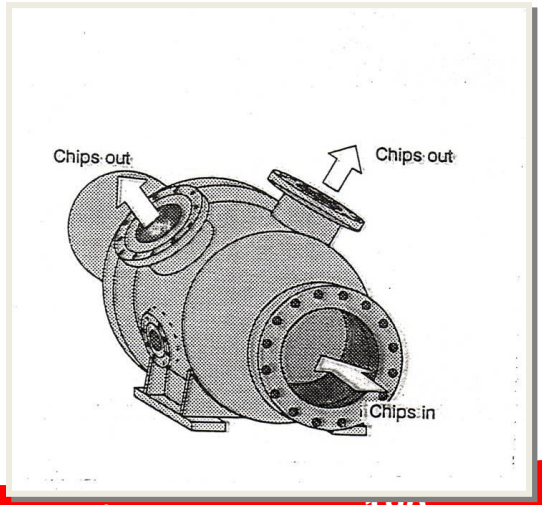
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Continuous digester

Blow unit (C43)



- The blow unit is equipped with blow nozzles and an agitator. ***The agitator can work as a consistency indicator as amperage of the motor varies with the consistency***. The agitator keeps the blow nozzles clean from agglomerated chips



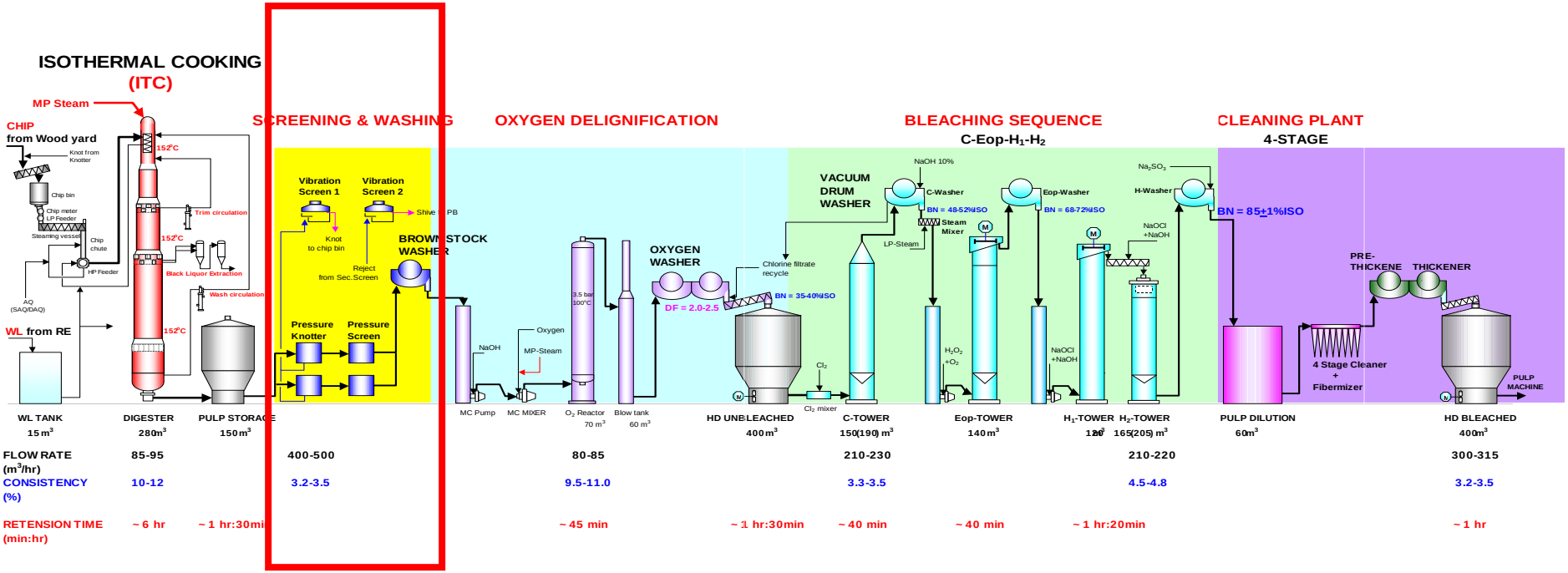
- The pulp discharge is controlled by flow control valves FRC219 or FRC220. The parallel FRC219 and FRC220 can be flushed with high pressure liquor via valves HS222 and HO223.

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PULP SCREENING process



PULP SCREENING process



	WL TANK	DIGESTER	PULP STORAGE					HD UNBLEACHED	C-TOWER	Eop-TOWER	H1-TOWER	H2-TOWER	PULP DILUTION	HD BLEACHED
	15 m^3	280 m^3	150 m^3					400 m^3	150(190) m^3	140 m^3	120 m^3	165(205) m^3	60 m^3	400 m^3
FLOW RATE (m^3/hr)		85-95		400-500		80-85			210-230			210-220		300-315
CONSISTENCY (%)		10-12		3.2-3.5		9.5-11.0			3.3-3.5			4.5-4.8		3.2-3.5
RETENSION TIME (min:hr)		~ 6 hr	~ 1 hr:30min			~ 45 min		~ 1 hr:30min	~ 40 min	~ 40 min	~ 1 hr:20min			~ 1 hr

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Effect of reject rate on screening Efficiency

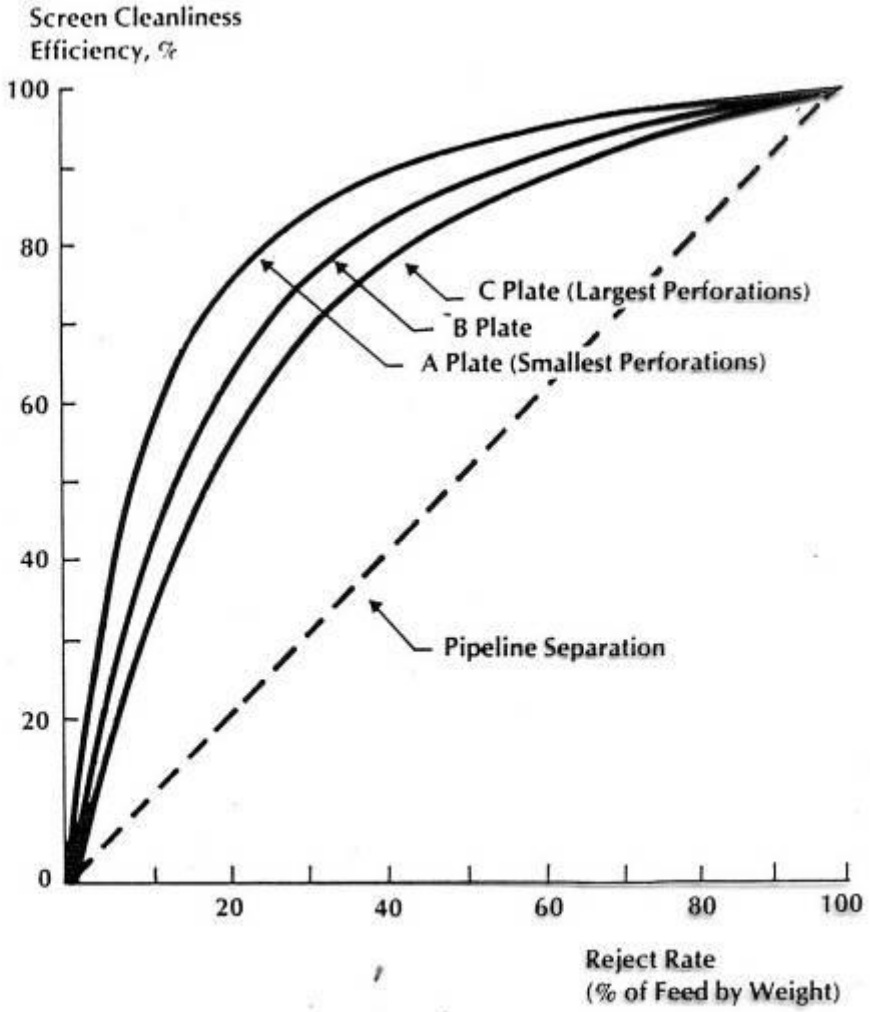


Figure 6.36 Effect of reject rate on screening efficiency.

Screen Plate Parameters

- ★ Screen Size
- ★ Screen Pattern – Slot / Hole
- ★ Slot / Hole Size
- ★ Opening area , %
- ★ Profile

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Screen plate pattern designs

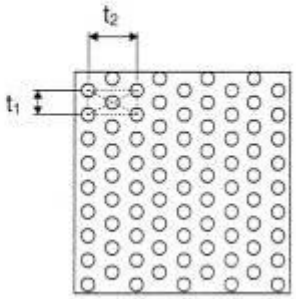
Hole / Slot / Bar Profile

= proportion of the slot or hole area compared with the area of the whole screen basket in percent.

Example: Fiedler HF 500,
i.d. = 495 mm, height = 500 mm

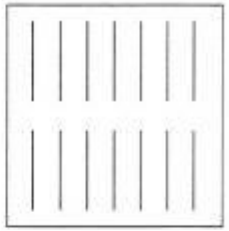
• drilled:

holes 1,6 mm
spacing 2,8 mm : 5,6 mm (t_1 : t_2)
open area = 25 %



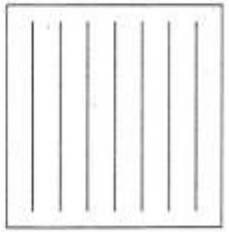
• milled:

slots 0,25 mm
spacing 3,2 mm
open area = 4,4 %

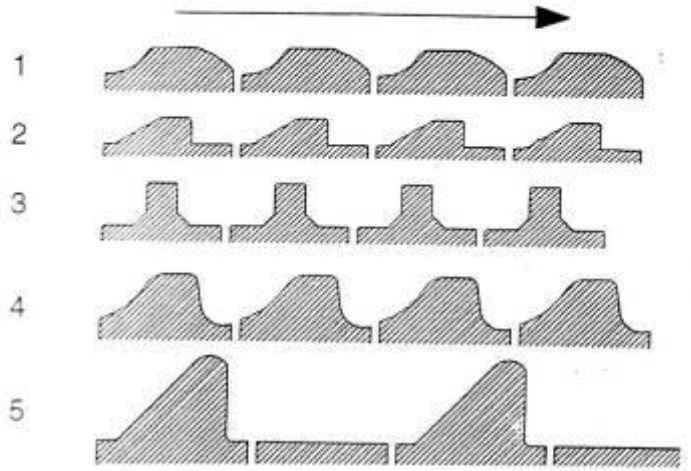


Bar-Screen:

slots 0,25 mm
spacing 3,25 mm
open area = 6,7 %



April 1997 Jürgen Lehmann

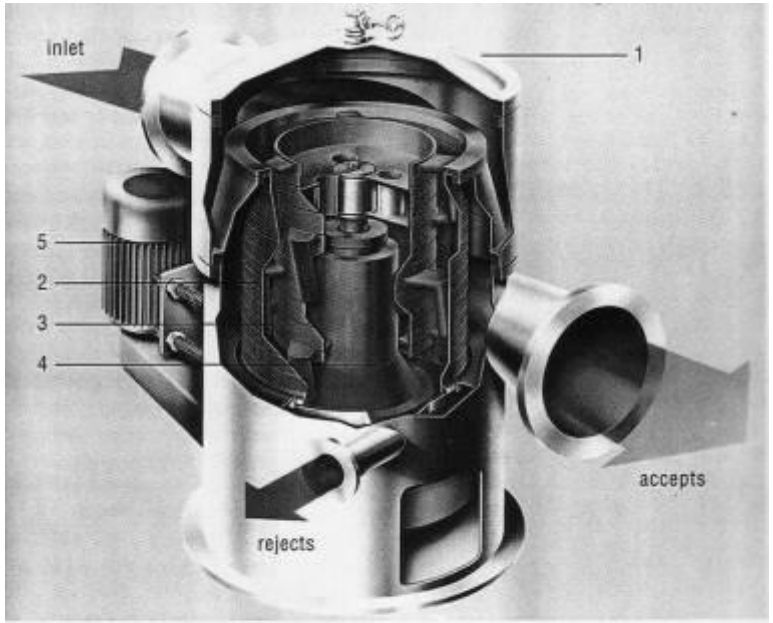


1 Fiedler High-Consistency-profile
2 APS-profile (Ahlstrom / CAE)
3 Lehmann-profile (Black Clawson)
4 Fiedler TOP-Screen profile
5 Voith-VV

April 1997 Jürgen Lehmann

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Screen plate pattern designs



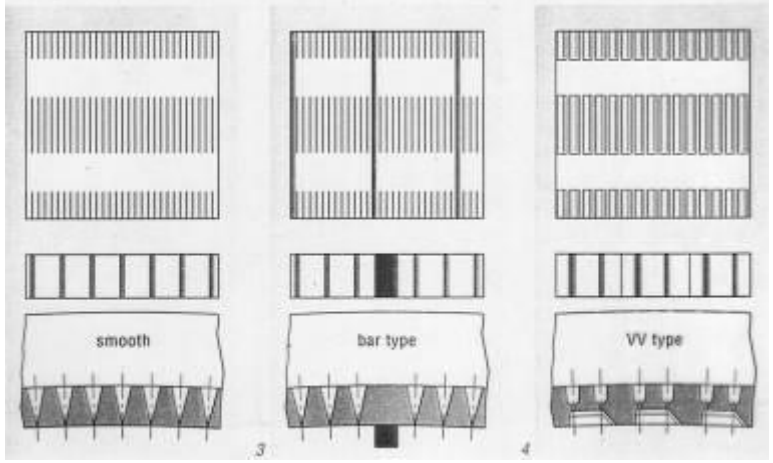
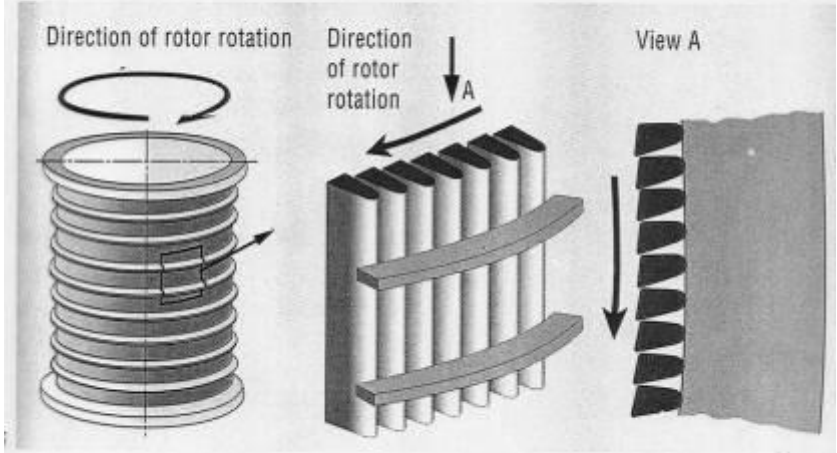
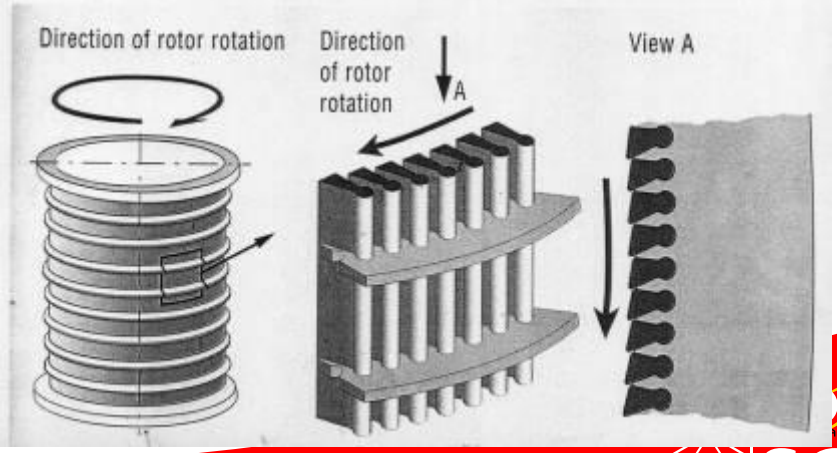


Fig. 5:
Principle of the conventional bar-type basket.

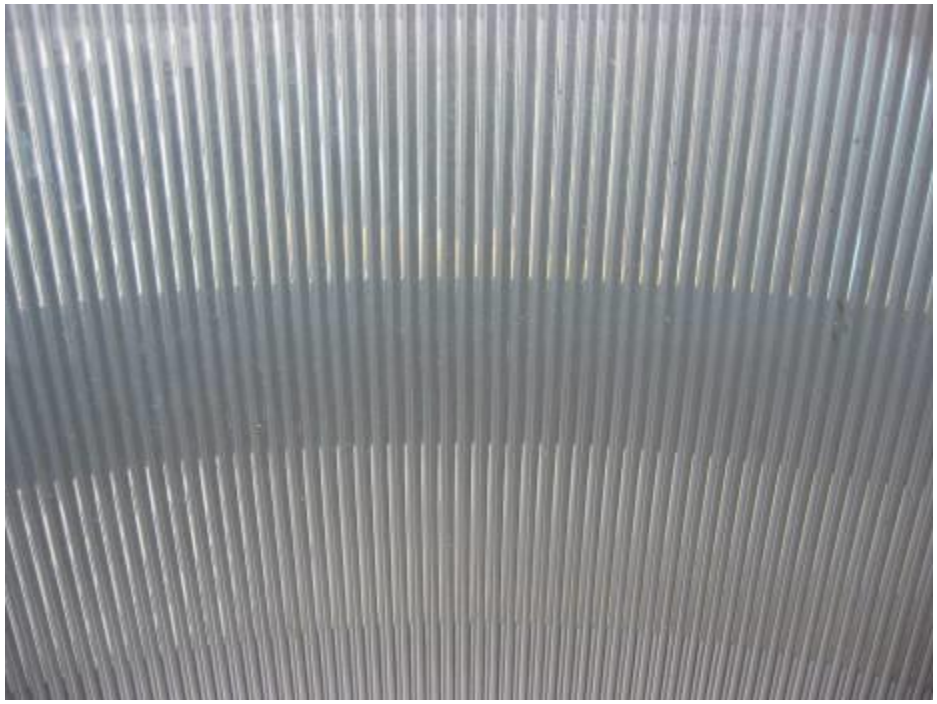
Fig. 6:
Principle of the C-bar screen basket.



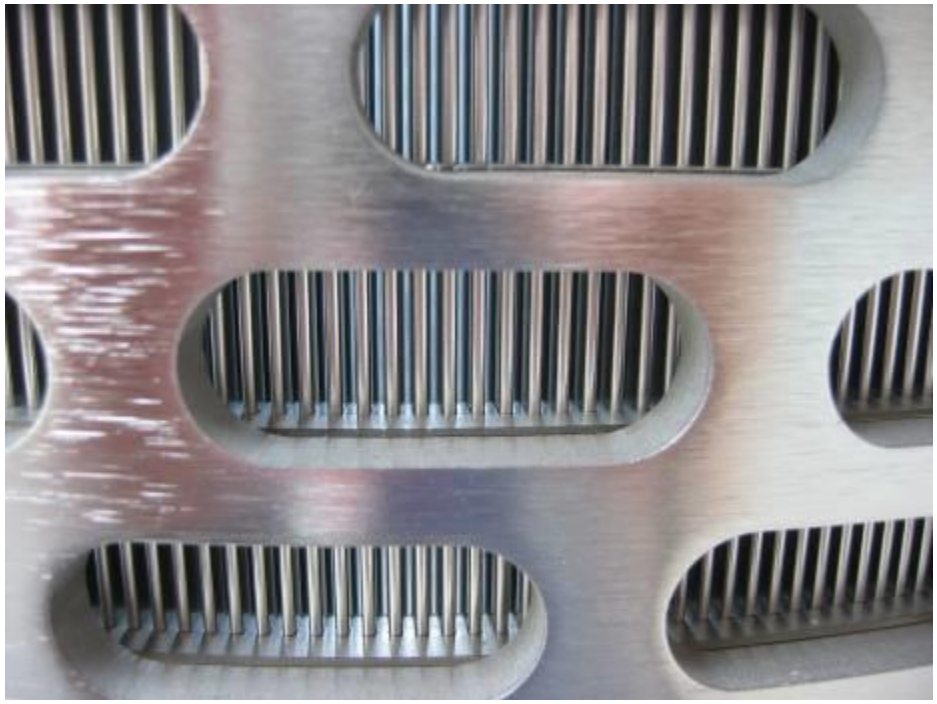


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Screen plate pattern designs



SLOT TYPE



BAR TYPE

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A Basic Guide to Open Area for SCREEN
Plate/Cylinder Selection

OPENING AREA Example :



Primary screen



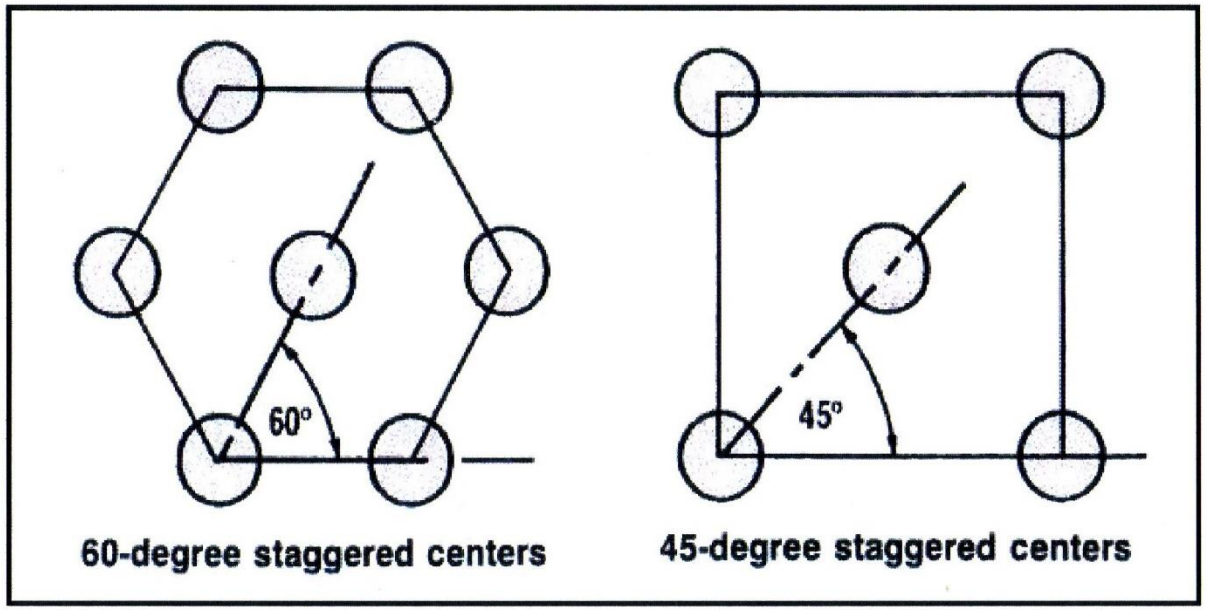
Knotter

A Basic Guide to Open Area for SCREEN

Plate/Cylinder Selection

Hole Patterns

1. *45*-degree staggered centers
2. *60*-degree staggered centers



60-degree staggered centers

45-degree staggered centers

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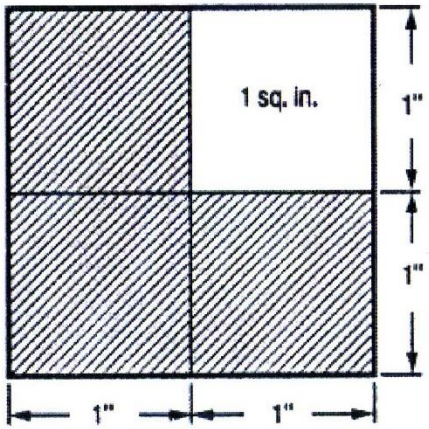
A Basic Guide to Open Area for SCREEN

Plate/Cylinder Selection

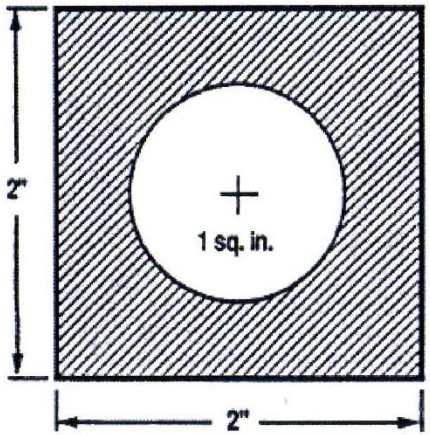
OPENING AREA

Opening area is a perforating trade term used to describe the number of holes per square inch.

The opening area is expressed as the percentage of open space in relation to the total area.



25% open area

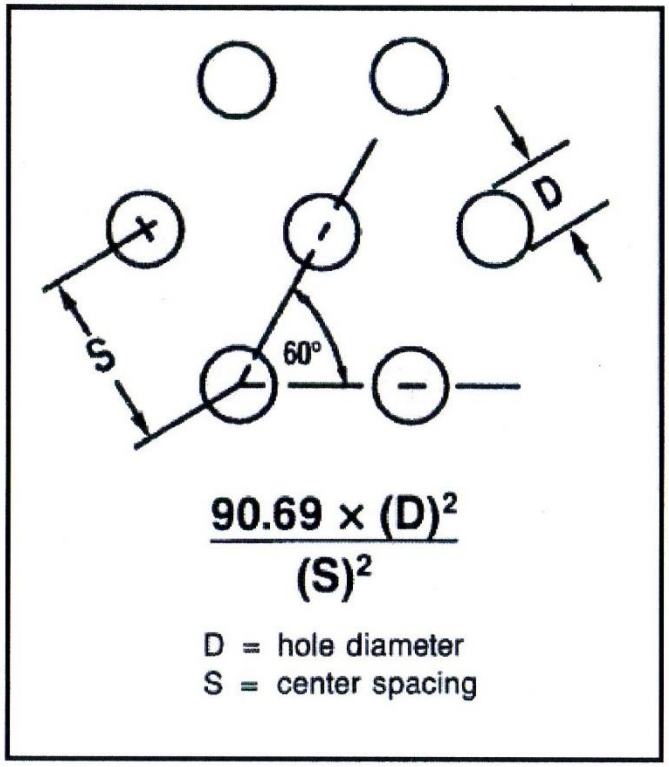


25% open area

A Basic Guide to Open Area for SCREEN

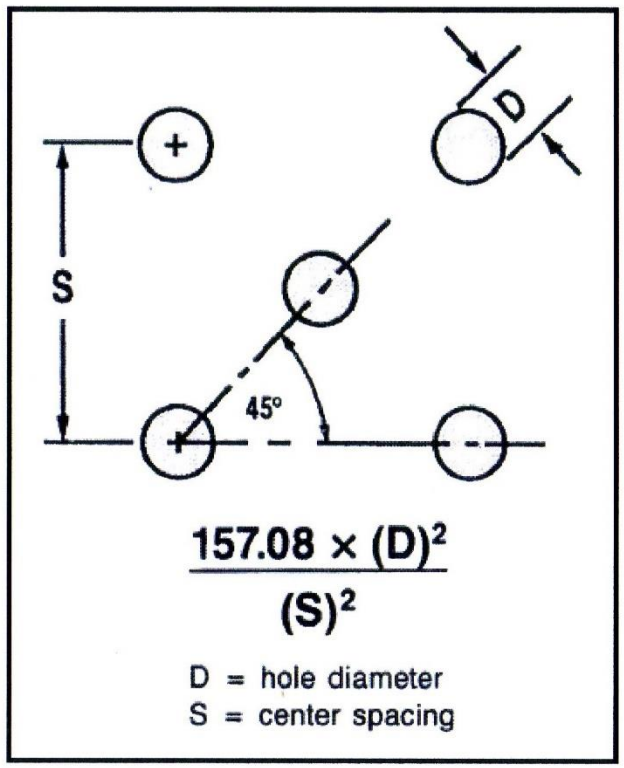
Plate/Cylinder Selection

OPENING AREA FORMULAR



$$\frac{90.69 \times (D)^2}{(S)^2}$$

D = hole diameter
S = center spacing



$$\frac{157.08 \times (D)^2}{(S)^2}$$

D = hole diameter
S = center spacing

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A Basic Guide to Open Area for SCREEN

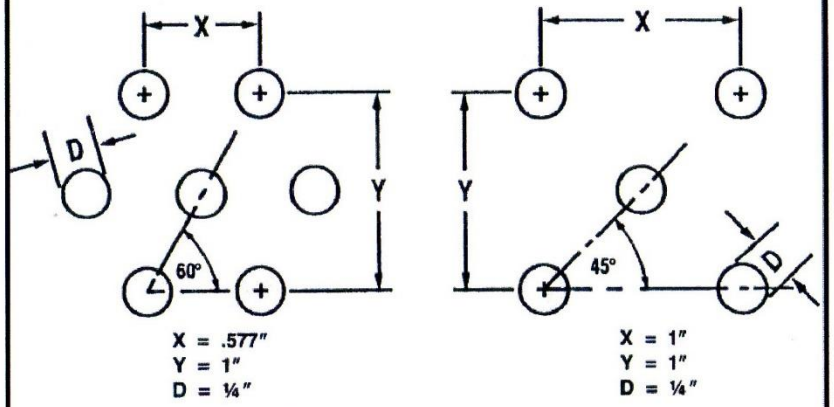
Plate/Cylinder Selection

OPENING AREA

Example :

$$\frac{157.08 \times (D)^2}{(X \times Y)}$$

D = hole diameter
X = center spacing in horizontal direction
Y = center spacing in vertical direction



X = .577"
Y = 1"
D = ¼"

X = 1"
Y = 1"
D = ¼"

Therefore, the pattern on the left has an open area of:

$$\frac{157.08 \times (¼)^2}{(.577 \times 1)} = 17\%$$

The pattern on the right has an open area of:

$$\frac{157.08 \times (¼)^2}{(1 \times 1)} = 9.8\%$$

Since these patterns are exact 60-degree and 45-degree staggered-centers patterns, the open area can be double-checked by using the original formulas measuring the dimensions of "S", "X" and "Y":

$$\frac{90.69 \times (¼)^2}{(.577)^2} = 17\%$$

$$\frac{157.08 \times (¼)^2}{(1)^2} = 9.8\%$$

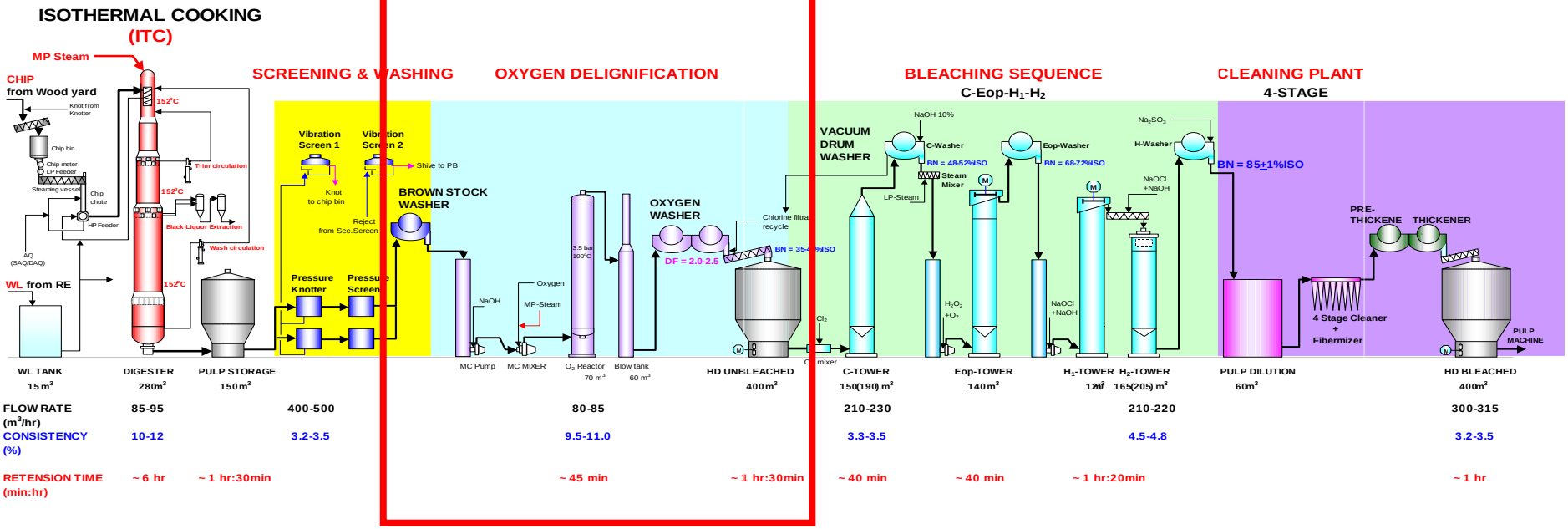
Rather than work directly on the screen plate, it is easier to get the needed center spacings from a pencil rubbing. Simply put a piece of paper on the screen plate and rub it with a No. 2 pencil. Measure the distances between ten holes in each direction and divide the total by ten. This will result in higher accuracy.

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PULP WASHING process and control



PULP WASHING process and control



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Topic outlines

1. **Objectives of pulp washing**
2. Principles of pulp washing
3. Washing equipments
4. Multi-stage counter-current washing
5. Washing parameters and efficiency

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Pulp washing Objective

- The purpose of brown stock washing is to displace and remove dissolved solids (both inorganic and organic) from the pulp. The two primary reasons why we must wash pulp are :

Alkali, Black Liquor

1. Remove residual liquor which would contaminate pulp pulp going to Bleaching process
2. Recover the maximum amount of spent cooking chemicals and dissolved wood solids from the fiber with minimum dilution, making chemical pulping economical.

Chemical recovery, Bleaching

Topic outlines

1. Objectives of pulp washing
2. **Principles of pulp washing**
3. Washing equipments
4. Multi-stage counter-current washing
5. Washing parameters and efficiency

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WASHING MECHANISMS

Washing - Simply bulk removal of liquor surrounding the pulp fibers.

Leaching - Dissolved solids & Chemicals interior of the pulp fibers diffuse into the surrounding liquor before they can be removed.

WASHING OPERATION

1. Displacement washing

2. Dilution/Extraction washing

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Principle of displacement washing

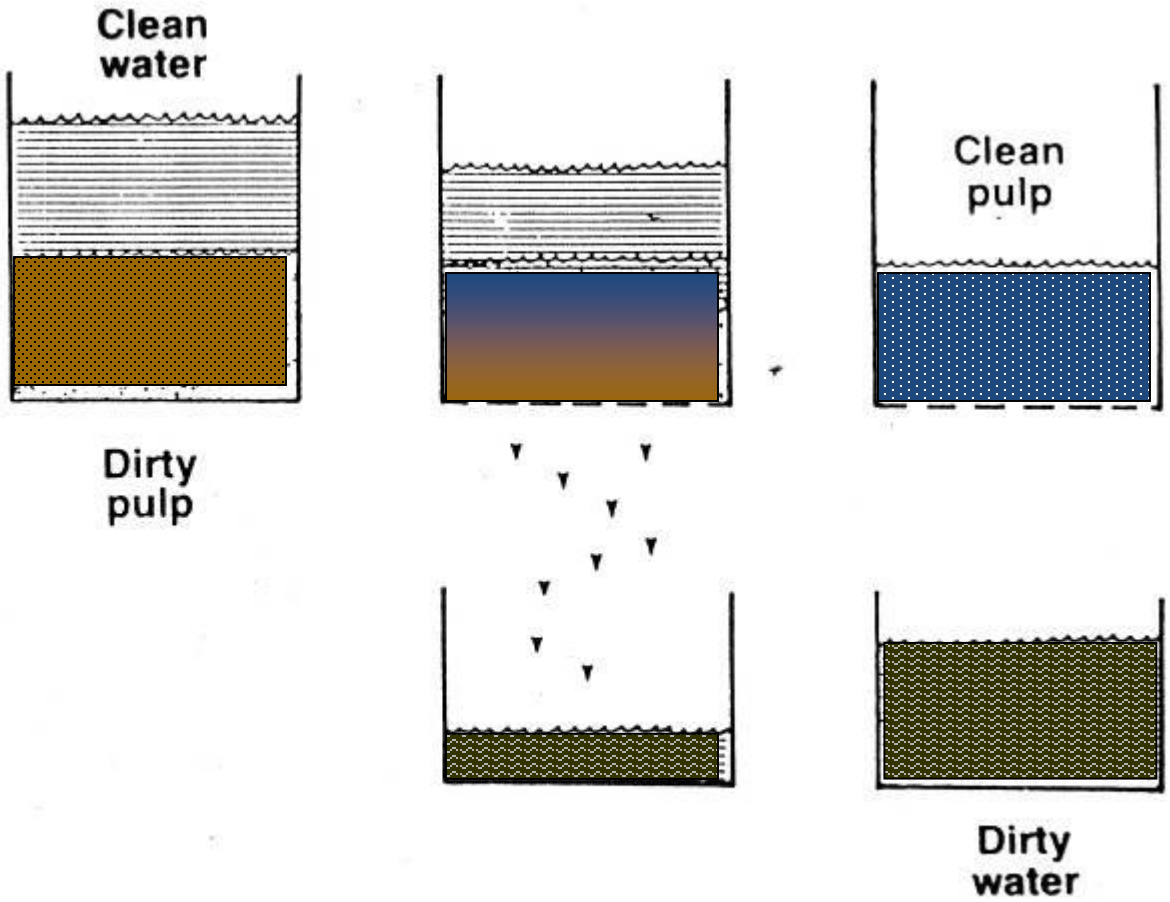


Figure 6.10 Principle of displacement washing.

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Principle of dilution/extraction washing

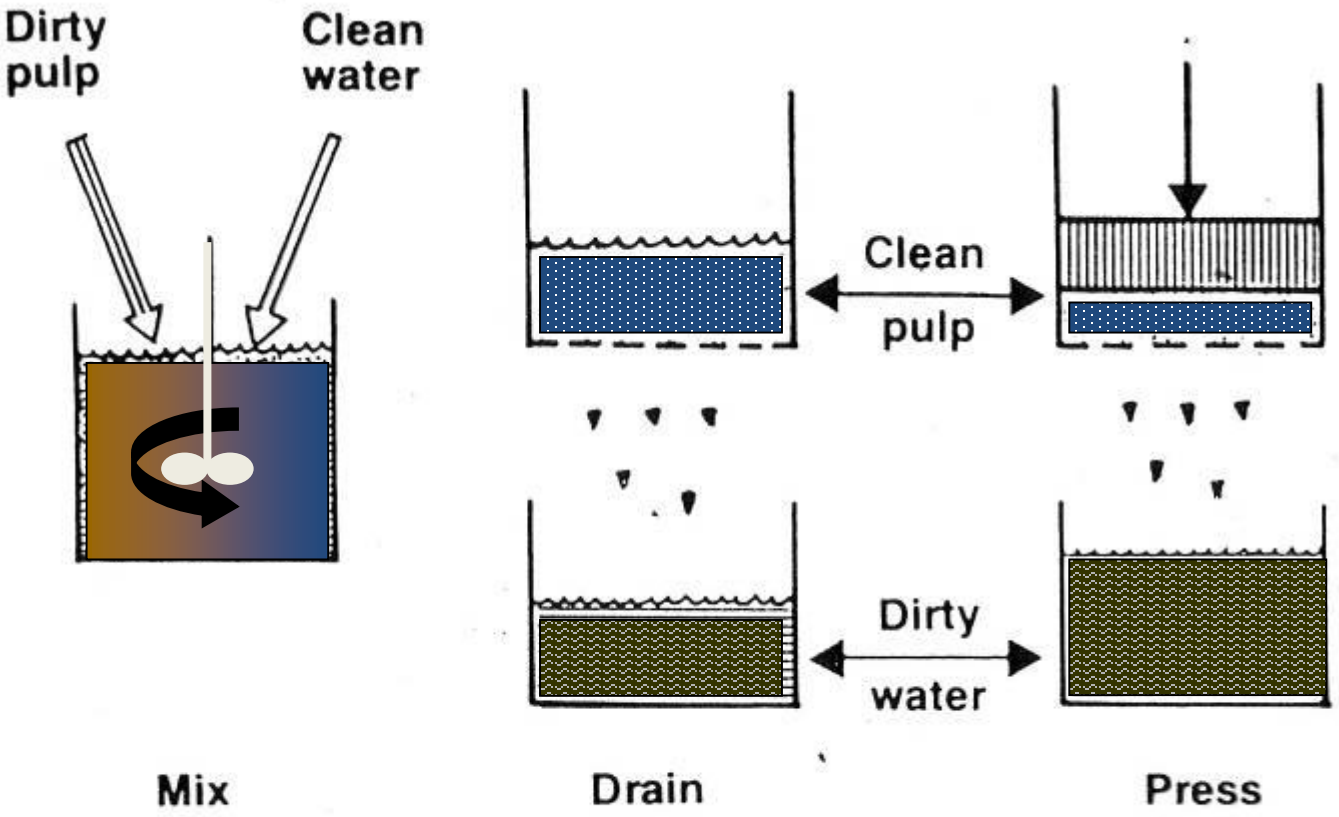


Figure 6.9 Principle of dilution/extraction.